TUGON: TECHNOLOGY-UNIFIED GOVERNMENT OPERATIONS NETWORK - A MOBILE-BASED AI-POWERED COMPLAINT & INCIDENT MANAGEMENT WITH VEHICLE DISPATCH SYSTEM FOR THE MAYOR'S ACTION TEAM OF THE MUNICIPALITY OF LILIW, LAGUNA

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A Capstone Project
Presented to the Faculty
of the College of Computer Studies
Laguna State Polytechnic University

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In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Science in Information Technology

LAGUNA STATE POLYTECHNIC UNIVERSITY
COLLEGE OF COMPUTER STUDIES
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Philippines

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The Capstone Project entitled "**TUGON: TECHNOLOGY-UNIFIED GOVERNMENT OPERATIONS NETWORK - A MOBILE-BASED AI-POWERED COMPLAINT & INCIDENT MANAGEMENT WITH VEHICLE DISPATCH SYSTEM FOR THE MAYOR'S ACTION TEAM OF THE MUNICIPALITY OF LILIW, LAGUNA,"** prepared and submitted by **JUSTINE CARL V. GRINDULO**, **MAZEL CATHERINE P. JOYA**, and **GEOFF P. SANDOVAL** in partial fulfillment of the requirements for the degree of **BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY**, Major in **WEB AND MOBILE APPLICATION DEVELOPMENT,** is hereby recommended for approval and acceptance.

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ABSTRACT

The Mayor's Action Team (MAT) of Liliw, Laguna faces operational challenges due to manual complaint processing, undocumented verbal vehicle dispatching, and lack of digital location sharing or family safety features. To resolve these inefficiencies, this study presents the design, development, testing, and evaluation of TUGON (Technology-Unified Government Operations Network), a mobile-based AI-powered complaint and incident management system integrated with a vehicle dispatch platform. The system centralizes emergency reporting, automates responder dispatch and provides real-time GPS tracking capabilities for the municipality.

Developed using an Agile methodology, TUGON features a Flutter mobile app and a Next.js administrative web dashboard. Ten core features underwent Black-Box testing including AI-powered urgency classification using Gemini 3.1 Flash, GPS tracking, automated dispatch queues, Face ID geofenced attendance and an interactive incident heatmap. User evaluation was conducted among 203 respondents comprising 181 citizens, 18 MAT responders, 3 dispatchers and 1 administrator using the Technology Acceptance Model (TAM).

Black-Box testing achieved a 100% success rate across 480 test steps proving operational readiness and system reliability. Additionally, the TAM evaluation yielded a grand mean of 4.94 ("Strongly Agree") across six core constructs. Findings confirm that TUGON successfully modernizes local emergency response workflows while maintaining high user acceptability and functional stability.

Keywords: TUGON, Mayor's Action Team, Complaint and Incident Management, Vehicle Dispatch, Artificial Intelligence, Technology Acceptance Model (TAM), Agile Methodology.

TABLE OF CONTENTS

Commented [JG4]: Okay na po ba tong format ng TOC?

Preliminary Pages **Page**

Title Page i

Vision, Mission, Quality Policy, Goals, and Objectives ii

Approval Sheet iii

Acknowledgements iv

Abstract v

Table of Contents vi

List of Tables xiii

List of Figures xv

List of Equations xvi

List of Appendices xvii

Definition of Terms xviii

Technical Terms xviii

Operational Terms xx

CHAPTER I

INTRODUCTION AND BACKGROUND 1

1.1 Project Context 1

1.2 Research Objectives 5

1.2.1 General Objective 5

1.2.2 Specific Objectives 6

1.3 Conceptual Framework 7

1.3.1 Components of the Conceptual Framework 7

1.3.2 Conceptual Framework Diagram 8

1.4 Project Purpose 10

1.4.1 Intended Beneficiaries 12

1.5 Scope and Limitations of the Study 15

1.5.1 Scope of the Study 15

1.5.2 Limitations of the Study 17

CHAPTER II

REVIEW OF RELATED LITERATURE 19

2.1 Studies and Literature 19

2.1.1 Selection of Related Literature 19

2.1.2 Review and Analysis of Studies 20

Digital Transformation in Local Government Operations 20

Mobile-Based e-Government and Citizen Service Delivery 23

Complaint Management Systems in Local Government 25

Communication Systems for Government-Citizen Engagement 28

Citizen Complaint Prioritization and Workflow Automation 31

Public Service Response Efficiency and Operational Optimization 34

AI and NLP for Complaint Classification 36

Filipino Language Processing and Multilingual NLP 39

Notification Systems and Real-Time Alerting in Public Service Platforms 42

AI-Enabled Decision Support for Local Government Services.....46

Vehicle Dispatch and Fleet Management in Government Operations.....49

GPS-Based Tracking and Location-Aware Public Service Systems.....52

Real-Time Monitoring and Accountability in Public Service Systems.....55

Geospatial Technologies for Local Disaster Preparedness.....58

Disaster Response, Heatmaps, and Emergency Monitoring.....62

Anomaly Detection in Vehicle and IoT Systems.....65

Backend Infrastructure and Cloud-Based Systems.....69

Data Privacy and Security in Government IS.....73

Agile Methodology in Government System Development.....77

Technology Acceptance and User Adoption in Government Systems.....80

2.1.3 Thematic Organization.....84

2.1.4 Relevance to the Present Project.....84

2.1.5 Citation and Referencing.....85

2.2 Related Systems and Technology Gap Analysis.....85

2.2.1 Identification of Related Systems.....85

2.2.2 System Feature Comparison.....90

2.2.3 Analysis of Existing Systems.....91

2.2.4 Technology Gap.....94

2.2.5 Proposed Response to the Gap.....96

2.3 Synthesis.....97

2.3.1 Integration of Findings 97

2.3.2 Identified Gap 99

2.3.3 Contribution of the Proposed Project 101

CHAPTER III

RESEARCH METHODOLOGY 108

3.1 Materials and Methods 108

3.1.1 Research and Fact-Finding Activities 108

3.1.2 System Development Activities 109

3.1.3 Resources and Technologies 109

3.2 Research Design 109

3.2.1 Rationale for the Selected Research Design 110

3.3 Population of the Study 110

3.4 System Testers 111

3.4.1 Profile of System Testers 111

3.5 Respondents for User Evaluation 112

3.5.1 Distribution of Respondents 112

3.6 Sampling Design 113

3.7 Data Collection Instrument 114

3.7.1 Instrument Development 114

3.7.2 Instrument Validation 115

3.7.3 Administration of the Instrument 115

3.8 Project Design 115

3.8.1 System Requirements 116

3.8.2 System Design 116

3.8.3 System Architecture 116

3.9 Software Development Model 117

3.10 System Architecture 120

3.10.1 Hardware and Software Requirements 121

3.10.2 System Components and Modules 121

3.10.3 Database Structure 122

3.10.4 User Interface Design 123

3.10.5 System Communication and Interaction 124

3.11 System Testing and Evaluation Procedure 125

3.11.1 Testing Plan 125

3.11.2 Test Cases and Test Scenarios 125

3.11.3 Testing Data 126

3.12 System Testing 127

3.12.1 Test Cases 128

3.12.2 Testing Instrument 128

3.12.3 Validation of Testing Instrument 128

3.12.4 Interpretation of Test Results 128

3.13 System User Evaluation 129

3.13.1 User Evaluation Instrument 130

3.13.2 Instrument Source and Validation 130

3.13.3 User Evaluation Procedure 131

3.13.4 Evaluation Criteria 131

3.14 Statistical Treatment 132

Frequency and Percentage Distribution 132

Mean Formula 133

Weighted Mean 133

Standard Deviation Formula 135

CHAPTER IV

RESULTS AND DISCUSSIONS 136

4.1 General Presentation of Results 136

4.1.1 Presentation of Objective-Based Results 137

4.2 System Architecture and Development Results 138

4.2.1 System Overview 138

4.2.2 System Architecture 140

4.2.3 Hardware and Software Requirements 141

4.2.4 Database Structure 143

4.2.5 User-Interface and System Modules 144

4.2.6 Functional and Non-Functional Requirements 153

4.3 System Testing Results 155

4.3.1 Test Results by Requirements 166

4.3.2 Testing Summary 168

4.3.3 Discussion of Testing Results 170

4.4 System User Evaluation Results 171

4.4.1 Respondent Profile 173

4.4.2 Evaluation Results by Criteria 174

4.4.3 Statistical Results 178

4.4.4 Discussion of User Evaluation 180

4.5 Discussion and Interpretation of Findings 182

CHAPTER V

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS 185

5.1 Summary 185

5.2 Conclusions 186

5.3 Recommendations 187

References 189

LIST OF TABLES

Commented [JG5]: Tama po ba labeling ng table and figure numbers, based na po sya sa chapter kung san sya located?

Page

Table 2.1. System Feature Comparison of Related Systems and TUGON.....90

Table 3.1. IT Experts (Technical Evaluators).....111

Table 3.2. MDRRMO Experts (Domain Evaluators).....112

Table 3.3. Distribution of Respondents for User Evaluation.....112

Table 3.4. Five-Point Rating Scale.....131

Table 4.1. Hardware Requirements.....142

Table 4.2. Software Requirements.....143

Table 4.3. Requirement-to-Results Matrix.....154

Table 4.4. Black-Box Testing Results for TC-01.....156

Table 4.5. Black-Box Testing Results for TC-02.....157

Table 4.6. Black-Box Testing Results for TC-03.....158

Table 4.7. Black-Box Testing Results for TC-04.....159

Table 4.8. Black-Box Testing Results for TC-05.....161

Table 4.9. Black-Box Testing Results for TC-06.....162

Table 4.10. Black-Box Testing Results for TC-07.....163

Table 4.11. Black-Box Testing Results for TC-08.....164

Table 4.12. Black-Box Testing Results for TC-09.....165

Table 4.13. Test Results by Requirements.....166

Table 4.14. Testing Statistical Outcome.....169

Table 4.15. Level of Acceptability in Terms of Quality Factors 174

Table 4.16. Level of Acceptability in Terms of Perceived Usefulness..... 175

Table 4.17. Level of Acceptability in Terms of Perceived Ease of Use 175

Table 4.18. Level of Acceptability in Terms of Attitude Toward Using TUGON..... 176

Table 4.19. Level of Acceptability in Terms of Behavioral Intention to Use TUGON..... 177

Table 4.20. Level of Acceptability in Terms of Experience 178

Table 4.21. Statistical Results 179

LIST OF FIGURES

Page

Figure 1.1. Conceptual Framework of TUGON 8

Figure 3.1. System Architecture 116

Figure 3.2. Agile Development Methodology 118

Figure 3.3. Black-Box Testing Model (adapted by Will Kenton, 2025) 127

Figure 3.4. Technology Acceptance Model (TAM) 129

Figure 4.1. System Context Diagram 139

Figure 4.2. System Architecture 140

Figure 4.3. User Interface of Admin and MAT Officers - Web 147

Figure 4.4. User Interface of Responder - Mobile 150

Figure 4.5. User Interface of Citizen - Mobile 152

Figure 4.6. Distribution of Respondents 173

LIST OF EQUATIONS

Page

Equation 1. Slovin's Formula 113

Equation 2. Frequency and Percentage Distribution 132

Equation 3. Mean Formula 132

Equation 4. Weighted Mean Formula 132

Equation 5. Standard Deviation Formula 135

LIST OF APPENDICES

APPENDIX

A. TECHNICAL BACKGROUND

Transcribed Interview

Story Board

Flowchart

Use Case Diagram

System Architecture

System Context Diagram

Data Flow Diagram

Data Dictionary

Project Schedule

Input / Output / Report Screenshots

Hardware and Software Resources

System Testing Questionnaire

System Simulation Testing

User Acceptance Questionnaire

User's Manual

B. COMMUNICATION LETTERS AND FORMS

Communication Letter

ISO Forms

Rating Sheets

Summary of Recommendations

C. CERTIFICATE OF AGREEMENT / MOA / MOU

Memorandum of Agreement

D. CURRICULUM VITAE OF SYSTEM TESTERS

E. CURRICULUM VITAE OF RESEARCHERS

F. 10 PAGER

Turnitin Certification from Adviser (IMRAD)

DEFINITION OF TERMS

The researchers describe the terminology used in the development of the system, which can be beneficial for a thorough understanding of the key terms used in the study.

TECHNICAL TERMS

Agile Methodology	Agile Methodology is a software development approach that emphasizes iterative development, collaboration, continuous feedback, and adaptability to changing requirements.
Artificial Intelligence (AI)	Artificial Intelligence (AI) refers to computer systems capable of performing tasks that normally require human intelligence, such as decision-making, classification, and pattern recognition.
Attitude Toward Using	Attitude Toward Using refers to an individual's positive or negative feelings about performing the target behavior of using a system.
Backend-as-a-Service (BaaS)	Cloud computing architecture that provides managed backend services like real-time database synchronization and authentication, reducing the need for custom server engineering.
Behavioral Intention to Use	Behavioral Intention to Use refers to the strength of an individual's intention to perform a specified behavior, such as using a system.
Black-Box Testing	Black-Box Testing is a software testing method that evaluates system functionality based on input and output behavior without examining the internal source code.

Citizen User Tracking	A location-aware mechanism that enables users to continuously share or view real-time geographic position data via mobile GPS components.
Complaint and Incident Management	A software workflow system designed to capture, categorize, prioritize, assign, and track reported public concerns and emergency events.
Family Safety	Family Safety refers to a mobile application feature that allows family members to share their real-time locations and communicate emergency alerts within a private family group.
Flutter and Dart	The cross-platform UI software development kit (Flutter) and programming language (Dart) used to build mobile applications for iOS and Android.
GPS Tracking	GPS Tracking is a technology that uses satellite signals to determine and monitor the geographic location of a device or vehicle in real time.
JWT (JSON Web Token)	A secure, URL-safe method for representing claims between two parties, used in software systems to handle user authentication and ensure data privacy compliance.
Natural Language Processing (NLP)	Natural Language Processing (NLP) is a branch of artificial intelligence that enables computers to understand, interpret, and process human language.
Perceived Ease of Use	Perceived Ease of Use is the degree to which a person believes that using a particular system would be free of effort.

Perceived Usefulness	Perceived Usefulness is the degree to which a person believes that using a particular system would enhance his or her job performance.
Role-Based Access Control (RBAC)	Role-Based Access Control (RBAC) is a security mechanism that restricts system access based on assigned user roles and permissions.
Supabase and PostgreSQL	The real-time backend platform (Supabase) and relational database (PostgreSQL) used to manage structured data synchronization across networked applications.
Technology Acceptance Model (TAM)	Technology Acceptance Model (TAM) is a theoretical framework used to predict and explain user acceptance of information technology based on Perceived Usefulness and Perceived Ease of Use.
Vehicle Dispatch Management	An operational module responsible for coordinating, scheduling, assigning, and monitoring fleet vehicles and operators responding to active service requests.

OPERATIONAL TERMS

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Agile Methodology	In this study, Agile Methodology refers to the iterative development approach used in designing, developing, testing, and refining TUGON through continuous stakeholder consultation and incremental system improvement.
Artificial Intelligence (AI)	In this study, AI refers to the technologies integrated into TUGON for complaint classification, urgency prioritization, anomaly detection, and decision-support functions.

Attitude Toward Using	In this study, Attitude Toward Using refers to the overall positive or negative evaluation of MAT personnel and citizens toward the use of TUGON in their daily operations.
Backend-as-a-Service (BaaS)	In this study, BaaS refers to the cloud backend services used in TUGON to support real-time data synchronization, user authentication, and serverless operations without custom server management.
Behavioral Intention to Use	In this study, Behavioral Intention to Use refers to the likelihood that MAT personnel and citizens will use and continue using TUGON for their complaint, incident, and dispatch activities.
Black-Box Testing	In this study, Black-Box Testing refers to the testing procedure used to evaluate the functionality and correctness of the major features of TUGON.
Citizen User Tracking	In this study, Citizen User Tracking refers to the feature in TUGON that enables citizens to share and monitor their live location or report location through an interactive map.
Complaint and Incident Management	In this study, Complaint and Incident Management refers to the module in TUGON responsible for receiving, classifying, prioritizing, and managing citizen complaints and emergency incident reports.
Family Safety	In this study, Family Safety refers to the feature in TUGON that enables citizens to create a family circle, share live locations with family members, and send emergency alerts during critical situations.
Flutter and Dart	In this study, Flutter and Dart refer to the cross-platform development framework and programming language used to develop the TUGON mobile application for citizens and responders.

GPS Tracking	In this study, GPS Tracking refers to the real-time monitoring of municipal vehicles and citizen-reported locations within TUGON.
JWT (JSON Web Token)	In this study, JWT refers to the secure token-based authentication mechanism used in TUGON to authorize API requests and safeguard user session data.
Natural Language Processing (NLP)	In this study, NLP refers to the AI-based complaint classification mechanism used in TUGON to categorize and prioritize complaints written in Filipino and Taglish.
Perceived Ease of Use	In this study, Perceived Ease of Use refers to the extent to which MAT personnel and citizens believe that using TUGON is easy to learn, navigate, and operate.
Perceived Usefulness	In this study, Perceived Usefulness refers to the extent to which MAT personnel and citizens believe that using TUGON enhances their effectiveness in complaint management, vehicle dispatch, and emergency response.
Role-Based Access Control (RBAC)	In this study, RBAC refers to the access management mechanism used in TUGON to control system permissions for citizens, drivers, MAT officers, and administrators.
Supabase and PostgreSQL	In this study, Supabase and PostgreSQL refer to the real-time database management infrastructure implemented in TUGON to store and sync operational records across web and mobile clients.
Technology Acceptance Model (TAM)	In this study, Technology Acceptance Model (TAM) refers to the framework used in evaluating the user acceptance and satisfaction of TUGON.

Vehicle Dispatch Management	In this study, Vehicle Dispatch Management refers to the module in TUGON responsible for assigning, monitoring, and tracking municipal vehicles and drivers through GPS-enabled monitoring.

CHAPTER I

INTRODUCTION AND BACKGROUND

This chapter presents the project context, research objectives, conceptual framework, project purpose, and scope and limitations of the study. It establishes the rationale for developing TUGON as a response to the operational gaps of the Mayor's Action Team of the Municipality of Liliw, Laguna.

1.1 Project Context

Effective local government administration is a matter of institutional responsibility, public safety and citizen welfare. Local Government Units (LGUs) around the world are inundated with thousands of reports from citizens each day; they manage emergency deployments and operate fleets of government vehicles, tasks that without the structure of digital systems can lead to delays, misrouted information and accountability gaps that directly impact the communities they serve. Yigitcanlar et al. (2024) reported an exponential growth in AI applications in local governance since 2015 and found public service delivery, emergency response and resource management as the most impactful areas of municipal AI deployment globally. Van Noordt and Misuraca (2022) further affirm that AI-powered platforms and GPS-based fleet management systems are the norm tools used now for improving institutional responsiveness in government operations worldwide.

The citizens look up to the LGUs for safety of the public and the effective response. The MAT is the front-line unit at the municipal level that receives citizen complaints and incident reports, coordinates emergency response and dispatches government vehicles to address concerns of the public. The MAT is an integral part of

the day-to-day governance of the Municipality of Liliw, Laguna, dealing with everything from life-and-death emergencies such as rescue operations and flooding incidents to everyday issues such as stray animals, road obstructions and requests for municipal aid. Through direct site visits, observational assessment, and key informant interviews with the MAT personnel and its head Honorable Krich Carlo V. Monfero, the preliminary investigation revealed that there was no digital infrastructure which can support the frontline functions of the MAT.

Yet, the Municipality of Liliw, Laguna's Mayor's Action Team (MAT), the front-line unit that responds to an average of 20 to 30 complaints and incidents a week, is still beset by persistent and compounding operational challenges. A preliminary investigation involving site visits and key informant interviews with MAT personnel and their head Hon. Krich Carlo V. Monfero, Observational evaluation revealed that reports were received through informal channels, manually logged without automated urgency classification, vehicles were dispatched verbally without GPS monitoring or trip logging and citizens had no way to share their precise location digitally. Without a family safety mechanism, we cannot ensure that citizens and their families stay coordinated and informed during emergencies. Such conditions are observed in a wider trend in ASEAN, where the gap between national digital platforms and municipal operational capacity is the largest, as reported by Chen and Kimura (2024) and Rahmansya et al. (2025).

Examples of platforms that have demonstrated the feasibility of technology-driven governance at the national level are eGov PH, QC mySMS and the LTFRB fleet monitoring system. But these are restricted to large, well-resourced urban units and small municipalities are without locally relevant solutions (Adiazmil & Perdana, 2024; Ofori, 2024). Unlike these systems, the proposed platform combines a number of

features into a single platform specifically designed for a provincial Philippine LGU, including NLP-based complaint/incident classification in Filipino and Taglish, real-time GPS vehicle tracking, citizen location sharing, family safety and family circle features, and a direct government-citizen communication module design principles that Jou et al. (2024) identified as critically important for mobile government adoption in Philippine provincial settings.

The investigation identified the following problems in the operation: (1) Manual processing of complaints and incidents. Complaints of citizens and incident reports are received only through informal channels (mainly telephone calls and walk-in reports) and are manually logged without any digital record keeping system. There is no set way to sort or prioritize concerns by urgency automatically. A critical emergency such as a flood rescue and a minor concern such as a request for a gas tank repair can be given the same weight depending on the availability and judgment of the officer on duty. (2) No formal vehicle dispatch protocols. There is no formal dispatching protocol, GPS monitoring or trip logging system - municipal vehicles are dispatched verbally. This creates significant accountability gaps as there is no way to track the use of vehicles outside of official duties, and no verifiable audit trail for any use of vehicle. (3) Citizens are unable to send a digital exact location of complaint or incident. MAT Responders/Drivers have to rely on verbal description, and this is often not complete and not accurate, particularly in an emergency situation where precise navigation is important. (4) There is no centralized monitoring platform. There isn't a single platform where MAT officers can see active complaints, where dispatched vehicles were, and geographic data in real time. This causes a major lack of coordination, especially in multi-incident responses. (5) Absence of a family safety and emergency alert mechanism. Citizens have no structured way to share their location with family

members or send emergency alerts during critical situations. This limits family-level coordination and safety awareness during emergencies.

The study directly contributes to SDG 16, Peace, Justice and Strong Institutions and SDG 11, Sustainable Cities and Communities. Sengupta and Sengupta (2022) state that SDG 11 is not only related to physical infrastructure but also includes the accessibility of governance and equitable public service delivery at the community level. The United Nations Department of Economic and Social Affairs (2024) noted that digital technologies could directly be used to strengthen the transparency and accountability of public institutions. Thus, this study contributes to both aims at the most relevant level of governance in the Philippines, the municipal level, by bringing AI-powered public service management to the municipal level where the impact on citizens is most immediately felt.

The main research problem of this study is: How to design and develop an AI-powered mobile-based system to streamline the complaint and incident management with vehicle dispatch operations of the MAT of the Municipality of Liliw, Laguna? To address this issue, TUGON: Technology-Unified Government Operations Network, a Mobile-Based AI-Powered Complaint and Incident Management with Vehicle Dispatch System for the MAT of the Municipality of Liliw, Laguna was developed by the proponents. TUGON, which is the Filipino word for “response,” is a one-stop platform that integrates AI-powered complaint and incident classification with Gemini 3.1 Flash, real-time GPS-based vehicle tracking, citizen location sharing and complaint pinning, an administrative dashboard, family safety and family circle features, and a direct government-citizen communication module. The mobile application is developed using Flutter and Dart, the web dashboard is built using Next.js and TypeScript, and the real-time database management is done using PostgreSQL via Supabase. It is

compliant with data privacy compliance under RA 10173 with Role-Based Access Control and JWT Authentication. Unlike the existing platforms such as Quezon City's mySMS and the LTFRB Fleet Monitoring System that separately deal with complaint management and vehicle monitoring respectively for large cities with significant resources, TUGON provides one, integrated solution suited to the linguistic, institutional and operational realities of a small provincial Philippine municipality. The TUGON seeks to enhance the operational efficiency, responsiveness, accountability and disaster preparedness capabilities of the MAT of the Municipality of Liliw through the integration of AI-powered complaint classification, GPS-based vehicle accountability, family safety and family circle features and structured government-citizen communication efforts. In addition, TUGON aspires to be a model of technology-driven local governance in small Philippine municipalities.

1.2 Research Objectives

This section presents the main and specific goals of the capstone project using the TUGON system. These objectives are clear and realistic, aiming to solve the operational problems of the Mayor's Action Team (MAT) in Liliw, Laguna. From the design phase up to the final user evaluation, every step is aligned with the research problem to produce reliable results. These goals are divided into one general objective and several specific objectives.

1.2.1 General Objective

The general objective of this study is to design, develop, test, and evaluate TUGON: Technology-Unified Government Operations Network, a mobile-based AI-powered emergency response and incident management with vehicle dispatch system for the Mayor's Action Team (MAT) of the Municipality of Liliw, Laguna.

1.2.2 Specific Objectives

Specifically, this study aims to:

1. To design and develop an emergency response and incident management system (TUGON) that centralizes complaint and incident reporting, responder dispatch, and real-time tracking for the Mayor's Action Team (MAT) of the Municipality of Liliw, Laguna.
2. To test the functionalities of the developed TUGON system based on the following:

   2.1 AI-Powered Incident and Complaint Classification

   2.2 AI KYC Verification and Identity Management

   2.3 Automated Dispatch Queue and Mission Management

   2.4 Real-Time GPS Tracking and ETA Calculation

   2.5 Face ID Attendance with Geofence Verification

   2.6 Patrol Management and Scheduling

   2.7 Evacuation Center Check-In/Check-Out

   2.8 Family Circle Safety and Live Location Sharing

   2.9 MAT-Citizen Real-Time Chat and Communication

   2.10 Incident Heatmap and Clustering Visualization

3. To evaluate the user acceptance of the developed TUGON system among citizens, MAT responders/officers and MAT admin of the Municipality of Liliw, Laguna using the Technology Acceptance Model (TAM) in terms of:

3.1 Quality Factors

3.2 Perceived Ease of Use

3.3 Perceived Usefulness

3.4 Attitude Toward Using

3.5 Behavioral Intention to Use

3.6 Experience/Actual Use

1.3 Conceptual Framework

This section outlines the conceptual framework that serves as the primary blueprint for the development and evaluation of the TUGON platform. Anchored on the Input-Process-Output (IPO) model, it theoretically and structurally maps out how the study translates the identified operational challenges of the Mayor's Action Team (MAT) of Liliw, Laguna, into a comprehensive technological solution. To provide a clear overview, the framework is divided into a detailed discussion of its core components followed by a visual diagram illustrating the entire research flow.

1.3.1 Components of the Conceptual Framework

This study's conceptual framework is built upon three interconnected components: Input, Process, and Output. The Input phase serves as the foundation, gathering critical data from preliminary investigations into the Mayor's Action Team (MAT) of Liliw, Laguna. It identifies the municipality's current operational gaps while establishing the specific technical requirements needed for AI integration and GPS tracking. Guided by Agile Methodology, the Process phase brings the system to life by developing the mobile and web-based modules. This involves integrating Natural Language Processing (NLP) for intelligent complaint classification and real-time GPS

tracking, alongside defining the evaluation protocols using functional testing and the Technology Acceptance Model (TAM). Finally, the Output is the fully realized TUGON system, which incorporates a continuous feedback loop that channels evaluation results back into the initial inputs for ongoing system refinement.

These inputs, processes, and outputs were carefully shaped by the synthesis of related literature and existing systems explored in Chapter II. This review highlighted a clear need for AI-powered NLP classification and real-time fleet tracking to directly address the manual, localized operational bottlenecks currently hindering municipal platforms.

1.3.2 Conceptual Framework Diagram

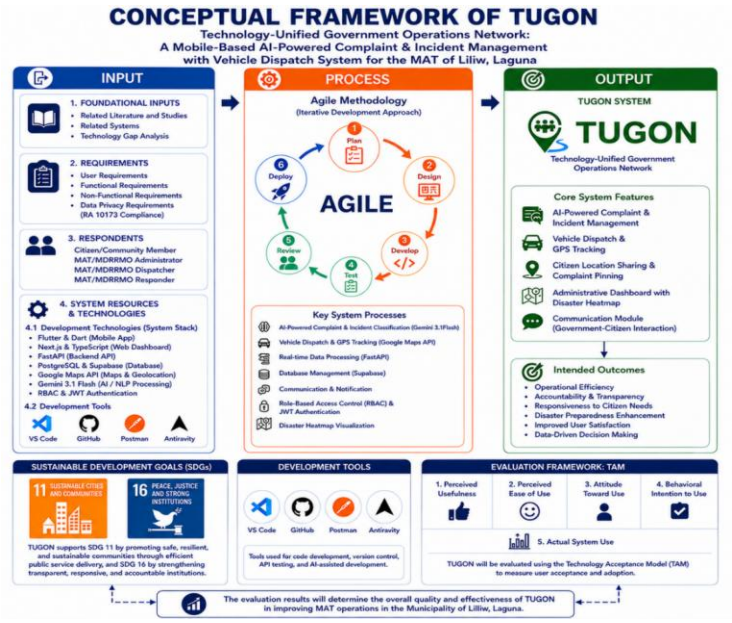


Figure 1.1. Conceptual Framework of TUGON

Image generated using ChatGPT by OpenAI with DALL·E (May 2026) from prompt: "Create a professional 1:1 IPO conceptual framework diagram for TUGON showing the Input, Process, and Output phases aligned with Agile methodology and Technology Acceptance Model (TAM) evaluation criteria." Diagram synthesized from related literature on AI-powered complaint management and GPS-based tracking.

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As illustrated in Figure 1.1, the study visualizes its research flow through an Input-Process-Output (IPO) model, specifically designed to map out solutions for the manual operational bottlenecks experienced by the MAT in the Municipality of Liliw, Laguna.

The journey begins with the Input phase, which diagnoses current operational inefficiencies through direct site visits, key informant interviews with MAT personnel, and hands-on observational assessments. This phase is heavily informed by theoretical foundations, such as contemporary research on transformer-based NLP models and citizen technology adoption, paired with a localized technology gap analysis. By evaluating existing platforms like Quezon City's mySMS and the LTFRB Fleet Monitoring System, the researchers established exactly what features were missing but critical for the MAT's daily operations.

The Process phase translates these needs into action using an Agile development approach, ensuring iterative progress and incorporating constant stakeholder feedback throughout the build cycle. Drawing from the study's literature synthesis, this phase integrates advanced core technologies to automate municipal workflows. It leverages NLP via Gemini 3.1 Flash for the zero-shot classification and urgency sorting of Tagalog and Taglish reports, while utilizing the Google Maps API for real-time fleet dispatching and citizen location sharing. Here, the ten major system features are actively developed across a robust, low-bandwidth-optimized technology stack featuring Flutter, Next.js, PostgreSQL via Supabase, and JWT RBAC. Additionally, this phase outlines the rigorous functional testing procedures conducted alongside IT experts to ensure all modules operate as intended.

The Output phase represents the culmination of this development: the fully functional TUGON platform. In this final stage, the system is rigorously evaluated by its four primary user groups, citizens, the system administrator, MAT dispatchers, and MAT responders. This evaluation applies the Technology Acceptance Model (TAM) across six specific constructs (Quality Factors, Perceived Ease of Use, Perceived Usefulness, Attitude Toward Using, Behavioral Intention to Use, and Experience/Actual Use) to gauge overall software quality and practical feasibility. To ensure the system adapts and improves, a feedback loop connects the Output back to the Input. This loop guarantees that user evaluations and functional conformance findings are continually used to refine the platform, ensuring TUGON remains highly effective for actual deployment in Liliw and serves as a viable, scalable model for other provincial municipalities across the Philippines.

1.4 Project Purpose

The study aims to design, develop and evaluate the TUGON: Technology-Unified Government Operations Network, a Mobile-Based AI-Powered Complaint and Incident Management with Vehicle Dispatch System for the MAT of the Municipality of Liliw, Laguna. The aim of this study is to offer a structured, intelligent and accessible platform that increases the capacity of the MAT to receive, classify and respond to reports from the citizens, while ensuring the accountable and efficient deployment of municipal vehicles. The project seeks to transform the MAT operations from a reactive, manually operated process to a proactive, data driven, and technology supported system for public service delivery through the integration of Artificial Intelligence and Natural Language Processing for automated report classification and urgency sorting, GPS-based real-time tracking for both citizens and municipal vehicles, an administrative monitoring dashboard, family safety and family circle features, and a direct

communication module between MAT officers and citizens. TUGON directly contributes in the promotion of transparent and responsive local governance in the Municipality of Liliw, Laguna in support of the United Nation's Sustainable Development Goal 16 (Peace, Justice and Strong Institutions) to build accountable, inclusive and effective institutions at all levels.

The present work aims at filling the significant gap of complaint management efficiency by providing MAT with an NLP-based classification system that could automatically classify incoming reports according to their urgency level. This would enable life-threatening emergencies to take precedence over minor issues, regardless of the number of reports. The Complaint and Incident Management module will allow MAT officers to go beyond manual judgement to determine which concerns need immediate action and will significantly minimize risk of delayed responses to truly critical situations. The Vehicle Dispatch Management module also contributes to operational accountability, as drivers must record their trips, associate their accounts with specific vehicle plate numbers, and be tracked by GPS while in service, thus providing an auditable and transparent record of all municipal vehicle movements. The Citizens Tracking module allows the citizens to know the exact location of their report on an interactive map, thus eliminating the need for verbal descriptions and enabling MAT Responders/Drivers to reach the site of the report directly and efficiently. The Family Safety module allows citizens to create private family circles, share their live locations with family members, and send emergency alerts during critical situations, strengthening community-level safety awareness and coordination. The Communication Module also aids in bridging the gap between the government and its constituents, by facilitating direct real-time messaging between MAT officers and

citizens, instead of the informal and often unreliable phone-based coordination that currently defines MAT operations.

Moreover, the system is meant to be practical enough to be implemented in a provincial municipality with its resource limitations. The mobile app is designed to be optimized for the different levels of digital literacy of citizens and an administrative web dashboard is developed to be user-friendly for MAT officers and staff. The development, testing and evaluation of TUGON adds to the growing body of knowledge on AI-driven e-governance in developing countries, and the study provides a scalable and replicable model for other municipalities in Laguna and the Philippines to adopt in improving their public safety and service delivery systems. The following persons will be benefited by this study and the value that TUGON expects to provide to each:

1.4.1 Intended Beneficiaries

Citizens

The citizens are the main beneficiaries of TUGON. The mobile application offers citizens a direct, accessible and effective channel to report complaints and provide their exact location to the MAT, replacing the current practice of phone calls and walk-in reports. The system enables citizens to follow up on the status of their reports, get updates and follow-up communications from MAT officers and get a faster and more accountable government response. The system's Family Safety feature also provides an additional layer of safety awareness by enabling citizens to share their live locations with family members and send emergency alerts during critical situations.

Mayor's Action Team (MAT) Officers and Personnel

TUGON provides MAT officers with a centralized web-based dashboard that aggregates all active complaints and incident reports, dispatched vehicle locations, real-time GPS feeds, and live operations data in one unified platform. The AI-powered complaint and incident classification system alleviates officers' manual workload in sorting and prioritizing incoming concerns, enabling them to direct their attention and resources to addressing the most urgent situations. The built-in communication module also allows officers to coordinate directly with citizens without using informal channels, improving the clarity and efficiency of every response operation.

MAT Vehicle Drivers

TUGON is a structured and transparent dispatch system that provides MAT vehicle drivers with clear responsibilities before, during, and after each trip. The GPS tracking features provide a formal and accountable system for vehicle utilization, reducing uncertainty in assignments and protecting drivers from unsubstantiated claims of abuse.

Municipal Government of Liliw, Laguna

TUGON supports the municipal government in terms of operational transparency, data-driven decision-making, and public trust. The administrative dashboard and analytics capabilities of the system offer complaint data, vehicle utilization reports, and response time data. These enable municipal administrators to evaluate the performance of the MAT, optimize resource allocation, and identify common problems that could entail long-term policy solutions. With TUGON, a technology solution designed for the municipality of Liliw, Laguna, the town is also a model for digital governance adoption by similarly sized municipalities in the province.

Researchers

The development of TUGON also provided the proponents with first-hand experience in full-stack development of mobile and web applications, integration of AI and NLP models, design of GPS-based systems, and deployment of technology for government-facing applications. The research process also gave the proponents a deeper understanding of the operational realities faced by LGUs and the technical and contextual considerations necessary for the development of systems that are functional and useful and adoptable in the real-world environment of the public sector.

Future Researchers

This study may serve as a reference and basis for future researchers who are interested in AI-powered e-governance, mobile-based public service systems and complaint management technology in the Philippine setting. The results, approach, architecture and evaluation results of this study may serve as a reference for future work on similar systems for other LGUs, especially for those in provincial and rural areas where the gap between the existing digital governance solutions and the actual operational needs is still large. Future research can extend the NLP models, GPS tracking architecture, and Technology Acceptance Model (TAM) evaluation framework used in this study to develop more advanced or localized versions of government service systems.

1.4.2 Expected Benefits

The expected benefits of TUGON include: (1) faster and more accountable complaint and incident response through AI-powered urgency classification; (2) improved vehicle dispatch accountability via GPS tracking, trip logging (3) enhanced citizen; government communication through real-time chat and location sharing; (4)

strengthened family safety coordination during emergencies; (5) data-driven decision-making for the Municipal Government through analytics and heatmap visualization; and (6) a scalable model for digital governance adoption by other small municipalities. These benefits are realistic and directly aligned with the system's ten major functionalities and are evaluated through user acceptance testing.

1.5 Scope and Limitations of the Study

This section describes the operational boundaries and technical constraints of the TUGON system. The scope explicitly defines the core functionalities, target users, and the specific deployment environment of the mobile-based complaint management and vehicle dispatch platform designed for the Mayor's Action Team in the Municipality of Liliw, Laguna. Conversely, the limitations identify the technical, environmental, and time-bound constraints encountered during the development and testing phases. Establishing these parameters ensures a transparent and realistic assessment of the system's capabilities, clearly distinguishing the intended features from areas outside the current study's purview.

1.5.1 Scope of the Study

This study encompasses the complete lifecycle, design, development, testing, and evaluation, of the TUGON (Technology-Unified Government Operations Network) platform, a mobile-based, AI-powered complaint and incident management system equipped with vehicle dispatch capabilities for the Mayor's Action Team (MAT) of Liliw, Laguna. The research is operationally bounded within the jurisdiction of the MAT office, which serves as the primary setting and intended deployment environment. Functionally, the study covers ten major system modules: (1) AI-powered incident classification, (2) KYC verification, (3) an automated dispatch queue, (4) real-time GPS tracking and ETA calculation, (5) Face ID attendance with geofencing, (6) patrol

management, (7) evacuation center check-in and check-out, (8) family circle safety features, (9) real-time chat, and (10) an interactive incident heatmap.

To ensure a comprehensive evaluation, the research involves four distinct user groups: citizens of Liliw who submit geolocated complaints via the mobile app, municipal drivers whose trips are logged and tracked via GPS, MAT officers managing dispatch operations through a web dashboard, and a system administrator overseeing overarching operations and reports. Conducted during the capstone development period (A.Y. 2025–2026, up to May 2026), the study involved a total of 203 participants. This sample includes 1 system administrator, 3 MAT dispatchers and officers, 18 MAT responders and drivers, and 181 citizens drawn from a total population of 39,976 using stratified random sampling with Slovin's Formula at a 7% margin of error. The evaluation itself is twofold: it includes rigorous functional Black-Box testing for all ten system features and a user acceptance assessment based on the Technology Acceptance Model (TAM). This TAM evaluation measures six specific constructs: Quality Factors, Perceived Ease of Use, Perceived Usefulness, Attitude Toward Using, Behavioral Intention to Use, and Experience/Actual Use.

Technologically, the scope of implementation relies on Flutter and Dart for the mobile application, ensuring deployment compatibility across both Android and iOS environments, Next.js and TypeScript for the administrative web dashboard, and Supabase with PostgreSQL for real-time database management. The architecture integrates the Google Maps API for location services and utilizes JWT alongside Role-Based Access Control (RBAC) to ensure secure authentication and strict compliance with the Data Privacy Act (RA 10173). Throughout its operation, the system processes specific data points, such as citizen reports, GPS tracks, dispatch logs, and attendance records, while facilitating core workflows like complaint intake, AI classification,

dispatch assignment, GPS monitoring, patrol and evacuation management, and citizen communication. Finally, the scope acknowledges the guided use of Generative Artificial Intelligence (GenAI) tools during the research process; under strict researcher supervision and validation, these tools assisted in interface prototyping, workflow conceptualization, documentation refinement, content generation, and software development.

1.5.2 Limitations of the Study

While the TUGON system introduces a comprehensive technological intervention for the Mayor's Action Team (MAT) of the Municipality of Liliw, Laguna, the researchers acknowledge certain limitations that shape the interpretation of its findings. Primarily, the study's conclusions are contextually bound to the specific operational setting of the Liliw MAT, meaning they may not be directly generalizable to larger cities or municipalities with different population sizes, organizational structures, and technological resources. The availability of local data and access to respondents were strictly confined to the identified sample within the academic timeframe. Specifically, the user acceptance evaluation is limited to a sample of 203 respondents, reflecting actual system use solely during the evaluation period rather than capturing the full, long-term spectrum of potential users. Additionally, access to users residing in remote barangays with limited internet connectivity was constrained during the data gathering phase. Furthermore, while the TUGON mobile application was developed for both Android and iOS environments using Flutter, the user acceptance evaluation and functional testing phases were conducted exclusively on Android devices to establish baseline operational functionality.

From a technical standpoint, the study focuses on functional validation rather than exhaustive enterprise-level security or load assessments. The research does not

perform advanced penetration testing, large-scale stress testing under high concurrent loads, or long-term integration testing with external local government unit (LGU) systems. System performance is instead verified entirely through Black-Box functional testing, simulation testing of the ten major features, and performance benchmarking conducted under controlled conditions. Furthermore, the research timeline is strictly limited to the capstone development period of the 2025–2026 academic year. Consequently, continuous large-scale operational deployment, ongoing system maintenance, and longitudinal impact assessments on MAT operations remain beyond the project's scope. Practical deployment is also inherently constrained by external factors, particularly internet and network dependency, as well as the availability of essential municipal hardware and infrastructure like GPS devices and server hosting.

Finally, the study acknowledges the integration of Generative Artificial Intelligence (GenAI) tools to assist in interface prototyping, workflow conceptualization, documentation drafting, and software development. Because GenAI tools can occasionally produce incomplete or contextually inaccurate outputs, a strict mitigation protocol was enforced. All AI-generated content and development outputs were thoroughly reviewed, validated, and refined by the researchers to guarantee accuracy and strict alignment with the study’s overarching objectives, as detailed in Section 3.1.3.

CHAPTER II

REVIEW OF RELATED LITERATURE

This chapter presents the scholarly foundation of the capstone project. It establishes what has already been studied, developed, and implemented regarding mobile-based e-government systems, AI-powered classification, and GPS tracking. Furthermore, it identifies the knowledge, functional, technological, and process gaps that the project intends to address for the Mayor's Action Team (MAT) of the Municipality of Liliw, Laguna. The review is not presented as a collection of unrelated summaries; instead, the reviewed literature and existing systems are analyzed, compared, synthesized, and directly related to the development of the TUGON platform.

2.1 Studies and Literature

This section presents relevant scholarly literature, research studies, theories, concepts, and technologies related to the capstone project. The materials reviewed have a clear connection to the research problem, the objectives of the MAT, the proposed AI-powered solution, and the major system components.

2.1.1 Selection of Related Literature

The literature prioritized for this study is directly related to the research problem and relevant to the proposed system technologies, specifically artificial intelligence (AI), Natural Language Processing (NLP), and GPS tracking. To ensure the scholarly foundation is current and credible, the selected sources were published between 2021 and 2026 and were drawn from reliable academic journals, government reports, and institutional publications. The selected works are highly useful in establishing the

methods and technologies underlying the e-governance adoption required in developing contexts.

2.1.2 Review and Analysis of Studies

Previous studies are discussed analytically to identify their purpose, methodology, major findings, and relevance to the TUGON project.

Digital Transformation in Local Government Operations

For many citizens in developing countries, the LGU is the closest representation of the state, the office to which they go to report a flooded road, to seek assistance in the event of a calamity, or lodge a complaint about a broken streetlight. But even as the frontlines for government, LGUs are the most underserved tier of digital transformation. While national agencies might have functional online portals and automated workflows, the town hall down the street still operates on paper logs, verbal instructions, and phone calls. Gasco Hernandez (2024) reviewed thirty years of research on digital transformation in local governments in Local Government Studies and concluded that although local governments have made visible progress in the use of digital tools in the last thirty years, the pace and scope of transformation at the municipal level still lags far behind national and regional governments. Their longitudinal analysis found that the most resilient obstacle is not a lack of technology but a lack of institutional frameworks that embed digital transformation into the everyday governance logic of LGUs, which reframes the problem from a technical to a structural one. Successful local governments in digital transformation do not do so by deploying individual digital tools, but by cultivating an organizational culture that sees technology-enabled service provision as a core governance function rather than an administrative add-on (Gasco Hernandez, 2024).

The real meaning of this gets clearer when Oghuvbu, Gberevbie and Oni (2022), is brought into the discussion. In their research on the challenges and prospects of e-governance in Nigeria, the authors found that government employees who are used to manual workflows do not resist digital platforms due to a lack of capability but due to a lack of familiarity with the platforms, and that even digitally powered projects come to a halt at the implementation stage, regardless of their level of funding without purposeful capacity-building and stakeholder engagement. This internal organizational limitation reported by Oghuvbu et al. (2022) is precisely the institutional culture problem pointed out by Gasco Hernandez (2024) at the strategic level. One study identifies the absence of a culture of digital governance at the organizational level, and the other illustrates its implications at the operational front line. In sum, they suggest that successful digitalization of local government needs to be done simultaneously on institutional strategy and on human capacity, not one without the other.

Lafioune et al. (2023) took a different approach to the question, looking at digital transformation in municipalities for infrastructure planning and management in their study published in Sustainable Futures. They looked at how municipalities can develop structured, strategic approaches for digital transformation that are more than just ad hoc adoption of tools but integrated digital governance architectures. They found that municipalities that create explicit strategic and organizational structures for digital transformation that specify what digital tools they will adopt, how they will integrate with existing workflows, and who will be responsible for their maintenance, achieve much longer lasting digital transformation outcomes than those that adopt technologies reactively to immediate operational demands. This argument of a strategic framework expands both Gasco Hernandez's (2024) cultural diagnosis and Oghuvbu et al.'s (2022) capacity argument, which argues that a digital governance culture and trained staff is

necessary, but not sufficient, without a deliberate organizational architecture that structures how technology integrates into municipal operations. Adiazmil and Perdana (2024) drew upon a systematic review of e-government adoption in Southeast Asia and other developing regions to develop a framework that incorporates a design dimension. They found that mobile-first platforms and modular system architectures consistently outperform comprehensive top-down implementations in rural and provincial settings. Their finding echoes the argument of Lafioune et al.'s (2023) strategic framework that a sequenced, structurally grounded digital adoption delivers more sustainable outcomes than an ambitious yet underprepared implementation.

But the study of Ofori (2024) took the analysis further by introducing the problem that the other studies had not explored enough, the role played by socio-cultural factors and urban-rural disparities in the success or failure of digital government systems to take root at the local level. Ofori, viewing e-government progress through the lens of SDG 16.6, found that the accountability and transparency goals of sustainable development are most stalled right at the municipal tier, where governance is closest to citizens, but institutional capacity is thinnest. This finding links the broader digital transformation literature to a governance imperative that resonates directly with TUGON: the municipalities that stand to benefit the most from digital transformation are often those least able to pursue it themselves, and in the absence of targeted, contextually sensitive solutions designed to their particular institutional context, the digital governance gap between well-resourced cities and resource-constrained provincial municipalities will only grow.

All five studies have a common blind spot despite their different geographic settings and analytical lenses: they describe the conditions for and barriers to local government digital transformation but offer few documented examples of AI-powered,

integrated service systems that have been purpose-built for and evaluated within a small provincial municipality. The literature overwhelmingly focuses on city-level or national-level cases, which leaves smaller LGUs largely unexamined, particularly in provincial Philippine contexts. TUGON is meant to begin addressing that gap, by applying the mobile-first, modular, strategically grounded and context-sensitive principles that the literature espouses, while producing a concrete municipal case study that other similarly situated local governments across the Philippines can learn from.

Mobile-Based e-Government and Citizen Service Delivery

There is something fundamentally different about a government service that fits in your pocket. Mobile government applications have changed the basic equation of public service delivery, instead of requiring citizens to travel to an office, wait in line, and speak to an officer in person, a well-designed mobile platform puts that same interaction within arm's reach, available at any hour and from any location with a data connection. Castilla, Pacheco, and Franco (2023), in their study published in SoftwareX, examined how mobile government applications reshape citizen access to public information, finding that mobile platforms significantly reduce the time and physical effort required for citizens to engage with government services. Their analysis documented that the most effective mobile government applications share three consistent design characteristics: they are intuitive enough for users with limited digital experience, they are optimized for low-bandwidth environments, and they provide clear, actionable feedback after every citizen interaction. These findings establish a practical design baseline for what mobile government must deliver to be genuinely accessible rather than merely technically available.

Whether that baseline is actually being met in practice, however, is a more complicated question, and one that Jou et al. (2024) examined directly in the Philippine

context. Their study of the eGov PH mobile application, published in Sustainability, found that perceived ease of use, service quality, and the presence of locally relevant features are the strongest predictors of citizen adoption among Filipino users. Encouragingly, their findings aligned with Castilla et al.'s (2023) design principles: Filipinos are more likely to adopt mobile government platforms when those platforms feel designed for them rather than adapted from somewhere else. Less encouragingly, however, Jou et al. (2024) also found that adoption rates in provincial municipalities lag significantly behind urban centers, not because provincial citizens are less willing to use digital government services, but because the platforms they are given do not adequately account for the connectivity limitations and varying digital literacy levels of non-urban populations. This urban-provincial divide is a gap that Castilla et al. (2023) did not directly address, making Jou et al.'s Philippine-specific findings an important corrective to the more generalized design principles the former study established.

Kurniasih, Setyoko, and Huda (2024), studying mobile government adoption in Indonesia through the lens of the Peduli Lindungi health application, brought another dimension into view that both Castilla et al. (2023) and Jou et al. (2024) touched on but did not fully develop: the role of trust. Their study found that citizens who trust the government institution behind a mobile application are significantly more likely to adopt and continue using it, regardless of how well-designed the interface is. This introduces an important tension with Jou et al.'s (2024) emphasis on ease of use, a platform can be perfectly intuitive and locally relevant, but if citizens do not trust the institution operating it, adoption will remain low. Desmal, Alsaeed, Hamid, and Zulait (2023), in their IEEE conference paper on leveraging social media in mobile government, offered a partial resolution to this tension by demonstrating that

integrating familiar communication channels, particularly social media and messaging features, into government mobile platforms measurably improves citizen engagement and trust by making the government feel more responsive and accessible. Where Kurniasih et al. (2024) identified trust as a prerequisite for adoption, Desmal et al. (2023) pointed to communication design as a practical mechanism for building that trust over time.

Read together, these four studies paint a consistent but still incomplete picture. They agree that mobile government works best when it is simple, locally relevant, low-bandwidth friendly, and backed by institutions that citizens already trust or are actively working to earn that trust. What they do not address, however, is the specific challenge of designing a mobile government platform for a small provincial municipality in the Philippines. one that must simultaneously serve citizens with varying digital literacy, operate under inconsistent network conditions, and represent a local government institution that may not yet have an established track record with digital services. TUGON's citizen-facing mobile application is designed with precisely these conditions in mind, applying the accessibility principles documented by Castilla et al. (2023), the local relevance imperatives identified by Jou et al. (2024), and the trust-building communication features recommended by Desmal et al. (2023) and Kurniasih et al. (2024) within a single platform purpose-built for the operational realities of Liliw, Laguna.

Complaint Management Systems in Local Government

When a citizen walks into a government office to report a broken road, a flooded street, or a missing streetlamp, what happens next depends almost entirely on the system, or lack of one, that the government has in place to receive, record, and act on that concern. In many LGUs across developing countries, that system is little more than a

logbook and a phone call. Shofiana, Riadi, and Evanita (2025), in their study published in Journal La Multiapp, examined this exact reality by developing and evaluating a web-based public complaint information system for a local government in Indonesia. Their findings confirmed that even a basic digital complaint platform, one that allows citizens to submit concerns online and enables officers to track and respond to them through a structured interface, produces measurable improvements in service delivery speed, complaint traceability, and citizen satisfaction compared to purely manual processes. Critically, however, Shofiana et al. (2025) also found that the effectiveness of their system was limited by its inability to automatically prioritize complaints by urgency, meaning that officers still had to manually determine which concerns required immediate attention. This manual prioritization bottleneck, they concluded, represents the next frontier that complaint management systems in local government must address.

Javith Kausar et al. (2026) took on precisely that frontier. Their study in the International Journal for Research in Applied Science and Engineering Technology proposed an NLP-enhanced complaint management system designed specifically for municipal services, using BERT-based models as language models to automatically classify incoming complaints by type and assign urgency levels without human intervention. Where Shofiana et al. (2025) documented the limitations of systems that still rely on officer judgment for prioritization, Javith Kausar et al. (2026) demonstrated that those limitations can be systematically resolved through AI-powered classification, and that the resolution comes with tangible operational benefits, including faster complaint resolution times and more consistent urgency ranking across high-volume complaint periods. The contrast between these two studies is instructive: one shows what structured but non-intelligent complaint management can achieve; the other shows what becomes possible when intelligence is added to the structure.

The question of whether AI-powered complaint handling actually improves outcomes beyond operational efficiency, specifically, whether it changes how respondents perceive the quality of the government's response, was examined by Alyahya, Agag, and Abdelwahab (2026) in a study published in the Journal of Retailing and Consumer Services. Using a quasi-experimental difference-in-differences design, they found that AI-enabled complaint systems significantly improve complainant satisfaction and perceived institutional competence, with the most pronounced effects occurring in high-volume environments where manual processing had previously created visible backlogs. This finding adds an important dimension to what Javith Kausar et al. (2026) established: AI-powered complaint management is not just operationally efficient, it is also perceptually transformative, changing how citizens experience and evaluate the government's responsiveness. Laylo and Sonza (2026), studying AI-segmented complaint management in the context of electric cooperatives in the Philippines, corroborated this from a local standpoint, finding that automated complaint categorization and routing reduced average resolution time and generated higher citizen confidence in the service provider's accountability, even among users who were initially skeptical of digital complaint channels.

That the Philippine government itself has begun moving in this direction is significant. In 2025, the Anti-Red Tape Authority partnered with a private technology firm to deploy an AI-powered complaint management system for national government agencies, a development that the Philippine Information Agency (2025) reported as a direct response to longstanding inefficiencies in manual complaint handling across public institutions. This national-level adoption signals an institutional acknowledgment that AI-driven complaint management is no longer experimental, it is becoming expected. Yet what the national initiative and the broader literature share is

a common limitation: the focus remains overwhelmingly on large institutions, urban governments, and national agencies. The specific challenge of deploying an intelligent complaint management system in a small provincial municipality, one where complaint volumes are lower but operational capacity is also significantly more constrained, remains largely undocumented. TUGON addresses this gap by bringing AI-powered complaint classification and urgency sorting to the municipal level in Liliw, Laguna, extending the operational and perceptual benefits documented by Alyahya et al. (2026), Laylo and Sonza (2026), and Javith Kausar et al. (2026) to a government context that the existing literature has not yet reached.

Communication Systems for Government-Citizen Engagement

One of the most persistent frustrations citizens have with their local government is not necessarily that problems go unsolved, it is that they never hear back. A complaint is filed; a concern is raised and then silence. The absence of meaningful two-way communication between government and citizens is not a new problem, but it has become an increasingly studied one as digital tools have made direct, real-time engagement technically feasible in ways that were simply not possible a generation ago. Shin et al. (2024), in their systematic analysis of digital tools for citizen participation published in Government Information Quarterly, established that the most effective government-citizen communication platforms share a defining characteristic: they are not broadcast systems that push information outward, but dialogue systems that create structured channels for citizens to speak and feel heard. Their review documented that digital tools which enable genuine bidirectional exchange, where citizens can submit concerns and receive individualized, timely responses, consistently produce higher satisfaction and greater civic engagement than those that merely digitize one-way announcements.

The practical reality of how governments actually use digital communication, however, often falls short of this ideal. Bangero (2025), examining how local governments use official Facebook pages for e-government communication in a study published in Government Information Quarterly, found that while social media has dramatically expanded the reach of local government communications, most LGUs use these platforms predominantly for broadcasting official announcements rather than for responsive citizen engagement. Citizen's comment, tag, and message, and frequently receive no reply, or a generic one that does not address their specific concern. Benlahcene, Ramdani, and Safi (2024), studying government-citizen communication in Algerian provincial governments, arrived at a strikingly similar finding in a very different geographic context: the structural gap between what citizens want from government communication, personalized, timely, accountable responses, and what governments typically deliver through their digital channels is wide and largely unacknowledged by the institutions themselves. Together, Bangero (2025) and Benlahcene et al. (2024) paint a consistent picture of local governments that have adopted the tools of digital communication without fully embracing its possibilities, using platforms designed for dialogue primarily as sophisticated notice boards.

Alenzi and Miskon (2024), in their literature review of government-to-citizen communication via social media published in the International Journal of Academic Research in Business and Social Sciences, identified the underlying reason for this gap: most government communication strategies were designed for traditional, asynchronous channels, press releases, official notices, physical bulletin boards, and have simply been transplanted onto digital platforms without being fundamentally reimagined for the interactive environment those platforms create. This finding connects directly to Shin et al.'s (2024) emphasis on dialogue design: the problem is not

the platform but the communication philosophy behind it. Benlahcene, Awang, and colleagues (2024), in a separate systematic review of citizen e-participation through e-government services published in Cogent Social Sciences, reinforced this argument by demonstrating that citizen participation rates are significantly higher on platforms explicitly designed for interaction, where citizens can track the status of their submissions, receive direct replies, and feel that their input influences government action, compared to platforms where participation is nominally possible but structurally discouraged by slow or absent responses.

Huanca (2025), examining digital platforms of public entities and their actual use by citizens in a systematic review published in Lex Localis, brought a sobering perspective to this body of literature by documenting that even well-designed government communication platforms suffer from low utilization when citizens lack confidence that their engagement will produce a meaningful government response. This creates a circular problem that the previous studies collectively describe but do not fully resolve; governments communicate poorly because they have not built responsive systems, and citizens disengage because governments communicate poorly, which in turn reduces the pressure on governments to improve their communication systems. Breaking this cycle, Huanca (2025) argued, requires platforms that make government responsiveness visible and verifiable, where citizens can see that their complaint was received, that someone is acting on it, and that the government is accountable for following through.

Across all six studies, the convergence is clear: effective government-citizen communication requires structured, bidirectional, and accountable channels, not social media pages managed as digital bulletin boards, but purposefully designed systems where every citizen interaction receives a traceable, timely government response. What

the literature does not yet address, however, is the deployment of such a purpose-built communication system within a small provincial Philippine LGU operating without any existing digital infrastructure. TUGON's Communication Module is designed to fill this gap directly, providing MAT officers and citizens with a structured in-platform messaging channel where complaint status updates, follow-up coordination, and government responses are formally logged and visible to both parties, building the kind of responsive, accountable government-citizen dialogue that the literature consistently identifies as the defining feature of effective digital public service communication.

Citizen Complaint Prioritization and Workflow Automation

Not all complaints are created equal. A citizen reporting a stray dog and a citizen reporting a rising floodwater during a typhoon may both be filing through the same government channel, but the urgency of those two situations could not be more different, and a system that treats them with equal weight is not just inefficient, it is potentially dangerous. The challenge of automatically distinguishing urgent from non-urgent complaints without requiring a government officer to manually read and evaluate every submission is one that has driven significant research interest in recent years. Ullah et al. (2023), in their study published in the Journal of Software Engineering and Applications, proposed a citizen-centric complaint reporting and analyzing mechanism that introduced structured complaint intake forms and rule-based classification logic to bring order to what is typically an unstructured, high-volume complaint workflow. Their system demonstrated that even without AI, structured complaint intake, requiring citizens to select complaint categories and provide standardized descriptions, meaningfully reduces the time officers spend sorting and routing incoming concerns. However, Ullah et al. (2023) also acknowledged a fundamental limitation of their rule-based approach: it relies on citizens accurately self-categorizing their complaints, which

is an unreliable assumption in real-world settings where citizens describe their concerns in informal, incomplete, or emotionally charged language.

It is precisely this limitation that AI-powered approaches are designed to overcome. Koh et al. (2025), in their cross-sectional study published in the Journal of Medical Internet Research, demonstrated that Large Language Models such as GPT-4o can classify unstructured complaint text into structured taxonomic categories with accuracy comparable to trained human reviewers, and at a fraction of the time. While their study was conducted in a healthcare complaint context, its implications extend directly to government complaint management: if an LLM can reliably classify a patient's free-text complaint about a hospital experience, it can equally classify a citizen's free-text report about a road obstruction or a flooding incident. Where Ullah et al. (2023) required citizens to do the classification work themselves through structured forms, Koh et al. (2025) showed that the system can do it for them, automatically, consistently, and without the accuracy loss that comes from citizen self-categorization. This represents a qualitative leap in what complaint management systems can deliver, shifting the cognitive burden of sorting from citizens and officers alike to the AI layer of the platform.

Dolas et al. (2026), in their study on a citizen grievance redressal platform with AI prioritization published in IJIRT, extended this line of research by demonstrating that AI-powered classification can be paired with an automated prioritization engine that ranks complaints not just by type but by estimated urgency, enabling the system to surface life-threatening concerns at the top of the officer's queue regardless of the order in which they were submitted. Their platform showed that combining classification with urgency ranking produces a more operationally useful output than classification alone, addressing a gap in Koh et al. 's (2025) study, which focused on categorical

accuracy without examining how classification translates into actionable prioritization in a real government workflow. Pallapu and Varshasri (2026), in their study on smart complaint prioritization using NLP and machine learning for e-governance published in IJRTI, reinforced and extended Dolas et al.'s findings by demonstrating that NLP-based urgency detection achieves higher consistency than officer-based manual prioritization, particularly during high-volume complaint periods when officer judgment is most likely to be influenced by cognitive fatigue and time pressure. Together, Dolas et al. (2026) and Pallapu and Varshasri (2026) make a compelling case that automated prioritization is not just a convenience feature, it is a public safety mechanism.

What emerges from reading these four studies together is a clear developmental trajectory: from structured intake without AI (Ullah et al., 2023), to AI-powered classification without prioritization (Koh et al., 2025), to classification with urgency ranking (Dolas et al., 2026), to NLP-driven smart prioritization in e-governance (Pallapu & Varshasri, 2026). Each study builds on the limitations of the previous, and together they describe exactly the kind of system that TUGON's Complaint and Incident Management module is designed to be. However, a gap runs consistently through all four studies: none of them addresses the challenge of complaint and incident classification in Filipino or Taglish, the mixed language in which citizens of provincial Philippine municipalities like Liliw, Laguna actually communicate their concerns. TUGON addresses this directly through its integration of the Gemini 3.1 flash model for Filipino language understanding and the facebook/bart-large-mnli zero-shot classifier for urgency detection, bringing AI-powered complaint and incident report prioritization to a linguistic context that the existing literature has not yet reached.

Public Service Response Efficiency and Operational Optimization

There is a quiet cost to inefficient public service delivery that rarely makes headlines but is felt daily by the citizens who depend on it, the emergency that took too long to reach, the complaint that was logged but never acted on, the government vehicle dispatched to the wrong location because no one had accurate real-time information. Improving the efficiency of public service response is not simply an administrative goal; it is a matter of equity and safety, particularly for communities in provincial areas where government resources are limited and response delays carry real consequences. Shen, Li, and Du (2023), in their study published in Environmental Impact Assessment Review, proposed an improved evaluation method for public service efficiency that centers the perspective of residents rather than government administrators. Their research found that conventional efficiency metrics, response times, complaint resolution rates, resource utilization figures, frequently diverge from how residents actually perceive and experience service quality, and that governments optimizing for administrative indicators alone risk improving their own performance metrics while leaving citizen satisfaction stagnant. This resident-centered framing is an important starting point: efficiency, properly understood, is not just about how fast the government moves but about whether citizens feel that the government moved meaningfully on their behalf.

Kulal et al. (2024), in their study published in the Journal of Open Innovation: Technology, Market, and Complexity, examined how AI specifically reshapes public service delivery efficiency, finding that AI-powered systems reduce administrative overhead, accelerate decision-making, and enable government personnel to focus their attention on complex, high-stakes situations rather than routine sorting and routing tasks. Their findings build on Shen et al.'s (2023) resident-centered efficiency

framework by demonstrating that AI does not just make government faster, it makes government more responsive in the ways that residents actually notice and value. Where Shen et al. (2023) identified the gap between administrative and citizen-perceived efficiency, Kulal et al. (2024) pointed to AI as the mechanism most capable of closing it. However, Kulal et al. (2024) also noted that the efficiency gains from AI are most pronounced in environments where the baseline process is highly manual and repetitive, precisely the conditions that characterize the MAT of Liliw, Laguna, where every complaint is currently sorted by hand and every dispatch decision is made through verbal communication without data support.

Zhao (2024), in a study presented at the International Conference on Artificial Intelligence and Computer Engineering, approached public service response efficiency from a data mining perspective, applying the Apriori algorithm to identify association patterns in government service request data that can inform proactive resource deployment. Their research demonstrated that historical complaint data contains exploitable patterns, certain types of complaints cluster in specific locations, at specific times, and under specific environmental conditions, and that governments equipped with the analytical tools to surface these patterns can shift from reactive response to anticipatory resource positioning. This is a meaningful extension of what Kulal et al. (2024) established: where the latter showed that AI improves response efficiency in real time, Zhao (2024) showed that data-driven analysis of historical patterns can improve efficiency before complaints are even filed, by positioning resources where and when they are most likely to be needed.

Odigie, Odesanya, and Tamirepi (2024), in their comprehensive analysis of data-driven decision-making for public service optimization published in the World Journal of Advanced Research and Reviews, synthesized findings across multiple

public service sectors, water, mental health, and homelessness, to establish that data-driven operational models consistently outperform intuition-based management in government contexts, not just in efficiency metrics but in equity outcomes. Their study found that government units operating with real-time data dashboards make faster and more consistently equitable resource allocation decisions than those relying on periodic reports and officer judgment, a finding that directly supports the design rationale for TUGON's administrative dashboard. Taken together, these four studies trace a coherent argument: resident-perceived efficiency requires AI and data support (Shen et al., 2023; Kulal et al., 2024), real-time data enables proactive rather than reactive deployment (Zhao, 2024), and data-driven decision-making produces more equitable public service outcomes at the operational level (Odigie et al., 2024).

What none of these studies addresses, however, is the deployment of an integrated AI and real-time data system specifically within the constrained operational environment of a small provincial municipality in the Philippines, one where a single team of officers manages complaints, vehicle dispatch, emergency coordination, and citizen communication simultaneously and without digital support. TUGON responds to this gap by bringing together AI-powered complaint and incident classification, real-time GPS vehicle monitoring, and a centralized administrative dashboard into a single platform designed for exactly this operational context, translating the efficiency principles documented across these four studies into a practically deployable system for the MAT of Liliw, Laguna.

AI and NLP for Complaint Classification

Sorting complaints has always been a fundamentally human task, someone reads what a citizen wrote, makes a judgment about what kind of problem it describes and how urgent it is, and routes it to whoever is responsible for addressing it. For

decades, this worked well enough when complaint volumes were low and the channels through which they arrived were limited. But as digital platforms have multiplied the volume and velocity of incoming citizen concerns, the manual sorting model has begun to break down in ways that are difficult to patch without fundamentally rethinking the process. Natural language processing offers that rethinking. Jiang et al. (2023), in their systematic review of NLP adoption in governments published in Applied Sciences, documented that NLP has moved steadily from experimental deployment in well-resourced national agencies to a more broadly applicable tool for public sector text classification tasks, with complaint routing and service request categorization emerging as among the most consistently successful application domains. Their review found that the core advantage of NLP in government settings is not just speed but consistency, human sorters fatigue, apply varying standards across shifts, and make different judgments about the same complaint depending on context, while a well-trained NLP model applies the same classification logic to every submission regardless of volume or time of day.

Trung et al. (2022), in their study presenting the MACCHIEF algorithm for machine learning-based complaint classification published in the proceedings of a computer science conference, approached this consistency argument from a technical standpoint by demonstrating that ensemble machine learning models combining multiple classification algorithms outperform single-model approaches in complaint categorization tasks, particularly when complaint text is informal, abbreviated, or grammatically inconsistent, as citizen-submitted complaints characteristically are. Their findings both supported and extended Jiang et al.'s (2023) conclusions: NLP works in government complaint contexts, but the specific architecture of the classification model matters enormously, and models that account for the messiness of

real-world citizen language perform substantially better than those trained on clean, structured text. Saha et al. (2024), in their study on automated complaint classification and routing using NLP and machine learning published in IJIRT, built directly on this insight by demonstrating that combining NLP-based text classification with automated routing logic, where classified complaints are not just categorized but immediately directed to the appropriate response queue without officer intervention, produces the greatest operational efficiency gains. Where Trung et al. (2022) focused on improving classification accuracy, Saha et al. (2024) showed what happens downstream when that accurate classification is connected to an automated workflow: complaint resolution times drop, officer workload decreases, and the risk of misrouted or overlooked concerns is substantially reduced.

Thayumanavan (2025), in a study presented at the IEEE International Conference on Sustainable Computing and Data Communication Systems, shifted the technical conversation toward pre-trained transformer models by demonstrating that fine-tuned BERT and similar pre-trained architectures significantly outperform traditional machine learning classifiers in telecom complaint categorization tasks, particularly in handling domain-specific language and abbreviated customer descriptions. This finding creates an important tension with Trung et al.'s (2022) ensemble approach: while ensemble models improve on single-model performance, pre-trained transformers fine-tuned on domain-specific data appear to outperform even well-constructed ensembles when sufficient domain data is available for fine-tuning. Thayumanavan (2025) and Saha et al. (2024) together suggest that the optimal NLP complaint classification architecture pairs a powerful pre-trained language model with an automated routing and prioritization workflow, combining classification intelligence

with operational automation in a way that neither study fully achieved independently but that both point toward as the direction the field is moving.

What these four studies collectively establish is a maturing body of evidence that NLP-powered complaint classification is technically viable, operationally beneficial, and increasingly indispensable for government and institutional complaint management at scale. Where they converge, the argument is compelling: automated classification is more consistent than manual sorting, pre-trained models handle informal language better than rule-based systems, and classification gains its greatest value when connected to automated routing and prioritization workflows. Where they diverge is instructive, particularly on the question of architecture, but the gap none of them addresses is the one most relevant to TUGON: complaint classification in Filipino and Taglish, the mixed-language register in which citizens of provincial Philippine municipalities like Liliw, Laguna naturally express their concerns. Existing studies work almost exclusively in English or other well-resourced languages, leaving the classification challenge for Philippine civic complaints largely unaddressed. TUGON responds to this directly by deploying the Gemini 3.1 flash model for Filipino language understanding alongside the facebook/bart-large-mnli zero-shot classifier, bringing the classification capabilities documented across these studies to a linguistic and governmental context that the literature has not yet reached.

Filipino Language Processing and Multilingual NLP

Language is not a neutral container for information, it carries the texture of how people think, feel, and communicate in their daily lives. When a citizen of Liliw, Laguna files a complaint, they are unlikely to write in formal Filipino or standard English. They will write the way they speak: a mix of Filipino, Tagalog, English, and local expressions that linguists call code-switching, and that everyday Filipinos simply

call normal. This linguistic reality presents a fundamental challenge for any AI-powered complaint classification system deployed in a Philippine provincial context, because the overwhelming majority of NLP models used in government systems globally are trained on English or other high-resource languages and perform poorly, sometimes catastrophically so, when confronted with the informal, multilingual character of Filipino civic communication. Coronia (2022), in a master's thesis from De La Salle University examining the clustering of Philippine languages in multilingual neural machine translation, documented precisely this challenge, finding that Philippine languages, including Filipino, Tagalog, and their regional variants, cluster in ways that distinguish them structurally from both English and other Southeast Asian languages, making direct application of general-purpose multilingual models to Philippine language tasks unreliable without specific architectural accommodations. Coronia's analysis established that Philippine language NLP is not simply a translation problem that general multilingual models can solve by proximity, it is a distinct technical challenge that requires language-specific model design or fine-tuning to address adequately.

Montalan et al. (2025), in their arXiv preprint introducing Batayan, a Filipino NLP benchmark specifically designed for evaluating Large Language Models on Filipino language tasks, extended Coronia's (2022) foundational observation into a rigorous empirical framework. Their study systematically evaluated multiple state-of-the-art LLMs on Filipino language benchmarks and found that even the most capable general-purpose models show measurable performance degradation on Filipino tasks compared to their English-language performance, with the degradation most pronounced in tasks involving informal language, code-switching, and domain-specific vocabulary. Where Coronia (2022) established the structural distinctiveness of

Philippine languages as a theoretical challenge, Montalan et al. (2025) quantified the practical consequences of that distinctiveness for real-world NLP deployment, providing empirical evidence that deploying an English-optimized or general multilingual model for Filipino complaint classification without language-specific adaptation will produce systematically less accurate results than the same model achieves in English. Together, these two studies make a strong case that Filipino NLP is an active and unresolved technical frontier, not a solved problem that existing general-purpose models can be assumed to handle adequately.

Sarzaeim et al. (2023), in their systematic review of machine learning and NLP in smart policing published in Computers, approached the language challenge from a different angle, not as a linguistic research question but as a public safety deployment concern. Their review documented that NLP-assisted complaint and incident classification systems deployed in public safety contexts frequently underperform in communities where the local language or dialect is underrepresented in the model's training data, with minority-language communities experiencing systematically lower classification accuracy than majority-language ones. This finding, while not specific to Filipino, has direct implications for TUGON's deployment context: if a complaint classification model is not adequately trained on Filipino and Taglish text, Citizens who submit complaints in their natural language will receive lower-quality automated classification than would be achievable with a language-appropriate model, a disparity that translates directly into slower or less accurate urgency sorting. Where Coronia (2022) and Montalan et al. (2025) established the linguistic challenge from a Filipino NLP research perspective, Sarzaeim et al. (2023) connected that challenge to the operational consequences it produces in public safety complaint systems, completing

the argument for why language-specific model selection is not a secondary technical detail but a primary design requirement.

What these three studies together make clear is that deploying effective AI-powered complaint classification in a Philippine provincial government context is inseparable from the question of Filipino language processing, and that the general-purpose NLP models most commonly used in government complaint systems are not adequate for this context without deliberate language-specific adaptation. This is precisely the gap that TUGON addresses through its dual-model NLP architecture: the Gemini 3.1 Flash, a general-purpose multilingual large language model developed by Google, supports Filipino and Taglish inputs as part of its broad multilingual capability, enabling zero-shot complaint classification and urgency detection without requiring a locally annotated training dataset. This combination directly responds to the technical challenges documented by Coronia (2022) and Montalan et al. (2025) and addresses the public safety implications identified by Sarzaeim et al. (2023), bringing linguistically appropriate AI-powered complaint classification to a community whose language the broader NLP literature has only recently begun to adequately serve.

Notification Systems and Real-Time Alerting in Public Service Platforms

When there's an urgent event, a flood, a missing person, a road closed by a fallen tree, how fast the right people are notified can be the difference between a problem that is addressed and a problem that becomes a disaster. Notification systems of public service platforms are not just a nice to have feature but an important part of the chain from the occurrence of an event to the triggering of a government response. Sadiq et al. (2023) in a comprehensive literature review of public alert and warning systems in the United States published in Natural Hazards documented that despite decades of investment in alert infrastructure, there are significant gaps in the extent to

which public warning systems reach the people who need them, when they need them, and in a form, they can act on.

Their review found that the most serious failures in public alerting are not technical, but structural, systems that send alerts without confirmation of receipt, that use channels that are not available to some segments of the population, and that have no mechanism for two-way communication between alert senders and recipients of alerts. Sadiq et al. (2023) concluded that structural gaps are not unique to the United States but reflect a larger pattern of how governments have historically designed alerting systems for the convenience of the institution rather than the needs of the citizen.

Mowbray et al. (2024) in a systematic review of mobile alerting for public emergencies published in the Journal of Contingencies and Crisis Management, brought the conversation directly into the mobile era by examining how citizens respond to emergency alerts delivered through mobile devices. Their review found that mobile alerting is significantly more effective than traditional broadcast channels, radio, television, sirens, at reaching citizens during emergencies, especially younger populations and those in areas with limited broadcast infrastructure. Yet there is a consistent finding that Sadiq et al. (2023) pointed towards but did not fully elaborate, that Mowbray et al. (2024) also identified: the effectiveness of a mobile alert is strongly moderated by the citizen's prior relationship with the alerting institution. Those who have received correct, timely and actionable alerts from a government platform before respond faster and more appropriately to new alerts from the same source. People who experience false alarms, delayed notifications, or notifications not followed by visible government action are progressively less responsive over time. This erosion of trust is central to TUGON's notification design: every alert and status update the system

delivers to citizens has to be accurate and follow-up action has to be visible, because every interaction either builds or chips away at the notification credibility that will determine whether future alerts will be heeded.

In their research on design and implementation of real-time IoT-based public safety alert and emergency response systems, Zhang, Zhang, and Sun (2025) tackled the notification problem from the perspective of technical architecture. They showed that real-time alerting systems with IoT integration, where data from sensors, reports on incidents, and GPS information are automatically processed and transformed into targeted push notifications to specific responders or groups of citizens, perform significantly better than broadcast-style alert systems in terms of response initiation time and resource deployment accuracy. Their system architecture, which is similar to TUGON's design in several meaningful ways, demonstrated that linking complaint intake directly with automated notification triggers, such that a newly submitted high-urgency complaint immediately generates an alert to the on-duty MAT officer without requiring manual review, substantially reduces the time between complaint submission and response initiation by removing the human-in-the-loop bottleneck that characterizes manual notification workflows. Building on the technical findings of Zhang et al., Villarreal, MacPherson-Krutsky, and Painter (2025) added an equity dimension that the architecture-focused study did not address in their study of barriers and best practices for inclusive emergency alerts published in the International Journal of Disaster Risk Reduction: effective public safety notifications must be designed to reach all citizens, including those with limited digital literacy, inconsistent internet access, and varying language proficiency. The study's findings are directly relevant to TUGON's rollout in Liliw, Laguna, where citizens' digital literacy and connectivity vary considerably across barangays. Alert systems optimized solely for the technically

adept systematically underperform in exactly the communities most vulnerable during emergencies.

In their study on notification, recommendation, and monitoring systems published in the International Journal of Human-Computer Interaction, Marquès et al. (2022) added a behavioral dimension to this technical and equity-focused literature by showing that notifications are most effective when they are contextually relevant, timely, and actionable, providing the recipient with not just an alert, but a clear next step they can take in response. In their study, they found that generic notifications that do not contain specific, actionable content result in significantly lower engagement and follow-through compared to those that tell the recipients exactly what has happened and what they should do next. This behavioral insight links the technical architecture documented by Zhang et al. (2025) to the citizen trust argument of Mowbray et al. (2024): well-timed, contextually specific, action-oriented alerts build the institutional credibility that makes future alerts more effective, while vague or poorly timed alerts erode it. Collectively, these five studies present a coherent and multi-layered argument for how public service notification systems must be designed to be truly effective, technically real-time, institutionally trustworthy, inclusively accessible, and behaviorally actionable.

However, none of these studies address the deployment of an integrated real-time notification system within a small provincial Philippine municipal government platform that simultaneously notifies MAT officers of newly submitted high-urgency complaints, notifies citizens of status updates on their filed concerns, and triggers driver dispatch notifications, all within a single unified system operating under inconsistent network conditions and serving users of varying digital literacy. TUGON's notification architecture directly addresses this gap, embedding automated push notifications for

complaint status changes, officer alerts for newly submitted urgent concerns, and direct messaging triggers within its Communication Module. This translates the alerting principles documented by Sadiq et al. (2023), Mowbray et al. (2024), Zhang et al. (2025), Villarreal et al. (2025), and Marquès et al. (2022) into a real-time, citizen-centered notification system purpose-built for the operational conditions of the MAT of Liliw, Laguna.

AI-Enabled Decision Support for Local Government Services

A government that is armed to anticipate problems is different from one that reacts to problems. For most LGUs the former is reality, officers respond to what is coming through the door, dispatch based on what they know now, and make decisions that are only as good as the information immediately available to them. AI-powered decision support systems are changing this equation by providing government personnel with real-time data analysis, pattern recognition, and automated recommendations that amplify their ability to make faster, more informed, and more consistently sound decisions. The current basis for this argument was laid out by Chen et al. Journal of Organizational and End User Computing. (2025). Artificial Intelligence-Based Intelligent Decision Support System in E-Government Services: They documented that AI decision support systems deployed across multiple e-government contexts consistently improve the speed, accuracy, and consistency of government decision-making; especially in high-volume operational environments where officers have to process large amounts of incoming information simultaneously. Their research also found that the best AI decision support systems in government are those integrating multiple data streams, such as complaint data, location data, historical patterns, and resource availability, into a single decision interface that offers officers

actionable recommendations rather than raw data, which lowers cognitive load while preserving human control over final decisions.

At the International Conference on Digital Economy and Computer Science, Liu and Li (2025) examined AI-based public policy decision support systems and validated and extended the results of Chen et al. (2025) that AI decision support systems attain their greatest operational value when designed to enhance, not supplant, officer judgment by exposing relevant data, highlighting anomalies, and offering prioritized options, but ultimately deferring to human judgment for final decisions. The study demonstrated that officers employing AI decision support tools made faster decisions, used more relevant information before acting, and had more confidence in their decisions than officers who did not have data support. This augmentation design principle articulated by Liu and Li (2025) provides an important qualification to Chen et al.'s (2025) efficiency argument: AI decision support is not simply about processing speed but about the quality of human judgment that the system enables. The two studies together suggest that the most consequential benefit of AI in government operations is not that it replaces human decisionmakers but that it makes them measurably better at their jobs.

Cities. David et al. (2024) reviewed local government AI strategy and discussed how municipal governments globally have formally responded to the adoption of AI through policy frameworks and strategic documents. Their international review revealed local governments that have developed explicit AI governance strategies that describe what AI will and will not be used for, who is responsible for AI-assisted decisions, and how citizens will be informed of AI's role in service delivery, experience significantly more successful and sustainable AI deployments than those that have adopted AI tools without corresponding governance frameworks. This finding expands

upon both Chen et al.'s (2025) efficiency argument and Liu and Li's (2025) augmentation design principle: technical soundness and human-AI complementarity matter but are not sufficient without institutional governance that makes AI's role in government decision-making visible and accountable to the public. Crucially, David et al. (2024) also observe that smaller municipalities are significantly underrepresented in the existing AI governance frameworks. Most policy documents are designed around the capacities and resources of large city governments, leaving smaller LGUs with no clear guidance on how to responsibly adopt AI tools within their more constrained institutional environments.

Most of the studies (Chen et al., 2025, Liu and Li, 2025; David et al., 2024) have focused on the institutional perspective of the public support to AI-enabled local government services. Senadheera et al. (2026) bringing the citizen perspective to their analysis in the paper on SSRN. Their research indicated that governance context is a strong moderator of public support for AI-enabled government services, citizens are much more likely to accept AI-assisted government decisions when they understand the governance framework around the use of AI, when they perceive the AI as supporting, rather than supplanting, human accountability, and when the AI is applied to service domains, they perceive as high-need. This finding introduces a democratic legitimacy aspect to David et al.'s (2024) governance framework argument: institutional AI governance is important not only for technical accountability but also for maintaining the public trust that AI-enabled government services rely on to work effectively. The OECD (2025), in its report on governing with artificial intelligence in core government functions, synthesized global evidence across all these dimensions to establish that AI decision support is most transformative in exactly the operational areas where human capacity is most stretched, complaint triage, resource dispatch, emergency coordination,

and real-time situational monitoring, and that governments deploying AI in these functions consistently report reductions in response times, improvements in resource allocation efficiency, and increases in officer capacity to handle complex multi-incident situations.

The argument across these five studies converges on a point that is directly relevant to TUGON: AI decision support in local government is technically validated, institutionally necessary, and increasingly supported by the citizens it serves, but the evidence base is overwhelmingly drawn from large, well-resourced governments, leaving small provincial municipalities without documented models they can adapt. TUGON provides a solution to this gap by applying AI-enabled complaint classification, anomaly detection, and real-time administrative supervision within the limited institutional scope of the MAT of Liliw, Laguna, providing a documented Philippine municipal case study that adds to the evidence base urgently needed by smaller local governments in their journey of responsible and publicly accountable AI adoption as collectively identified by the OECD (2025), David et al. (2024), and Senadheera et al. (2026).

Vehicle Dispatch and Fleet Management in Government Operations

A government vehicle sitting idle when it should be responding to an emergency or being used for personal errands when it should be on official duty, is not just an administrative inconvenience, it is a failure of public trust and a misallocation of resources that citizens have collectively paid for. Fleet management in government operations has long been recognized as a governance challenge as much as a logistical one, and the research reflects this dual nature. Chiparo, Tukuta, and Musanzikwa (2022), in their study of vehicle fleet management practices and service delivery in state-owned enterprises in Zimbabwe published in the Journal of Transportation Technologies,

established that poor fleet management practices, including the absence of formal dispatch protocols, inadequate vehicle utilization records, and the lack of structured accountability mechanisms for drivers, directly and measurably reduce the quality of public service delivery. Their findings documented that organizations with formalized fleet management systems, where every vehicle deployment is logged, authorized, and traceable, consistently outperform those relying on informal verbal dispatch arrangements in both operational efficiency and service reliability. Critically, Chiparo et al. (2022) found that the accountability gap is most severe in government institutions where fleet oversight is assigned informally rather than institutionally, a description that maps directly onto the current operational reality of the MAT of Liliw, Laguna, where drivers are assigned verbally and no formal dispatch record exists.

Chikazhe et al. (2023), in their study of fleet management systems and perceived service quality in Zimbabwe's public health sector published in the International Journal of Public Sector Management, extended Chiparo et al.'s (2022) accountability argument by demonstrating that fleet management quality does not just affect internal operational efficiency, it shapes how citizens perceive and experience public service quality as a whole. Their study found that citizens who experience reliable, timely vehicle-supported service delivery, ambulances that arrive within expected timeframes, government responders who navigate accurately to incident locations, report significantly higher confidence in the government institution responsible, while delayed or misdirected responses erode institutional trust in ways that take considerable time and effort to rebuild. This finding adds a citizen-facing dimension to what Chiparo et al. (2022) established from an operational perspective: fleet management is not an internal administrative matter but a direct determinant of

the government-citizen relationship, making its improvement not just an efficiency goal but a public trust imperative.

Bolaños et al. (2022), in their study published in Transportation Research Interdisciplinary Perspectives, approached fleet management from a technical design perspective by proposing a fleet management and control system specifically engineered for the constraints of developing country contexts, inconsistent network connectivity, limited institutional technical capacity, and constrained budgets that make enterprise-grade fleet management solutions impractical. Their system, built on Intelligent Transportation Systems services, demonstrated that cloud-connected GPS fleet monitoring can be implemented effectively even in low-resource environments, directly challenging the assumption that sophisticated fleet accountability requires sophisticated infrastructure. Where Chiparo et al. (2022) and Chikazhe et al. (2023) documented the governance and citizen trust dimensions of fleet management, Bolaños et al. (2022) provided the technical validation that meaningful fleet accountability is achievable in exactly the kind of resource-constrained developing country context that characterizes provincial Philippine municipalities like Liliw, Laguna. Punith, Nithya, and Deepa (2022), in their IEEE conference paper on IoT-enabled smart fleet management, extended this technical validation further by demonstrating that integrating IoT sensors with cloud-based dashboards enables administrators to monitor fleet efficiency metrics, idle time, route adherence, trip duration, driver behavior patterns, through a centralized web interface that requires no specialized infrastructure beyond a stable internet connection and mobile-equipped vehicles.

Read together, these four studies construct a compelling and multi-layered argument for why structured, technology-supported fleet management matters in government operations: it is a governance accountability mechanism (Chiparo et al.,

2022), a citizen trust determinant (Chikazhe et al., 2023), and a technically achievable goal even in resource-limited developing country environments (Bolaños et al., 2022; Punith et al., 2022). What they do not address, however, is the integration of fleet management with complaint management and machine learning-based anomaly detection within a single unified platform designed for a small provincial municipal government, the specific combination that TUGON proposes. Each study treats fleet management as a standalone system, optimized independently of the complaint intake, citizen communication, and emergency coordination functions that the MAT of Liliw, Laguna must manage simultaneously. TUGON fills this integration gap by embedding its Vehicle Dispatch Management module within a broader operational platform, enabling MAT officers to correlate active complaints with available vehicles, monitor dispatch status in real time, and receive automated alerts when vehicle movement patterns deviate from expected operational behavior, capabilities that no standalone fleet management system in the existing literature fully provides.

GPS-Based Tracking and Location-Aware Public Service Systems

Knowing where something is, a vehicle, a citizen in distress, a government responder navigating to an emergency, is deceptively simple as a concept and surprisingly consequential as an operational capability. Before GPS-based tracking became practically accessible, government field operations relied on radio communication, verbal location descriptions, and the personal familiarity of drivers with their service area. These methods worked well enough in stable, low-complexity situations but broke down predictably during emergencies, high-volume operational periods, and situations where precise navigation to an unfamiliar or difficult-to-describe location was critical. Raikar et al. (2023), in their IEEE conference paper on fleet tracking and geofencing using IoT presented at the International Conference on Secure

Cyber Computing and Communication, demonstrated that GPS-based tracking systems equipped with geofencing capabilities, the automatic detection and alerting of vehicle movement outside designated operational boundaries, substantially improve fleet accountability by making unauthorized use immediately visible to administrators rather than discoverable only after the fact. Their study established that the combination of real-time location streaming and boundary violation alerts shifts fleet oversight from a retrospective auditing activity to a continuous monitoring function, fundamentally changing the accountability dynamic between government institutions and their vehicle operators.

Jeevan et al. (2024), in their study presented at the IEEE International Conference on Intelligent Cyber Physical Systems and Internet of Things, extended the GPS tracking architecture by incorporating two-factor authentication and IoT integration to address a vulnerability that Raikar et al. (2023) did not fully resolve: knowing where a vehicle is tells an administrator very little if it is unclear who is operating it. Their research demonstrated that linking GPS location data to authenticated driver identity, requiring drivers to verify their credentials before a trip is formally logged and tracked, creates a verifiable chain of accountability that connects every recorded vehicle movement to a specific responsible individual. This identity-location linkage is a meaningful technical advancement over the geofencing approach of Raikar et al. (2023): where the latter flags when a vehicle goes somewhere it should not, Jeevan et al.'s (2024) approach simultaneously records who took it there, producing an audit trail that is actionable for disciplinary and administrative accountability purposes in ways that location data alone is not.

Gade et al. (2024), in their study on a real-time tracking system for city bus location and passenger count published in the International Journal of Advanced

Research in Science, Communication and Technology, approached GPS tracking from a citizen-facing public service perspective rather than a fleet accountability one. Their system demonstrated that real-time GPS location data, when made accessible to citizens through a mobile interface, significantly improves the citizen experience of government-managed transport services by reducing uncertainty and enabling informed decision-making. While their study focused on public transit rather than government emergency response vehicles, its core finding, that location awareness is as valuable to citizens on the receiving end of a service as it is to administrators managing the delivery of that service, translates directly to TUGON's citizen-facing design. A citizen who can share their precise GPS location with the MAT, and a MAT officer who can see exactly where a dispatched vehicle is relative to that location, together produce a more efficient and more reassuring emergency response experience than verbal coordinate exchange and manual navigation ever could.

Kumar et al. (2023), in their study on a smart assistance and governing system for disabled individuals published in the E3S Web of Conferences, examined GPS tracking in the context of real-time location monitoring for vulnerable populations, a context that, while different in application domain from municipal fleet management, surfaced a technical insight with broad relevance: GPS-based location awareness systems are most operationally effective when location data is not just streamed passively but actively integrated with response triggers and administrative alerts. Their system demonstrated that continuous location monitoring combined with threshold-based alerting, where specific location events automatically notify designated responders, produces faster and more reliable response outcomes than systems where location data is available but must be actively checked by an administrator. This active integration principle directly informs TUGON's administrative dashboard design,

where real-time vehicle GPS feeds are displayed alongside active complaint locations, enabling MAT officers to visually correlate vehicle positions with incoming emergency reports rather than monitoring each data stream in isolation.

Together, these four studies trace a coherent evolution in GPS-based public service tracking: from boundary-based fleet accountability (Raikar et al., 2023), to identity-linked vehicle monitoring (Jeevan et al., 2024), to citizen-facing location transparency (Gade et al., 2024), to alert-integrated real-time location systems (Kumar et al., 2023). Each study addresses a different dimension of what comprehensive location-aware public service management requires, and together they describe the full capability profile that TUGON's GPS tracking architecture is designed to deliver. What none of them addresses, however, is the deployment of an integrated GPS tracking system within the specific context of a Philippine provincial municipal government, one that must simultaneously track vehicles, receive citizen location pins, and display both on a unified administrative map without the benefit of dedicated technical staff or enterprise-grade infrastructure. TUGON bridges this gap by combining driver GPS tracking with citizen location sharing and a real-time administrative map interface, translating the location-awareness capabilities documented across these studies into a single operationally coherent platform for the MAT of Liliw, Laguna.

Real-Time Monitoring and Accountability in Public Service Systems

Accountability in government has traditionally been a retrospective exercise, audits conducted after the fact, reports compiled at the end of a quarter, complaints reviewed long after the situation that generated them has passed. This temporal gap between action and accountability is not just an administrative inconvenience; it is a structural weakness that allows problems to compound, resources to be misused, and citizens to remain uninformed about what their government is actually doing on their

behalf. Real-time monitoring systems are changing this fundamental dynamic by compressing the gap between when something happens and when it becomes visible to those responsible for overseeing it. Naga Srinivasa Rao Balajepally (2024), in a study published in the Journal of Information Systems Engineering and Management, examined how real-time data integration in public sector software transforms government transparency and citizen trust. Their findings established that government systems equipped with live data dashboards, where complaint statuses, resource deployment, and operational metrics are updated continuously rather than periodically, produce measurably higher levels of citizen-perceived transparency than those relying on batch-processed or manually updated reporting. Crucially, Balajepally (2024) also found that real-time visibility is not just valuable to citizens but operationally transformative for government personnel, enabling officers to make faster and more contextually informed decisions when they can see the full current state of their operational environment rather than working from information that may already be hours old.

Hidayat (2025), in their study on digital governance and transparency published in the Journal of Governance and Regulation, extended Balajepally's (2024) transparency argument by situating real-time data access within the broader framework of open data initiatives and democratic accountability. Their research found that governments which make operational data continuously accessible, not just to internal administrators but to the citizens and oversight bodies that those administrators serve, demonstrate measurably stronger accountability outcomes, with lower rates of resource misuse, faster identification of service delivery failures, and greater citizen confidence in institutional integrity. Where Balajepally (2024) focused on the internal operational benefits of real-time data for government decisionmakers, Hidayat (2025) emphasized

its external accountability function, the way that real-time visibility makes government behavior legible and contestable in ways that periodic reporting never fully achieves. Together, these two studies establish a dual accountability logic for real-time monitoring systems: they simultaneously improve what the government can do internally and what citizens can see externally, producing accountability gains on both sides of the government-citizen relationship.

Larcher et al. (2024), in their study on Backend-as-a-Service-enabled workflows in federated serverless infrastructures published in IEEE Transactions on Cloud Computing, approached real-time monitoring from the infrastructure layer, the technical architecture that determines whether real-time data delivery is even feasible at the performance levels that meaningful government accountability requires. Their research demonstrated that BaaS platforms, which provide managed real-time data synchronization through event-driven database replication, deliver the latency and reliability characteristics necessary for operational government monitoring systems without requiring the dedicated backend engineering teams that traditional real-time data architectures demand. This finding creates an important technical bridge between the accountability arguments of Balajepally (2024) and Hidayat (2025) and the practical deployment realities of small government institutions: the accountability benefits of real-time monitoring are only achievable if the underlying infrastructure can reliably deliver data updates fast enough to matter operationally. Larcher et al. (2024) established that BaaS architectures meet this technical threshold, directly validating the infrastructure choice that TUGON makes in using Supabase, a PostgreSQL-based BaaS platform, as its real-time data backbone.

What these three studies together establish is a layered argument for real-time monitoring in public service systems: the accountability case is compelling both

internally and externally (Balajepally, 2024; Hidayat, 2025), and the technical infrastructure to deliver it in resource-constrained government environments is now practically accessible (Larcher et al., 2024). The gap that runs through all three, however, is the absence of a documented implementation that brings these layers together in a small provincial government context, one where real-time monitoring must serve not just transparency goals but active operational coordination across complaint management, vehicle dispatch, and citizen communication simultaneously. TUGON's administrative dashboard addresses this gap directly, combining live complaint feeds, real-time vehicle GPS tracking, and citizen location data in a single unified monitoring interface powered by Supabase's BaaS real-time architecture, translating the accountability and transparency principles documented by Balajepally (2024) and Hidayat (2025) into an operationally deployable platform for the MAT of Liliw, Laguna.

Geospatial Technologies for Local Disaster Preparedness

The Philippines sits at the intersection of the Pacific typhoon belt and the Ring of Fire, making it one of the most disaster-prone countries in the world. For LGUs in flood-vulnerable municipalities, disaster preparedness is not a contingency plan, it is an operational necessity that shapes how government resources must be positioned, how response teams must be organized, and how citizens must be informed before, during, and after each typhoon season. The challenge has always been that effective disaster preparedness requires knowing where the risks are with enough precision and timeliness to act on that knowledge before a disaster unfolds. Geospatial technologies, particularly GIS-based mapping, remote sensing, and real-time spatial data visualization, have emerged as the most consequential tools for closing this gap between risk knowledge and preparedness action. Handoyo et al. (2024), in their study

published in Jàmbá: Journal of Disaster Risk Studies, examined the role of spatial disaster learning using geospatial technology in community and government disaster preparedness, finding that communities and government units equipped with spatially explicit disaster risk information, maps that show not just that flooding happens but where it is most likely to occur, at what depth, and under what weather conditions, demonstrate significantly stronger preparedness behaviors and faster emergency response times than those relying on generalized risk narratives without geographic specificity. Their study established that geospatial visualization is not merely an analytical tool but a communication instrument that makes abstract risk data actionable for both government planners and the citizens they serve.

Aggarwal, Kundu, and Sarma (2024), in their study on geospatial technology for effective disaster risk reduction published in the International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences, extended Handoyo et al.'s (2024) findings by examining best practices in geospatial capacity building across multiple disaster risk contexts. Their research found that the most effective government disaster preparedness programs combine geospatial risk mapping with real-time situational awareness tools, allowing emergency managers to update their understanding of where risk is concentrating as a disaster event unfolds, rather than relying solely on pre-event static maps. This distinction between static risk mapping and dynamic situational awareness is critical: Handoyo et al. (2024) established the value of knowing where risks exist in advance, while Aggarwal et al. (2024) demonstrated that this pre-event knowledge must be complemented by tools that track how the risk landscape changes in real time as weather conditions evolve. Together, these two studies make the case for the kind of dynamic, overlay-capable disaster heatmap that TUGON's Admin Management module provides, one that can

display pre-identified flood-prone areas within Liliw, Laguna while being updated as typhoons and heavy rain events develop.

Al-Omari et al. (2024), in their study on flood hazard mapping and risk assessment using remote sensing and GIS techniques published in the Civil Engineering Journal, and Diriba et al. (2024), in their study on flood hazard analysis using remote sensing, GIS, and AHP techniques published in Geomatics, Natural Hazards and Risk, both demonstrated that layered geospatial approaches, combining elevation data, land cover classification, drainage network analysis, and historical flood records, produce significantly more accurate flood hazard maps than single-variable approaches. While Al-Omari et al. (2024) focused on flood hazard delineation in an urban watershed context and Diriba et al. (2024) applied similar techniques to the Ethiopian Rift, both studies arrived at a consistent methodological conclusion: flood risk is spatially heterogeneous at the local level, meaning that even within a single municipality, some barangays face dramatically higher flood exposure than others under the same rainfall conditions. This sub-municipal spatial variability is directly relevant to TUGON's disaster heatmap feature, which is designed to display flood risk at the barangay level within Liliw, Laguna, giving MAT officers the geographic specificity they need to pre-position resources in the highest-risk areas before a typhoon makes landfall rather than responding reactively after flooding has already begun.

Khan et al. (2025), in their study on a fog computing-based real-time flood prediction and early warning system using machine learning and remote sensing data published in the Journal of Sustainable Development and Policy, pushed the technical frontier further by demonstrating that combining machine learning with real-time remote sensing data enables predictive flood early warning, not just mapping where flooding has historically occurred but forecasting where it is about to occur based on

current rainfall, soil saturation, and river level data. Their system achieved early warning lead times sufficient for meaningful emergency response preparation, representing a significant advancement over the static hazard mapping approaches documented by Al-Omari et al. (2024) and Diriba et al. (2024). Amatebelle et al. (2025), in their systematic analysis of remote sensing and GIS applications for flood disaster risk management published in the Journal of Spatial Science, synthesized the broader evidence base on geospatial disaster tools and confirmed that integration, combining static hazard maps, real-time weather overlays, and citizen-reported incident data within a single operational interface, consistently produces better disaster management outcomes than any single geospatial tool deployed in isolation.

Taken together, these six studies construct a clear and converging argument: effective local disaster preparedness requires geospatial tools that are specific enough to guide municipal-level resource positioning (Handoyo et al., 2024; Al-Omari et al., 2024; Diriba et al., 2024), dynamic enough to reflect evolving conditions during a disaster event (Aggarwal et al., 2024; Khan et al., 2025), and integrated enough to connect risk visualization with operational response coordination (Amatebelle et al., 2025). What the literature does not yet provide, however, is a documented implementation of a disaster heatmap integrated within a broader municipal government operations platform, one that places flood risk visualization alongside active complaint management and vehicle dispatch monitoring within the same operational interface used by frontline government responders. TUGON's typhoon and heavy rain disaster heatmap addresses this integration gap, giving the MAT of Liliw, Laguna a geospatially informed operational picture during disaster events that connects risk awareness directly to the complaint management and vehicle dispatch decisions officers must make in real time.

Disaster Response, Heatmaps, and Emergency Monitoring

How a government responds to an emergency, a typhoon or flash flood, for example, is a function not just of the resources it has, but of how fast and how well it can deploy those resources to where they are most needed. In fast-moving, high-pressure disaster scenarios, information can be the difference between a response that works and one that doesn't: Do emergency managers have a clear, real-time picture of where the danger is focused, where the victims are located and where their response assets are deployed? In their detailed review of public alert and warning systems in the United States, Sadiq et al. (2023) found that the most common failures in emergency response are not failures of resources but failures of information architecture, systems that cannot tell responders where to go, what is happening or how serious each incident is relative to others arriving at the same time. In their review, they determined that the biggest gap between having emergency management tools and using them effectively is at the local government level, where officers are often responding to multiple simultaneous incidents with limited staff and without a way to consolidate real-time information. The lack of local government information was highlighted by Sadiq et al. (2023), which is precisely the operational challenge that TUGON's disaster monitoring features aim to address.

In a paper about explainable AI in disaster risk management published in the International Journal of Disaster Risk Reduction, Ghaffarian et al. (2023) extended the information architecture argument by addressing a challenge that Sadiq et al. (2023) highlighted but were unable to fully resolve, namely, that emergency managers might be presented with AI-generated risk information without a full grasp of the logic behind it. Their review found that the operational value of AI-powered tools for disaster risk is maximized when they are designed to be explainable, when the system can tell you not

only what risk level it has calculated for a given area, but why, based on what data and analytical logic. This explainability requirement is particularly important in high-stakes emergency response contexts where an officer must be able to explain resource deployment decisions to supervisors and the public. While Sadiq et al. (2023) articulated the information gap that emergency monitoring systems must fill, Ghaffarian et al. (2023) articulated the governance conditions under which that information can be responsibly generated and used, a distinction that directly informs TUGON's disaster heatmap design philosophy of presenting risk overlays based on externally sourced PAGASA and PHIVOLCS data, rather than opaque proprietary risk calculations, to ensure that every risk visualization presented to MAT officers can be traced back to a recognized authoritative source.

Mohsin, Choudhury and Muyeed (2025) Transportation Engineering Automatic priority analysis of emergency response systems using IoT and machine learning. They took an operational workflow view for emergency monitoring and showed that emergency response systems can automatically prioritize the severity of an incident and suggest the deployment sequence of resources without manual officer assessment of each incoming report by integrating real-time sensor data with an ML-based priority analysis. Their results demonstrated that automated priority analysis decreases the time to initiate emergency responses by removing the assessment bottleneck that happens when officers must manually assess multiple simultaneous incidents during a disaster event. Here is a relevant link to the Ghaffarian et al. (2023) explainability argument: Mohsin et al. (2025) have demonstrated gains in operational efficiency from automated emergency prioritization, whereas Ghaffarian et al. (2023) have shown that such automation must remain transparent and interpretable to retain officer trust and accountability. Together, the two studies describe the design trade-off required for

effective AI-powered emergency monitoring systems: automated enough to speed up response, and explainable enough to remain trusted.

ResQ is an intelligent real time disaster management system using machine learning and IoT by Duraimurugan et al. (2024) in IEEE International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems, indicates that the combination of real time IoT sensor inputs with ML-based incident classification and geographic visualization is a disaster management platform that can simultaneously track multiple incident types, prioritize response and direct emergency teams to the highest need locations. The system enhanced the operational capabilities described by Mohsin et al. (2025) by adding geographic routing and multi-incident coordination features, demonstrating that real-time disaster monitoring systems are most functional when they unite incident detection, prioritization, and spatial navigation in one operational interface. In a separate study, Hossain (2026) examined the use of AI-enabled predictive analytics in disaster management and found that machine learning algorithms trained on historical disaster data can give preparedness recommendations before an actual disaster strikes, giving emergency planners advance intelligence on where to position supplies in anticipation of predicted impact zones. But while this predictive capability is beyond the scope of TUGON's initial implementation, it is the natural extension of the disaster heatmap functionality introduced by TUGON. The findings of Hossain (2026) support the basic premise of providing geographic disaster risk intelligence to MAT officers before and during emergency events.

The most forward-looking vision of this theme is brought by Gish and Rapaport (2026) in their work on agentic AI for unprecedented weather preparedness published in NPJ Digital Medicine. They recorded that AI systems capable of autonomous information gathering and analysis are beginning to demonstrate meaningful value in

weather-related emergency preparedness contexts, producing risk assessments and resource recommendations that are better than what human analysts can produce under time pressure. Their study had as its main finding that AI-augmented emergency monitoring leads to faster and more complete situational awareness than human monitoring alone. This supports the wider argument that underlies all six studies in this theme. The convergence is clear and consistent: emergency management systems most often fail not due to lack of resources but because information is fragmented, delayed or unavailable (Sadiq et al., 2023); AI-generated risk intelligence must remain explainable to be responsibly actionable (Ghaffarian et al., 2023); automated prioritization reduces response initiation time (Mohsin et al., 2025); integrated real-time platforms outperform single function tools (Duraimurugan et al., 2024); predictive analytics extend preparedness beyond reactive response (Hossain, 2026); and AI-augmented monitoring reliably improves emergency outcomes (Gish & Rapaport, 2026).

They all do not have a documented disaster monitoring feature within a municipal government operational platform for a small, resource-limited Philippine LGU. TUGON's typhoon and heavy rain disaster heatmap directly addresses this gap by providing the MAT of Liliw, Laguna a geographically anchored, real-time risk visualization layer overlaid with active complaint locations and vehicle dispatch status, giving frontline officers the integrated situational awareness that the literature consistently identifies as the defining capability of effective emergency response.

Anomaly Detection in Vehicle and IoT Systems

Government vehicle misuse is rarely dramatic. It does not announce itself, it looks, from the outside, exactly like normal vehicle use. A driver takes a longer route than necessary, makes a stop that has nothing to do with the assigned mission, or uses

the vehicle on a weekend when no official trip was logged. Without a system capable of distinguishing normal from abnormal vehicle behavior at scale, these deviations go undetected until someone notices something suspicious, which in manual oversight environments often means never. Anomaly detection algorithms address this surveillance gap by learning what normal looks like from historical data and automatically flagging observations that deviate from that learned pattern, not based on a human officer's intuition, but based on the statistical structure of the data itself. Pang et al. (2022), in their comprehensive review of deep learning for anomaly detection published in ACM Computing Surveys, established the foundational landscape of this field by documenting that deep learning-based anomaly detection methods have achieved state-of-the-art performance across a wide range of application domains, including transportation, cybersecurity, and industrial monitoring. Their review identified unsupervised anomaly detection, approaches that learn normal behavior from unlabeled data without requiring pre-labeled examples of anomalies, as the most practically deployable category of methods for real-world government and operational contexts where anomalous events are rare, unpredictable, and therefore rarely documented in labeled datasets.

Agyemang (2024), in a simulation study published in Scientific African examining unsupervised machine learning algorithms for anomaly detection, built directly on Pang et al.'s (2022) foundational review by empirically comparing multiple unsupervised anomaly detection algorithms, including Isolation Forest, One-Class SVM, and Local Outlier Factor, across diverse simulated operational datasets. Their findings confirmed that Isolation Forest consistently achieves the strongest balance of detection accuracy and computational efficiency among unsupervised methods, performing well even on datasets with high dimensionality and irregular distributions

without requiring manual threshold configuration. This self-calibrating characteristic, which Agyemang (2024) identified as Isolation Forest's most operationally significant advantage, is particularly relevant for TUGON's deployment context: MAT personnel do not possess machine learning expertise, and a vehicle anomaly detection system that requires expert parameter tuning to function correctly is not practically deployable in a small provincial LGU. Where Pang et al. (2022) established the theoretical superiority of deep learning approaches in well-resourced research settings, Agyemang (2024) provided empirical validation that simpler unsupervised methods like Isolation Forest remain the most practical choice for operational environments where computational resources and technical expertise are limited.

Cavallaro et al. (2023), in their review of metaheuristics and machine learning techniques for anomaly detection in big data published in Frontiers in Big Data, extended this comparative analysis by examining how anomaly detection algorithms perform specifically on time-series and sequential data, the data type most relevant to vehicle GPS tracking, where location readings are recorded continuously over time and anomalies manifest as deviations in movement patterns rather than as isolated data points. Their review confirmed that Isolation Forest is among the most robustly applicable algorithms for sequential operational data, and they documented that anomaly detection systems achieve the highest operational value in contexts where the cost of undetected anomalies is significant, a condition directly satisfied by TUGON's deployment context, where undetected unauthorized vehicle use represents both a financial loss and an accountability failure for the municipal government. Xu et al. (2023), in their study on Deep Isolation Forest published in IEEE Transactions on Knowledge and Data Engineering, pushed the technical frontier further by demonstrating that embedding deep neural network representations into the Isolation

Forest framework substantially improves detection accuracy on complex, high-dimensional datasets with non-linear anomaly patterns. While Agyemang (2024) and Cavallaro et al. (2023) validated classical Isolation Forest for operational deployment in resource-constrained environments, Xu et al. (2023) established that deep extensions of the same algorithm are available when greater detection precision is required, providing TUGON with a clear technical upgrade path as its vehicle movement dataset grows and its anomaly patterns become more complex over time.

Jesi et al. (2023), in their IEEE conference paper on sensor technology and machine learning for road safety and anomaly detection, demonstrated that ML-based anomaly detection integrated with GPS and sensor data is effective at identifying abnormal movement patterns in real-world transportation scenarios, not in controlled simulations but in messy, variable real-world operational data where vehicle behavior is naturally irregular and noise is high. Their study established that ML models consistently outperform rule-based threshold systems in transportation anomaly detection because they adapt to the genuine variability of vehicle movement data, reducing false positive rates that would otherwise overwhelm administrators with spurious alerts. Zahoor et al. (2025), studying Isolation Forest and One-Class SVM for IoT security in Scientific Reports, added a security dimension to the anomaly detection argument that the transportation-focused studies did not fully address: in IoT-connected systems where devices stream continuous location and status data, anomaly detection serves not just an operational accountability function but a security one, protecting the integrity of the data pipeline from both unintentional errors and deliberate manipulation. Their finding that Isolation Forest performs robustly in IoT environments with limited labeled training data directly validates its selection for TUGON's vehicle monitoring

context, where the system must detect anomalies from day one without the benefit of a pre-existing dataset of confirmed unauthorized trips.

What these six studies collectively establish is a technically robust, multi-perspective argument for Isolation Forest-based anomaly detection in TUGON's vehicle monitoring module: unsupervised methods are the only practical choice when labeled anomaly data is unavailable (Pang et al., 2022), Isolation Forest is the most operationally efficient and self-calibrating among them (Agyemang, 2024), it performs strongly on the sequential GPS data type that vehicle tracking generates (Cavallaro et al., 2023), deep extensions are available for increased precision as the system matures (Xu et al., 2023), it outperforms rule-based systems on real-world transportation data (Jesi et al., 2023), and it is validated for IoT-connected deployment environments (Zahoor et al., 2025). The gap that persists across all six, however, is the application of anomaly detection specifically to municipal government vehicle accountability in a developing country context, where the institutional, technical, and operational conditions of deployment differ substantially from the research and enterprise settings in which these algorithms have been most extensively validated. TUGON addresses this gap by deploying Isolation Forest-based anomaly detection within the actual operational environment of the MAT of Liliw, Laguna, generating documented evidence of its performance and practical utility in precisely the constrained, real-world government context that the existing literature has not yet examined.

Backend Infrastructure and Cloud-Based Systems

Every time a citizen accesses a digital government service, submits a complaint form, receives a status update, or views a vehicle’s location on a map, an underlying layer of infrastructure makes that interaction possible. This backend infrastructure is the difference between a system that responds in milliseconds or minutes, data that is

stored securely and remains available during peak usage, and a small government institution with limited technical staff that can operate and maintain the platform without dedicated engineering support. For many years, building this infrastructure has involved managing physical servers, writing custom backend code, and managing complex deployment environments that were just not within the capacity of small government institutions to sustain. The arrival of cloud computing and more recently Backend-as-a-Service platforms has turned this equation on its head. In their paper on the future of cloud computing published in the International Journal of Communications, Network and System Sciences (2023), Islam et al. made the foundational technical case by establishing that cloud platforms provide measurable benefits over on-premises infrastructure in three dimensions of most relevance to government service applications, availability, where cloud providers have uptime guarantees that most LGU-managed servers would be hard-pressed to meet; scalability, where cloud resources automatically scale up to service spikes in usage during emergencies or high-complaint-volume events; and security, where cloud providers have enterprise-grade data protection measures that small institutions would find difficult to replicate independently. However, the most significant practical constraint to cloud adoption in developing country settings, a dependency on internet connectivity, was also identified by Islam et al. (2023), a limitation system designers must plan around in environments such as Liliw, Laguna, where network connectivity is available but not uniformly reliable.

In a study of cloud based IoT systems for disaster management in urban areas, published in Risk Assessment and Management Decisions, Bhadani (2024) extends the foundational argument of Islam et al. (2023) into a domain that is directly relevant to TUGON, showing that cloud infrastructure is not just appropriate for routine government service delivery, but is especially consequential in contexts of disaster and

emergency management. Their research found that, in the conditions that matter most during emergencies, high concurrent usage, unpredictable data volumes, and the need to distribute real-time data across multiple response teams simultaneously, cloud-based IoT systems for disaster management outperform their on-premises counterparts. Bhadani (2024) identified cloud infrastructure elasticity, the capacity to automatically scale resources to meet demand surges, as a critical operational advantage in disaster response situations where system load is inherently unpredictable and failures during peak demand carry the highest costs. This finding adds a public safety dimension to Islam et al.'s (2023) technical efficiency argument: cloud infrastructure for government service platforms is not just a cost and complexity advantage but a resilience and safety advantage in the disaster contexts that TUGON's typhoon heatmap and emergency complaint features are specifically designed to support.

Gangal et al. (2025) furthered the architecture argument in their study on cloud-based disaster management frameworks for real-time emergency response presented at the IEEE International Conference on Communication, Security, and Artificial Intelligence. They showed that cloud platforms tailored for real-time emergency response need to integrate low-latency data streaming, distributed processing, and resilient failover mechanisms to ensure operational continuity amid the high-stress conditions of an active disaster event. Their framework proposed that the optimal cloud-based emergency response systems are those that are capable of ingesting real-time data from multiple sources (complaint reports, GPS location feeds, sensor data, and communication logs) into a single processing pipeline that then disseminates actionable information to all appropriate responders simultaneously. This multi-source integration architecture is very similar to TUGON’s design where complaint submissions, vehicle GPS streams, citizen location pins and officer communications all need to be fed to the

same backend infrastructure in real time. While Islam et al. (2023) made the general case for cloud infrastructure in government applications, and Bhadani (2024) validated it specifically for disaster management, Gangal et al. (2025) described the specific architectural requirements that cloud-based emergency response systems must meet, requirements that TUGON's backend must satisfy to reliably support MAT operations during the typhoon and flooding events that represent its most demanding operational scenarios.

The infrastructure argumentation was completed by Larcher et al. (2024) in their study on Backend-as-a-Service-enabled workflows in federated serverless infrastructures, where they identified BaaS as the cloud architecture most suitable for applications without dedicated backend engineering teams and needing real-time multi-user coordination, published in IEEE Transactions on Cloud Computing. Their work showed that BaaS platforms, which combine database management, authentication, real-time data synchronization, and API generation into a single managed service, provide the latency and reliability necessary for operational monitoring systems without the engineering complexity of custom real-time architectures. This finding synthesizes the more general cloud arguments of Islam et al. (2023), Bhadani (2024), and Gangal et al. (2025) into a concrete and operationally relevant design recommendation. For small government institutions requiring real-time multi-role digital platforms but without backend engineering capabilities, BaaS is not merely a convenient shortcut but the technically validated architectural choice that renders such platforms practically feasible. Where Islam et al. (2023) made the general case for cloud infrastructure, Bhadani (2024) validated it for disaster management contexts, Gangal et al. (2025) specified its real-time emergency response architecture requirements, and Larcher et al.

(2024) identified BaaS as the precise cloud model that meets all of these requirements with the lowest possible engineering burden.

Together these four studies constitute a layered and coherent infrastructure argument for TUGON's backend design decisions. Cloud platforms provide the availability, scalability, and security needed for government service applications, but also bring a connectivity dependence that must be architecturally controlled (Islam et al., 2023). Cloud infrastructure is particularly important for disaster and emergency management applications, where the resilience of the system directly affects public safety outcomes (Bhadani, 2024). Real-time emergency response systems require multi-source data integration architectures, which cloud platforms are uniquely capable of providing (Gangal et al., 2025). Specifically, BaaS is the cloud architecture that best fulfills the real-time, multi-user, low-engineering-overhead needs of TUGON's operational context (Larcher et al., 2024). However, what the existing literature has not yet documented is a specific BaaS deployment in a Philippine provincial LGU government service system, the exact context of TUGON's use of Supabase, a PostgreSQL-based BaaS platform. Thus, the deployment of TUGON provides this missing case study, translating the infrastructure principles validated by these four studies into a documented real-world deployment for the MAT of Liliw, Laguna and generating evidence on BaaS performance under the constrained connectivity and operational conditions of a provincial Philippine municipality.

Data Privacy and Security in Government IS

When citizens share their location, describe their emergency, or submit a personal complaint through a government platform, they are doing something that requires a degree of trust that governments have not always earned and do not always protect. The data that flows through government information systems is among the most

sensitive that any digital platform handles, it contains personal identifiers, location histories, descriptions of distressing situations, and records of interactions with government authority that citizens have a reasonable expectation will be handled with care. Yet the history of e-government globally is littered with data breaches, privacy violations, and institutional negligence that have eroded exactly the citizen trust that digital government platforms depend on to function. Azzahra et al. (2026), in their systematic literature review on data security and privacy for e-government published in JESICA, documented that data security and privacy concerns are among the most consistently cited barriers to citizen adoption of government digital platforms across multiple developing country contexts. Their review found that citizens who distrust a government platform's ability to protect their personal data are significantly less likely to engage with it fully, withholding location information, using incomplete contact details, or avoiding the platform altogether, behaviors that directly undermine the operational effectiveness of systems like TUGON that depend on accurate, complete citizen-submitted data to function as intended.

Li, Yang, and Lu (2024), in their study on a privacy risk identification framework for open government data published in Government Information Quarterly, approached the privacy challenge from a technical governance perspective, proposing a structured framework for identifying and categorizing privacy risks in government data environments using a mixed-method approach combining qualitative stakeholder analysis with quantitative risk modeling. Their research found that government data systems generate privacy risks not just through deliberate misuse but through structural design decisions, data retention policies, access control architectures, and data sharing arrangements that may individually seem reasonable but collectively create exposure that neither system designers nor citizens anticipated. This finding creates an important

extension of Azzahra et al.'s (2026) citizen trust argument: privacy risk in government IS is not solely a function of malicious actors or negligent administrators, but of systemic design choices that accumulate into structural vulnerabilities. Where Azzahra et al. (2026) documented the consequences of privacy failures for citizen adoption, Li et al. (2024) provided the analytical framework for identifying the design-level decisions that produce those failures before they become visible.

Wu (2023), in a study presented at the International Conference on Public Service, Economic Management and Sustainable Development, examined the tension between government data openness, the principle that government-held data should be accessible for public benefit, and citizen privacy protection, finding that this tension is most acute in systems that collect location data and personal identifiers alongside service request information. Their research established that the most effective government IS privacy frameworks are those that enforce strict data minimization principles, collecting only the data genuinely necessary for service delivery, and implement role-based access controls that limit data visibility to personnel with a legitimate operational need. Gao, Sun, and Wang (2023), in their study on personal information protection in government data openness published in the Journal of Global Information Management, extended Wu's (2023) framework by demonstrating that decision tree-based models can automate privacy risk classification in government data environments, enabling systems to dynamically evaluate whether a given data access or sharing request meets established privacy thresholds without requiring manual review by a privacy officer for every transaction. Together, Wu (2023) and Gao et al. (2023) describe a complementary approach to government IS privacy protection: structural data minimization and access control (Wu, 2023) combined with automated privacy risk classification (Gao et al., 2023), a combination that reduces both the

volume of sensitive data at risk and the likelihood that access decisions create unintended exposure.

What these four studies collectively establish is that data privacy and security in government information systems is simultaneously a citizen trust issue (Azzahra et al., 2026), a systemic design challenge (Li et al., 2024), an access governance problem (Wu, 2023), and an automated classification opportunity (Gao et al., 2023), and that addressing it adequately requires design decisions that span all four dimensions. TUGON addresses each of these dimensions directly: it implements Role-Based Access Control to enforce strict data access boundaries between its four user roles, ensuring that citizens can only see their own complaint records while MAT officers have dashboard-level visibility appropriate to their operational function; it complies with the Data Privacy Act of 2012 (RA 10173) through JWT-based authentication that governs every data transaction within the system; and it collects only the personal data operationally necessary for complaint management and vehicle dispatch, location, contact information, and complaint description, without storing data beyond its operational utility. The gap that the existing literature does not yet address, however, is the specific privacy design challenges of a government IS that simultaneously handles citizen location data, vehicle GPS streams, and real-time communication records within a single platform operating under Philippine data privacy law. TUGON's documented compliance architecture contributes a Philippine LGU-specific case study to a literature that has addressed government IS privacy primarily in Chinese, Indonesian, and broadly developing-country contexts, extending the evidence base to the specific legal and operational environment of provincial Philippine municipal governance.

Agile Methodology in Government System Development

Government technology projects have a troubled reputation. Across the world, large-scale government IT initiatives have been delayed by years, exceeded their budgets by orders of magnitude, and delivered systems that the people they were designed to serve found confusing, incomplete, or simply unusable. The root cause, as researchers and practitioners have increasingly recognized, is not primarily technical, it is methodological. Traditional waterfall development approaches, which attempt to define all requirements at the outset and deliver a complete system at the end of a long development cycle, are poorly suited to government contexts where requirements evolve as stakeholders engage with prototypes, operational conditions change during development, and the institutional users of the system need to be trained and consulted throughout the process rather than simply handed a finished product. Agile methodology emerged as a direct response to these limitations, replacing the single-delivery model with iterative development cycles that allow requirements to be refined, features to be tested incrementally, and stakeholder feedback to be incorporated continuously throughout the build process. Brown et al. (2018), in their case study of agile and lean practices in the United States federal government published by Carnegie Mellon University, established foundational evidence that agile methodologies can be successfully adapted to government development contexts, environments that have historically been resistant to iterative approaches due to procurement regulations, fixed scope contracting norms, and institutional risk aversion. Their study documented that federal agency that adopted agile practices experienced faster delivery of working software, higher stakeholder satisfaction with delivered features, and fewer costly late-stage requirement changes compared to projects managed under traditional waterfall approaches.

Neumann, Kirklies, and Schott (2024), in their comparative case study of agile adoption in government published in Public Management Review, extended Brown et al.'s (2018) foundational findings into the contemporary government context by examining how different public sector organizations navigate the institutional tensions that agile adoption introduces. Their study found that agile works best in government when it is adopted not as a rigid framework but as a flexible philosophy, one that preserves the core principles of iterative delivery and continuous stakeholder engagement while accommodating the governance structures, accountability requirements, and documentation obligations that distinguish public sector development from private sector software projects. Critically, Neumann et al. (2024) also found that agile adoption in government is most successful when it has explicit institutional endorsement from leadership and when development teams maintain direct, ongoing access to the operational staff who will ultimately use the system, a finding that directly reflects the approach taken in TUGON's development, which included key informant interviews with MAT personnel, site visits to the MAT office, and iterative stakeholder feedback throughout the build cycle. Where Brown et al. (2018) established that agile is viable in government, Neumann et al. (2024) identified the specific organizational conditions under which it succeeds.

Tai and Awasthi (2025), in their systematic literature review of agile government published in Government Information Quarterly, provided the most comprehensive synthesis of agile adoption evidence across public sector contexts to date, analyzing the phenomenon at macro, meso, and micro levels of analysis. Their review found that at the macro level, national digital government strategies increasingly mandate or encourage agile approaches for government IT projects; at the meso level, government agencies that adopt agile report stronger alignment between delivered

systems and actual operational needs; and at the micro level, individual development teams working on government projects experience higher morale and more effective collaboration under agile frameworks than under traditional project management approaches. Importantly, Tai and Awasthi (2025) also identified a consistent finding that both Brown et al. (2018) and Neumann et al. (2024) had noted but not fully elaborated: agile in government requires more deliberate stakeholder management than agile in private sector contexts, because government users, particularly frontline staff in public safety and service delivery roles, have operational demands that make sustained participation in development sprints difficult to sustain without institutional accommodation. This finding directly informed TUGON's development approach, which structured stakeholder engagement around the operational calendar of the MAT rather than a standard sprint schedule, ensuring that MAT personnel could participate in system feedback and testing without disrupting their frontline responsibilities.

Silva et al. (2025), in their study on agile project management in government software development published in the proceedings of the International Conference on Enterprise Information Systems, examined agile adoption specifically in the context of public policy software, systems that must be responsive to evolving policy requirements while maintaining stability for the government personnel and citizens who depend on them. Their study found that the most successful government software projects combine agile's iterative delivery model with a structured requirements traceability framework that ensures each development sprint produces features that are directly traceable to documented operational requirements, preventing the feature drift that can occur when iterative development loses sight of its original policy and operational objectives. This finding creates a productive dialogue with Neumann et al.'s (2024) flexibility argument: while agile in government must be adaptable, it must also be disciplined enough to

maintain alignment with its foundational operational requirements across multiple development iterations. Together, these four studies describe an agile practice for government system development that is iterative but accountable, flexible but traceable, and stakeholder-driven but institutionally grounded, precisely the development philosophy that TUGON follows, using Agile methodology to iteratively build and refine each of its five core modules while maintaining direct traceability to the operational requirements documented through its preliminary investigation of the MAT of Liliw, Laguna.

What these four studies share, despite spanning from 2020 to 2025 and examining agile across federal, national, and municipal government contexts in the United States, Europe, and Latin America, is a consistent blind spot: they address agile adoption in government institutions that already possess established IT governance structures, dedicated development staff, and prior experience with digital projects. The specific challenge of applying agile methodology in a capstone development context for a first-time digital government system in a small Philippine provincial municipality, where the development team is composed of student researchers, the institutional partner has no prior digital infrastructure, and the development cycle is constrained by academic timelines, remains undocumented in the existing literature. TUGON's application of Agile methodology in exactly this constrained context contributes a practical case study that extends the agile government literature into a development environment that is simultaneously more resource-limited and more representative of the conditions facing the majority of LGUs in developing countries.

Technology Acceptance and User Adoption in Government Systems

Building a government system is only half the task; the other half is knowing whether the people it was built for will actually use it. A platform can be technically

sound and still fail in practice if citizens and officers do not perceive it as useful, easy to use, or worth adopting over the manual routines they already know. The Technology Acceptance Model remains the most widely applied framework for explaining and predicting the gap between system availability and system adoption, particularly in e-government contexts where adoption cannot be mandated the way it can in a private workplace.

Tremblay-Cantin, Mellouli, Cheikh-Ammar, and Khechine (2023), in their literature review and high-level model of influential factors in e-government service adoption published in Digital Government: Research and Practice, synthesized decades of citizen-adoption research into a consolidated model of the factors that most consistently predict whether citizens accept and continue using e-government platforms. Their review confirmed that perceived usefulness and perceived ease of use, the two foundational constructs of TAM, remain the strongest and most replicated predictors of e-government adoption across studies, but that these core constructs are consistently supplemented by contextual factors such as trust in government, prior digital experience, and perceived risk, particularly in developing-country settings where citizens have less established experience with digital public services. This meta-level finding establishes the baseline that the empirical studies discussed below either confirm or extend: TAM's core constructs travel well across contexts, but their explanatory power is strongest when the model is adapted to the specific institutional and cultural conditions of the government being studied.

Anityasari, Pamungkas, and Sonhaji (2024) tested this adaptation directly in their empirical study of e-government adoption in Indonesia, published in the International Journal of Electronic Governance, using an extended TAM that incorporated trust and perceived risk alongside the original constructs. Their findings

validated Tremblay-Cantin et al.'s (2023) contextual argument almost precisely: while perceived usefulness and perceived ease of use remained significant predictors of behavioral intention, the extended constructs, particularly citizen trust in the implementing government institution, explained additional variance in adoption intention that the basic TAM model alone could not account for. This is a materially important finding for a first-time digital government platform such as TUGON, where Citizens have no prior experience with a MAT digital system and where trust in the institution behind the platform cannot yet be assumed.

Pradana and Salamat (2025), in their more recent analysis of public acceptance of e-government services using the TAM model published in the Majority Science Journal, took a narrower approach than Anityasari et al. (2024) by applying the original, unextended TAM framework, and found that perceived usefulness and perceived ease of use alone still produced strong, statistically significant predictive power over citizens' attitude toward use and behavioral intention to use e-government platforms. Read alongside Anityasari et al. (2024), this creates a productive tension rather than a contradiction: extended constructs such as trust add explanatory depth, but they do not displace the core TAM constructs as the primary drivers of acceptance, they supplement them. Xin, Dilanchiev, Ali, Irfan, and Hong (2022) reinforced this layered view in their study of citizens' attitudes and intentions to adopt e-government services published in Sustainability, finding that the pathway from perceived usefulness and ease of use to behavioral intention is mediated by attitude toward use, meaning that citizens do not adopt a platform simply because it is useful or easy to use, but because that usefulness and ease translate into a positive overall disposition toward the system, a disposition shaped as much by institutional credibility as by interface design.

Avazov and Lee (2022), in their empirical validation of the Unified Model of Electronic Government Acceptance (UMEGA) in Uzbekistan published in the Journal of Policy Studies, extended the discussion furthest by combining TAM constructs with elements of the Unified Theory of Acceptance and Use of Technology (UTAUT), including social influence and facilitating conditions, and found that this unified model outperformed either framework applied independently in predicting actual e-government adoption behavior, not just intention to adopt. Their study is instructive for TUGON in a specific way: it demonstrates that behavioral intention, the endpoint most TAM studies measure, does not automatically translate into actual system use, and that facilitating conditions such as device access and network reliability, both live concerns in a provincial municipality like Liliw, meaningfully affect whether stated intention converts into real-world adoption.

Taken together, these five studies establish a coherent and increasingly refined case for TAM as the appropriate framework for evaluating TUGON's user acceptance: perceived usefulness and perceived ease of use are the consistently validated foundation of technology acceptance in e-government contexts (Tremblay-Cantin et al., 2023; Pradana & Salamat, 2025), trust and perceived risk meaningfully extend this foundation in developing-country and first-time-adoption settings (Anityasari et al., 2024), attitude toward use mediates the path from these perceptions to behavioral intention (Xin et al., 2022), and facilitating conditions ultimately determine whether that intention becomes actual, sustained use (Avazov & Lee, 2022). What none of these studies addresses, however, is the application of TAM to a multi-role, AI-powered, GPS-integrated government operations platform deployed in a small provincial Philippine municipality, where four distinct user groups, citizens, dispatchers, responders, and a system administrator, must each form their own acceptance judgments about the same system

under markedly different conditions of prior digital experience and institutional trust. TUGON's evaluation addresses this gap directly by applying TAM's core constructs, perceived usefulness, perceived ease of use, attitude toward use, and behavioral intention to use, across each of these four respondent groups, extending the technology acceptance literature into a government context, and a linguistic and institutional setting, that it has not yet reached.

2.1.3 Thematic Organization

The literature is organized using a thematic approach to provide a clearer understanding of the topic. Sources are grouped according to common technologies, processes, and governance issues, spanning twenty specific themes. These themes include digital transformation, mobile e-government, complaint management, AI/NLP for classification, GPS-based vehicle tracking, real-time monitoring, disaster preparedness, backend cloud infrastructure, and the Technology Acceptance Model (TAM).

2.1.4 Relevance to the Present Project

The reviewed literature directly informs the understanding, design, development, and evaluation of the proposed system. The analysis establishes a clear technological and contextual gap: while existing literature comprehensively covers national or large-city digital implementations, there is a distinct lack of integrated AI, GPS, and family safety platforms purpose-built for small, resource-constrained provincial municipalities. The literature review justifies the proposed features of TUGON, demonstrating that AI-powered complaint classification and real-time tracking are reasonable, evidence-based responses to the operational limitations currently experienced by the MAT in Liliw, Laguna.

2.1.5 Citation and Referencing

The APA 7th Edition format is used consistently for all in-text citations and references throughout the manuscript. Every cited source appears in the reference list, and all listed references are accurately cited in the text to uphold academic integrity and ensure the credibility of the research.

2.2 Related Systems and Technology Gap Analysis

This section evaluates existing software applications and platforms that share functional characteristics with the proposed TUGON system. By analyzing both foreign and local implementations, the project identifies current technological standards and determines specific operational gaps that the new system will address for the Mayor's Action Team (MAT) in Liliw, Laguna.

2.2.1 Identification of Related Systems

Ten related systems were selected based on relevance to complaint management, vehicle monitoring, civic reporting, and e-government. Each system was assessed for its core functions, technologies, and operational context to allow meaningful comparison with TUGON.

Quezon City mySMS (Service Management System)

The mySMS or Service Management System of the Quezon City Local Government is a web and mobile-based platform that allows residents to file service requests and complaints directly to the appropriate city government department. The system routes complaints to the relevant office based on the nature of the concern and allows officers to update the status of each request in real time. While mySMS represents a meaningful step in digitizing complaint management at the LGU level, its routing process remains entirely manual, relying on government operators to categorize

and assign complaints without any AI or NLP classification component. The absence of automated urgency sorting, GPS vehicle tracking, citizen location sharing, disaster heatmap, and direct communication features makes mySMS insufficient for the integrated operational management requirements of the MAT of Liliw, Laguna.

LTFRB Central Public Utility Vehicle Monitoring System (CPUVMS)

The Land Transportation Franchising and Regulatory Board (LTFRB, 2021) developed the Central Public Utility Vehicle Monitoring System as a GPS-based fleet monitoring platform for public utility vehicles operating under LTFRB franchises. The system logs trip data including routes, vehicle speeds, and idle times to support regulatory compliance monitoring among transport operators, demonstrating that GPS-based vehicle tracking is technically deployable within the Philippine government context. However, the CPUVMS is designed for large-scale national regulatory enforcement rather than municipal-level vehicle accountability. It does not incorporate machine learning for anomaly detection, lacks driver registration and trip accountability protocols suitable for small government fleets, and provides no integration with citizen complaint management or communication features, underscoring the need for a purpose-built solution like TUGON.

NYC / Pittsburgh 311 Civic Service Request Systems

The 311 service request systems deployed in major United States cities represent well-documented examples of large-scale digital complaint management in municipal government. These systems allow citizens to submit complaints and service requests through multiple channels and route them to the appropriate city department for resolution. However, these systems do not incorporate AI or NLP for automated complaint classification and urgency prioritization, do not support GPS-based vehicle

tracking or driver accountability, and are designed for large, well-resourced city governments. Their architecture is not transferable to the language constraints and resource conditions of a rural Philippine LGU such as the Municipality of Liliw, Laguna.

FixMyStreet

FixMyStreet, developed by the civic technology organization mySociety, is an open-source platform that enables citizens to report local issues by pinning their complaint location on an interactive map. The platform is recognized for its map-based complaint submission interface, which informs the design of TUGON's citizen location sharing and complaint pinning features. However, FixMyStreet lacks AI-powered complaint classification, urgency sorting, GPS-based vehicle tracking, driver accountability mechanisms, disaster heatmap capabilities, and a direct communication module between government responders and citizens. Its functionality is limited to complaint submission and routing, without the integrated operational management capabilities that the MAT of Liliw, Laguna requires.

eGov PH Super App (DICT Philippines)

The eGov PH Super App, developed under the Department of Information and Communications Technology (DICT) of the Philippines, provides citizens with access to a range of national government services through a single mobile application. As documented by Jou et al. (2024), the application has made measurable progress in improving digital government accessibility for Filipino citizens in urban areas. However, eGov PH does not provide AI-powered complaint management, GPS-based vehicle tracking, real-time driver accountability, disaster heatmap visualization, or MAT-specific communication modules. Its scope serves a broad national audience

rather than the specific operational requirements of a small municipal public safety team.

Quezon City Centralized Vehicle Management System

The Quezon City Government (2025) unveiled its centralized vehicle management system as an innovative GPS-based solution for tracking and managing the movements of city government vehicles. While this system represents a significant step forward for LGU-level vehicle accountability in the Philippines, its scope is limited to fleet tracking for a large, well-resourced city government. It does not integrate with a citizen complaint management system, does not include machine learning-based anomaly detection for unauthorized use, and is not designed for the scale of a small municipal fleet. TUGON builds upon this concept by extending GPS-based vehicle accountability to the specific scale and operational context of the Municipality of Liliw, Laguna.

SeeClickFix

SeeClickFix is a civic engagement platform widely used in the United States that allows citizens to report non-emergency local issues through a web and mobile interface, including geolocated complaint submission and status tracking. While SeeClickFix demonstrates the value of combining citizen-facing complaint submission with government-facing case management, it lacks AI-powered complaint classification and urgency sorting, GPS-based vehicle tracking, driver accountability features, disaster heatmap visualization, and Filipino language support. Its commercial focus on the US municipal market means it is neither designed for nor adaptable to the language and operational requirements of a Philippine provincial LGU.

Project NOAH Philippines (DOST-NDRRMC)

Project NOAH, developed by the Department of Science and Technology in collaboration with the NDRRMC, is a disaster risk management platform providing nationwide hazard mapping and flood risk visualization through an interactive web-based interface. The platform offers heatmap-style overlays showing flood-prone areas, storm surge zones, and landslide-susceptible regions, directly informing the design of TUGON's typhoon and heavy rain disaster heatmap feature. However, Project NOAH is a national-level disaster monitoring system rather than a municipal government operations platform. It does not include complaint management, vehicle tracking, driver accountability, administrative dashboard capabilities, or direct government-citizen communication modules.

DICT iGovPhil Portal

The iGovPhil portal, managed by the DICT, serves as the national e-governance infrastructure of the Philippines, providing access to national government services and supporting inter-agency data sharing. The portal represents the national-level ICT backbone for Philippine e-governance. However, iGovPhil is a national enterprise platform rather than a localized municipal operations system. It does not provide AI-powered complaint classification, GPS vehicle tracking, real-time dispatch management, disaster heatmaps, or direct citizen-government communication features specific to the operational context of a small municipal government team.

Google Waze

A widely used community-driven geographic and traffic application owned by Google. Waze utilizes crowdsourced data, allowing users to report real-time traffic incidents, road hazards, and localized emergencies. While the platform is highly

effective for real-time spatial mapping and providing immediate visibility on geographic events, it is inherently designed for commuter navigation rather than municipal administration. It lacks the institutional complaint-tracking lifecycle, automated AI categorization using natural language processing (such as RoBERTa), and the dedicated government vehicle dispatch management required by the Mayor's Action Team. Analyzing Waze provides valuable insights into intuitive, mobile-first GPS interfaces and real-time community reporting, which serve as foundational concepts adapted for the tracking and reporting modules of the TUGON platform.

2.2.2 System Feature Comparison

Table 2.1 presents a feature-by-feature comparison of the ten related systems against TUGON's ten major features. Features were selected based on TUGON's objectives: AI classification, GPS tracking, family safety, real-time chat, and heatmap visualization.

Table 2.1. System Feature Comparison of Related Systems and TUGON

SYSTEM	AI CLASSIFICATION	GPS TRACKING	FAMILY SAFETY	REAL-TIME CHAT	HEATMAP	RBAC
QC mySMS	No	No	No	No	No	Partial
LTFRB CPUVMS	No	Yes	No	No	No	Yes
NYC 311	No	No	No	Yes	Yes	Yes
FixMyStreet	No	No	No	Yes	No	Yes
eGov PH	No	No	No	Yes	No	Yes

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SYSTEM	AI CLASSIFICATION	GPS TRACKING	FAMILY SAFETY	REAL-TIME CHAT	HEATMAP	RBAC
QC Vehicle Mgmt	No	Yes	No	No	No	Yes
SeeClickFix	No	No	No	Yes	No	Yes
Project NOAH	No	No	No	No	Yes	No
iGovPhil	No	No	No	Yes	No	Yes
Google Waze	No	Yes	Yes	Yes	Yes	No
TUGON	Yes	Yes	Yes	Yes	Yes	Yes

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2.2.3 Analysis of Existing Systems

Following the structured comparison presented in Table 2.1, an analysis of the ten existing systems reveals several technological patterns in current civic engagement and municipal management software. The feature matrix highlights that certain baseline capabilities are widely recognized as standard across modern administrative platforms. Role-Based Access Control (RBAC) is implemented in seventy percent of the evaluated platforms, including LTFRB CPUVMS, NYC 311, FixMyStreet, eGov PH, QC Vehicle Management, SeeClickFix, and iGovPhil. RBAC serves as an established industry standard for maintaining data security and preserving administrative accountability across government agencies. Similarly, real-time communication and chat functionalities are present in sixty percent of the reviewed systems, namely NYC 311, FixMyStreet, eGov PH, SeeClickFix, iGovPhil, and Google Waze. Direct messaging

functionalities are commonly prioritized in these designs to facilitate bidirectional feedback between citizens and service administrators.

Despite the prevalence of standard administrative and communication tools, specific advanced capabilities remain either largely absent or fragmented across existing civic systems. Most notably, AI-powered classification is completely missing across all ten evaluated platforms, meaning these systems rely entirely on manual intake or citizen self-selection for complaint categorization. Family safety integration is absent in ninety percent of the public platforms reviewed, with only Google Waze incorporating consumer-oriented location sharing. Other capabilities are only partially implemented or restricted to specific functional domains. For instance, GPS vehicle tracking (found in LTFRB CPUVMS, QC Vehicle Management, and Waze) and spatial heatmaps (found in NYC 311, Project NOAH, and Waze) exist in specialized systems but are rarely integrated together within general municipal incident management software. Legacy implementations, such as QC mySMS, demonstrate only partial RBAC features due to scope restrictions and outdated system architectures.

The comparative analysis highlights several operational limitations inherent in current software solutions. Primary among these is functional siloing, where platforms are engineered for single, isolated operational roles. Fleet management applications track government vehicles but lack citizen complaint workflows, whereas civic reporting platforms process incident tickets without fleet monitoring or dispatch capabilities. Additionally, because current platforms lack automated classification mechanisms, administrative intake bottlenecks frequently occur. Incoming reports require manual evaluation, tagging, and routing by personnel, which leads to significant dispatch delays during high-volume incident periods. Furthermore, existing civic platforms often lack predictive and analytical visualizations; while disaster monitoring

applications utilize heatmaps for environmental risks, general civic management tools rarely integrate real-time spatial heatmaps to assist decisionmakers in identifying emerging incident clusters.

The evaluated systems vary significantly in their target user populations, governance frameworks, and operational environments. Large metropolitan platforms and national portals, such as NYC 311, eGov PH, and iGovPhil, are built for high population densities, national-scale transactions, and high-bandwidth urban infrastructure. Single-purpose regulatory tools like LTFRB CPUVMS cater exclusively to transit oversight officers, disconnected from municipal field response workflows. Commercial applications like Google Waze serve general commuters through crowdsourced traffic reports, lacking formal municipal ticketing or official government vehicle dispatch tracking. Consequently, these environments differ fundamentally from a provincial municipal context. For the Mayor's Action Team (MAT) in the Municipality of Liliw, Laguna, the operational environment requires a mobile-first, consolidated platform that operates effectively within provincial topography constraints while actively managing both field team dispatch and local civic reporting.

The operational gaps identified across existing systems present clear opportunities for structural improvement in municipal service management. Rather than relying on disparate, single-function applications, modern e-governance architectures benefit from an integrated, end-to-end framework. TUGON addresses these gaps by consolidating functionalities that are otherwise fragmented across single-purpose solutions. By utilizing natural language processing (RoBERTa) and LSTM networks, the system can automate initial complaint classification and support data trend analysis, reducing administrative triage bottlenecks. It couples real-time GPS vehicle tracking directly with incoming citizen reports to streamline dispatch operations. Operating

under a secure RBAC architecture configured exclusively by the System Administrator and opting to extend existing external system integrations rather than relying on disparate third-party modules, TUGON unifies civic reporting, automated intake, and field fleet dispatch into a cohesive ecosystem.

2.2.4 Technology Gap

Based on the evaluation of existing related systems, a distinct technology gap exists between currently available civic management platforms and the specific operational requirements of the Mayor's Action Team (MAT) in the Municipality of Liliw, Laguna. This gap is characterized by a lack of automation, functional siloing, insufficient decision support, and incompatibility with provincial operational environments.

The most prominent technological gap is the absence of automated triage mechanisms in current civic reporting platforms. Existing systems rely heavily on manual intake, meaning administrative personnel must read, categorize, and route every incoming citizen complaint. This creates a severe administrative bottleneck and inefficient processing, particularly during local emergencies. There is a documented need for a system that utilizes Natural Language Processing (specifically RoBERTa models) to automatically classify citizen reports upon submission, thereby eliminating manual triage delays and accelerating the response pipeline.

Current municipal technologies exhibit a significant lack of integration, typically splitting operations into siloed applications. Fleet management systems provide GPS tracking but lack civic complaint workflows, while civic engagement apps manage public reports but cannot dispatch or track government vehicles. The MAT requires a unified architecture where a classified complaint seamlessly transitions into

an active, GPS-monitored vehicle dispatch workflow. Furthermore, existing local systems often rely on fragmented, bolted-on third-party modules (such as separate SMS or Map APIs). The gap necessitates a platform that natively extends existing external system integrations within a cohesive environment, rather than forcing local government units to manage disjointed services.

Existing local platforms generally function as reactive repositories for complaints rather than proactive governance tools. While systems may generate basic statistical reports, they lack predictive monitoring capabilities. There is a missing functionality regarding the application of time-series forecasting and anomaly detection, specifically leveraging Long Short-Term Memory (LSTM) networks, to analyze complaint data trends. Without this intelligent decision support, local commanders lack the spatial and predictive insights needed to preemptively allocate dispatch vehicles to emerging incident hotspots.

Finally, current enterprise-grade solutions are predominantly designed for highly urbanized, high-bandwidth environments or lack rigid local governance structures. These systems are often too heavy or rigidly configured for a provincial municipality. A specific technological gap exists for a mobile-first (Flutter) and web-dashboard (Next.js) platform optimized for provincial connectivity constraints. Additionally, many legacy systems exhibit partial or flawed security hierarchies; the proposed environment requires strict, enterprise-grade Role-Based Access Control (RBAC) where configuration capabilities are strictly isolated to the System Administrator, preventing unauthorized access by field personnel or lower-level officers.

TUGON is specifically proposed to bridge this multi-faceted technology gap by synthesizing AI-driven automation, unified GPS dispatch, and predictive analytics into a single, localized platform built for the Municipality of Liliw.

2.2.5 Proposed Response to the Gap

TUGON responds by integrating Gemini 3.1 Flash zero-shot classification for Taglish, GPS/ETA with geofence Face ID, family circle live sharing, MAT-citizen chat, heatmap clustering, and Supabase/Postgres with RBAC/JWT, evaluated via Black-Box testing and TAM to provide an LGU-appropriate, accountable, and scalable solution.

Google Waze is a widely used GPS-based navigation and real-time traffic reporting application that enables users to share location data, report road incidents, and receive dynamic route recommendations. While Waze is not a government system, its real-time GPS tracking and user-generated location reporting provide a consumer-grade reference point for several of TUGON's technical features. However, Waze lacks complaint management features, government vehicle accountability modules, disaster heatmaps specific to Philippine hazard data, administrative dashboards for government use, and Filipino-language municipal government interfaces. As a consumer navigation tool, it cannot serve the operational management needs of the MAT.

The Technology Gap Analysis tells us that while each of the ten reviewed systems addresses one or more features relevant to municipal government operations, none provides the comprehensive, integrated solution that TUGON proposes. Systems with complaint management capabilities, such as QC mySMS and FixMyStreet, lack AI-powered classification, vehicle tracking, and disaster heatmap features. Systems with GPS vehicle monitoring capabilities, such as the LTFRB CPUVMS and the QC Vehicle Management System, do not integrate complaint management, citizen location

sharing, or communication modules. Disaster-focused platforms like Project NOAH provide valuable heatmap visualization without engaging with complaint or dispatch operations. No existing system reviewed combines all ten core features that TUGON proposes, and no existing system is specifically designed for the language, scale, and operational context of a small provincial Philippine LGU, confirming the existence of a clear technology gap that TUGON is designed to fill.

2.3 Synthesis

The synthesis of the reviewed literature and existing related systems provides a comprehensive foundation for the development of the TUGON platform. Rather than merely recounting preceding summaries, this section integrates the scholarly theories of digital governance with the practical technological realities observed in current software platforms. The convergence of these academic and technical materials validates the project's conceptual framework and justifies the specific technological interventions proposed to enhance the operational capabilities of the Mayor's Action Team (MAT) in the Municipality of Liliw, Laguna.

2.3.1 Integration of Findings

An evaluation of the related literature and existing systems reveals a clear trajectory in municipal e-governance, highlighting distinct successes, technological standards, and persistent operational gaps.

A consistent finding across both academic studies and existing platform evaluations is the necessity of modular, mobile-first architectures for effective citizen engagement. Furthermore, features such as real-time bidirectional communication and Role-Based Access Control (RBAC) are established as non-negotiable baselines for government software. To ensure absolute data integrity, administrative accountability,

and enterprise-grade security within these municipal environments, it is evident that access hierarchies must be strictly enforced, for example, by restricting RBAC policy configuration exclusively to the System Administrator, ensuring lower-tier users such as Barangay Officers cannot alter security protocols.

The review identified a critical divergence between the environments for which current systems are built and the operational realities of provincial local government units (LGUs). While national portals and metropolitan applications assume the presence of high-bandwidth infrastructure and expansive administrative teams, provincial operations require streamlined platforms optimized for connectivity constraints and limited personnel. Additionally, a structural difference exists between citizen reporting platforms and government fleet management systems; they operate as entirely separate domains, highlighting a disjointed approach to municipal incident response.

To bridge the identified gaps, several specific technologies emerge from the synthesis as highly applicable for a unified LGU platform:

- Application Frameworks: Flutter and Next.js provide the optimal environments for deploying responsive mobile applications and web-based administrative dashboards, respectively.
- AI and Automation: Natural Language Processing, specifically utilizing RoBERTa models, offers a proven, highly accurate approach to classifying unstructured citizen complaints and removing manual processing delays.
- Predictive Analytics: Integrating machine learning models such as LSTM, Prophet, and XGBoost is highly useful for anomaly detection and time-series

forecasting, enabling proactive LGU decision-making based on complaint volume trends.

Despite broader advancements in e-governance, the critical issue of administrative bottlenecks during incident intake has not been adequately addressed. Existing platforms continue to rely on manual triage, meaning human operators must read and categorize every incoming report before a MAT vehicle can be dispatched.

Furthermore, many local systems suffer from disjointed architectures characterized by bolted-on third-party modules. A significant remaining limitation is the over-reliance on separate, fragmented Map APIs and SMS gateway integrations, which create unnecessary maintenance overhead and system bloat. A more sustainable, unified approach, which remains largely unaddressed in current municipal systems, is structurally extending existing external system integrations directly within the primary architecture to ensure seamless geographic and communication workflows.

2.3.2 Identified Gap

The review of related literature and existing systems establishes several intersecting gaps that the TUGON platform is designed to resolve. Specifically, the synthesis supports the presence of a technological gap, a process gap, and a contextual gap within the domain of municipal e-governance.

Technological Gap

Existing municipal software does not sufficiently address the requirement for intelligent automation and streamlined infrastructure. Current platforms completely lack automated triage capabilities, relying on manual data entry rather than leveraging Natural Language Processing, such as RoBERTa models, to categorize text-based reports. Furthermore, they lack predictive monitoring frameworks that utilize models

like LSTM and XGBoost to analyze incident volume trends and detect spatial anomalies over time. Architecturally, existing local solutions often depend on disjointed, separate Map API and SMS integrations, which creates system bloat. There is a distinct technological need for a cohesive platform that natively extends existing external system integrations rather than bolting on disparate third-party services.

Process Gap

The existing workflow for civic incident management remains highly inefficient and functionally fragmented. Current e-governance implementations exhibit a profound disconnect between citizen complaint intake and field team dispatch. Because platforms operate in silos, separating civic reporting apps from government fleet tracking systems, the incident resolution lifecycle is broken. This fragmentation, combined with the administrative bottleneck of manual complaint triage, creates significant delays in deploying response vehicles during critical local events.

Contextual Gap

An existing solution has not been adequately applied to the specific target users and operational environment of a provincial municipality. Major civic platforms (like eGov PH or NYC 311) are structurally tailored for large metropolitan centers with high-bandwidth infrastructure, while commercial apps (like Waze) lack the formal governance structures required by local government units. There is a specific contextual gap for a platform tailored to the Mayor's Action Team (MAT) in the Municipality of Liliw, Laguna. This requires not only a mobile-first design suited for provincial connectivity but also an enterprise-grade security framework tailored to local governance, specifically, a strict Role-Based Access Control (RBAC) architecture where configuration privileges are assigned exclusively to the System Administrator,

preventing unauthorized access or policy modification by field operators or Barangay Officers.

By isolating these three specific gaps, the project definitively justifies the development of TUGON as a targeted, evidence-based solution.

2.3.3 Contribution of the Proposed Project

TUGON addresses the gap by providing a single platform that combines Gemini 3.1 Flash NLP for Taglish urgency classification, real-time GPS and geofence-verified attendance, family circle safety, and MAT-citizen chat with PostHog analytics and RBAC/JWT compliance. The project contributes a replicable model for provincial LGUs and empirical data on TAM-based acceptance in a low-resource municipal setting.

The twenty thematic areas reviewed in this chapter together constitute the theoretical, empirical, and technological foundation upon which TUGON: Technology-Unified Government Operations Network is developed and evaluated. And when the literature is not read as a collection of fragmented studies but as a conversation among researchers addressing different aspects of the same basic problem, the literature reveals a coherent and converging argument: local governments in developing countries require integrated, AI-enabled, mobile-first digital platforms to modernize public service delivery, and the particular combination of capabilities that TUGON provides has not been documented in a small provincial Philippine municipal context.

The starting point is the literature on digital transformation and e-governance. Adiazmil and Perdana (2024), Gasco Hernandez (2024), Oghuvbu et al. (2022), and Ofori (2024) all point out that while the benefits of digital government are well established, LGUs in developing countries are still the most underserved tier of

governance because the technical, financial and organizational barriers to adoption are highest at the municipal level, and because structural and cultural resistance to digital change is as consequential as the lack of technology itself. The technology acceptance literature adds a human component to this structural problem. This suggests that for being able to adopt the government systems successfully, attention should be given simultaneously to perceived usefulness, organizational readiness, institutional leadership, and citizen trust (Alfiani et al., 2024; Kurniasih et al., 2024; Jou et al., 2024; Lafioune et al., 2023). Investment in infrastructure and technical excellence are not sufficient. TUGON's development methodology, based on field investigation, key informant interviews with MAT personnel and iterative engagement with stakeholders, directly reflects these evidence-based prerequisites for meaningful adoption. This understanding is further elaborated in the literature on mobile government (Kurniasih et al., 2024; Desmal et al., 2023). Collectively, these studies suggest the success of mobile government platforms depends on simplicity, local relevance, low bandwidth friendliness, and communication designs that foster government-citizen trust over time (Castilla et al., 2023; Jou et al., 2024; Kurniasih et al., 2024; Desmal et al., 2023), principles that are embedded throughout the design of TUGON's citizen-facing mobile application.

The literature on complaint management and AI classification establishes a clear developmental trajectory which TUGON is designed to further. Shofiana et al. (2025) showed that even simple digital complaint platforms are better for service delivery than manual ones but identified automated urgency prioritization as the key gap that still needs to be filled. This gap is sequentially addressed by Javith Kausar et al. (2026), Ullah et al. (2023), Koh et al. (2025), Dolas et al. (2026) and Pallapu and Varshasri (2026), from structured intake without AI, to classification using LLMs, to

automated prioritization engines, to NLP-driven urgency detection in e-governance contexts. As Alyahya et al. (2026) and Laylo and Sonza (2026) show, such systems not only improve internal efficiency but also transform the way citizens perceive and trust government responsiveness. This trajectory is also technically reinforced in the AI and NLP classification literature: Jiang et al. (2023), Trung et al. (2022), Saha et al. (2024), and Thayumanavan (2025) show the operational indispensability of NLP-powered complaint classification in high-volume government environments, and the superiority of pre-trained transformer models overrule-based and ensemble approaches if they are adapted to the domain of language. The Filipino language processing literature closes the loop on why language adaptation is non-negotiable for TUGON: Montalan et al. (2025), Coronia (2022), Sarzaeim et al. (2023) show collectively that general-purpose multilingual NLP models systematically underperform on Filipino and Taglish inputs, and that the performance gap has direct operational consequences in public safety complaint classification. The direct technical response to this documented gap is TUGON's dual-model NLP architecture, Gemini 3.1 flash for Filipino language understanding and for zero-shot urgency detection.

A finding parallel to the complaint management research is corroborated by the literature on government communication and citizen engagement. Shin et al. (2024), Bangero (2025), Benlahcene et al. (2024), Alenzi and Miskon (2024), and Huanca (2025) agree that the chasm between citizens' expectations of government communications and what governments usually offer through digital channels is large, lasting, and rooted in communication philosophy, not technology availability. When governments use digital communication tools without rethinking their approach to communication, they reproduce the same broadcast-only dynamic they had before, and citizens disengage accordingly. The TUGON Communication Module aims to disrupt

this cycle by establishing a structured, in-platform messaging platform where every citizen engagement is met with a formal, traceable government response.

The literature on vehicle management and GPS also strongly supports TUGON's fleet accountability design. Vehicle accountability is both a governance imperative and a citizen trust determinant, as established by Chiparo et al. (2022) and Chikazhe et al. (2023). GPS fleet monitoring is technically feasible in resource-constrained developing country environments, validated by Bolaños et al. (2022) and Punith et al. (2022). Raikar et al. (2023), Jeevan et al. (2024), Gade et al. (2024), and Kumar et al. (2023) describe the progression from boundary-based geofencing to identity-linked authentication to citizen-facing location transparency to alert-integrated real-time monitoring, the full capability profile that TUGON's GPS tracking architecture provides. The anomaly detection literature complements this by making the technical case for vehicle behavior monitoring based on Isolation Forests: The need for unsupervised detection on unlabeled anomaly data is verified by Pang et al. (2022), whereas Agyemang (2024) confirms Isolation Forest as the most self-calibrating and operationally efficient among unsupervised methods, Cavallaro et al. (2023) validate its applicability on sequential GPS data specifically, Xu et al. (2023) demonstrate its deep learning extension for higher precision, and Jesi et al. (2023) and Zahoor et al. (2025) validate its performance in real-world transportation and IoT deployments, respectively. Collectively, these studies provide a multi-perspective technical underpinning not previously utilized in the context of Philippine municipal government vehicle accountability.

The heatmap feature of TUGON is justified from multiple angles at the same time by the literature on disaster preparedness and emergency monitoring (Handoyo et al., 2024; Aggarwal et al., 2024; Al-Omari et al., 2024; Diriba et al., 2024; Khan et al.,

2025; Amatebelle et al., 2025). The consensus is that local disaster preparedness is most effective when geospatial tools are spatially specific at the barangay level, dynamically updated to reflect changing weather conditions, and integrated with operational response coordination rather than used as isolated reference tools. Ghaffarian et al. (2023), Mohsin et al. (2025), Duraimurugan et al. (2024), Hossain (2026), and Gish and Rapaport (2026) further support that the heatmap visualization must be coupled with explainable AI-generated risk intelligence, automated incident prioritization, and integrated situational awareness platforms for emergency monitoring to be operationally effective. The literature on public service efficiency and AI decision support (Shen et al., 2023; Kulal et al., 2024; Zhao, 2024; Odigie et al., 2024; Chen et al., 2025; Liu and Li, 2025; David et al., 2024; OECD, 2025) fills in the operational picture, revealing that AI decision support is most transformative in the high-pressure, high-volume, multi-incident settings that the MAT of Liliw, Laguna experiences during typhoon seasons and emergency events, and that smaller municipalities are significantly underrepresented in the documented evidence base on AI government deployment.

TUGON's technical architecture and development approach is validated by peer-reviewed literature on infrastructure, privacy, agile development, and user acceptance evaluation. Together, Islam et al. (2023), Bhadani (2024), Gangal et al. (2025), and Larcher et al. (2024) demonstrate that BaaS cloud infrastructure is the correct and technically validated architectural choice for real-time, multi-role government service platforms in resource-constrained environments – and that, in the specific context of disaster management, cloud-based infrastructure is not just a convenience, but a public safety benefit. According to Azzahra et al. (2026), Li et al. (2024), Wu (2023), and Gao et al. (2023), data privacy in government IS must be

addressed at the design level through trust, structural access control, data minimization, and automated risk classification, all of which are handled by TUGON by its RBAC, JWT authentication, and RA 10173-compliant architecture. Neumann et al. (2024), Tai and Awasthi (2025) and Silva et al. (2025) together confirm Agile methodology as the suitable and evidence-based development approach for government systems, particularly those that necessitate iterative stakeholder engagement with operational frontline staff who cannot be expected to freeze their daily responsibilities for the sake of a development schedule. Tremblay-Cantin et al. (2023), Anityasari et al. (2024), Pradana and Salamat (2025), Xin et al. (2022), and Avazov and Lee (2022) show that the Technology Acceptance Model is the suitable, empirically validated, and methodologically well-established framework for assessing citizen and institutional acceptance of a multi-role, AI-based mobile and web-based government service system, and that perceived usefulness and perceived ease of use, extended by trust, attitude, and facilitating conditions, must be evaluated across distinct user roles to capture acceptance patterns that a single uniform construct would obscure.

The Technology Gap Analysis in this chapter further reinforces the implication in the thematic literature that there is no single unified platform that provides the full combination of AI-powered complaint classification, urgency sorting, GPS vehicle accountability, machine learning anomaly detection, citizen location sharing, disaster heatmap visualization, real-time administrative monitoring, structured government-citizen communication, Filipino language support, and small-scale LGU operational focus, whether Philippine local systems like QC mySMS and LTFRB CPUVMS, international civic platforms like FixMyStreet and SeeClickFix, national Philippine tools like eGov PH and iGovPhil, disaster monitoring systems like Project NOAH, or consumer GPS applications like Google Waze. Each of the systems reviewed addresses

a subset of these needs, in isolation. None of them covers all of them together, and none is built for the linguistic, institutional and operational conditions of a small provincial Philippine municipality.

The twenty thematic areas examined, and the ten related systems analyzed and the technology gap they collectively highlight, present a coherent and compelling rationale for TUGON. Each core design decision is backed by peer-reviewed evidence (2022–2026) on feasibility and operational effectiveness in analogous public service settings: transformer-based Filipino NLP for complaint classification; GPS vehicle tracking with driver authentication; Isolation Forest anomaly detection; mobile-first citizen interface; BaaS real-time backend infrastructure; and the Technology Acceptance Model for user acceptance evaluation. Apart from assisting in setting the conceptual and technological background for the research, the literature reviewed also affected the descriptive-developmental research design, Agile-based development methodology, Black-Box testing procedures, the choice of respondents from several groups of stakeholders, and the adoption of the Technology Acceptance Model as the main instrument to evaluate user acceptance and adoption of the system. TUGON fills a well-identified gap at the intersection of AI, municipal e-governance and public safety management in provincial Philippine local government by integrating these well-established capabilities into a single platform designed for the MAT of the Municipality of Liliw and presents a rigorously-supported case study that can be leveraged by other small municipalities across Laguna and the Philippines as a replicable model for technology-enabled local governance enhancement.

CHAPTER III

RESEARCH METHODOLOGY

This chapter presents the research methodology utilized in the development and evaluation of "TUGON: Technology-Unified Government Operations Network - A Mobile-Based AI-Powered Complaint & Incident Management with Vehicle Dispatch System for the Mayor's Action Team of the Municipality of Liliw, Laguna." It provides a comprehensive explanation of the materials, methods, research design, population, system testers, respondents, sampling design, data collection instruments, project design, software development model, system architecture, testing and evaluation procedures, business plan, and statistical treatments applied in this study.

3.1 Materials and Methods

This subsection details the systematic processes, activities, and resources utilized to investigate the operational requirements of the Mayor's Action Team (MAT) and to construct the TUGON system platform.

3.1.1 Research and Fact-Finding Activities

The researchers conducted extensive fact-finding activities to gather accurate operational data from the MAT of Liliw, Laguna. These activities included direct observational site visits at the MAT command center, key informant interviews with MAT Head Hon. Krich Carlo V. Monfero and duty dispatchers, and document analysis of existing manual incident and complaint logs and dispatch records. A technology gap analysis was also executed to benchmark existing municipal platforms against the specific needs of Liliw, Laguna.

3.1.2 System Development Activities

System development activities encompassed requirement analysis, UI/UX prototyping, database schema design, frontend mobile and web development, backend API construction, AI model integration, and system deployment. Development was executed iteratively using the Agile framework, allowing continuous refinement based on user feedback during early testing phases.

3.1.3 Resources and Technologies

The technological stack utilized in building TUGON includes Flutter and Dart for the mobile application, Next.js and TypeScript for the administrative web dashboard, PostgreSQL and Supabase for cloud database management and real-time synchronization, Redis as a caching layer to improve system responsiveness by storing frequently accessed data and as a rate-limiting tool to prevent API abuse and ensure system stability, Google Gemini 3.1 Flash for NLP-based incident and complaint classification, Google Maps API for real-time geolocation services, and JWT authentication with Role-Based Access Control (RBAC) for data privacy and security compliance.

3.2 Research Design

This study adopts a descriptive-developmental research design. Developmental research is a systematic study of designing, developing, and evaluating instructional programs, processes, and products (Nabayra, 2023). In the context of this study, the developmental approach guided the iterative engineering of the TUGON platform as an AI-powered civic technology artifact. To this end, the researchers followed an Agile methodology, enabling iterative development, continuous stakeholder feedback, and incremental system improvements throughout the project lifecycle. The developmental aspect covers the engineering, prototyping, and technical construction of the TUGON

platform. The descriptive aspect documents current frontline operational workflows, evaluates technical feature performance through Black-Box testing, and measures user acceptance using the Technology Acceptance Model (TAM).

The descriptive part of the study is used to find current trends and to document existing manual processes by using methods such as interviews and observation. According to Villamin et al. (2025), qualitative descriptive design aims to develop a comprehensive understanding of phenomena by being "data-generating" rather than merely "data-driven," employing methods such as in-depth interviews, focus group discussions, and participant observation. In this study, the descriptive approach was applied to capture the current manual incident and complaint handling and dispatch workflows of the Liliw MAT before the implementation of the TUGON system. The study integrates both approaches to ensure the creation of an innovative municipal solution that is empirically validated in terms of usability and government accountability.

3.2.1 Rationale for the Selected Research Design

A descriptive-developmental design is most appropriate because constructing an AI-powered municipal public safety system requires both an engineering process to create the artifact and a descriptive evaluation framework to empirically validate its operational viability, user acceptance, and practical impact on local governance in Liliw, Laguna.

3.3 Population of the Study

The population of the study consists of both internal operational stakeholders of the MAT and external citizen end-users in the Municipality of Liliw, Laguna. Internal stakeholders comprise 1 Administrator, 3 MAT Dispatchers/Officers, and 18 MAT

Responders/Drivers/Drivers. External stakeholders consist of the 39,976 total population of residents across the 33 barangays of Liliw, Laguna.

3.4 System Testers

System testers evaluated the functional completeness, software quality, and technical performance of the TUGON platform prior to deployment and user evaluation. Testing involved IT experts and MAT operational staff.

3.4.1 Profile of System Testers

Table 2 and Table 3 outline the demographic and professional profile of system testers who participated in functional and Black-Box testing of the TUGON system.

Table 3.1. IT Experts (Technical Evaluators)

Tester	Educational Background / Credentials	Current Position / Affiliation	Years of Relevant Experience
Jonas T. Santos	LSPU-SCC BSIT SMP Graduate	MISO IT Staff	3 years
Janssen R. Espino	LSPU-SCC BSIT Graduate	MISO IT Staff	1 year
John Marvin A. Tabo	LSPU-SCC BSIT Graduate	MISO IT Staff	3 years
Kevin D. Adlaon	Data Scientist Malaysia	MISO IT Staff	2 years
Wilmar T. Morota	Freelance Encoder	MISO IT Staff	4 years
Justine Jhay Lopez	LSPU SCC BSIT Graduate	MISO Programmer	1 year
Robin Paolo L. Bona	Union College BSIT Graduate	MISO IT Staff	6 years
Mharjun V. Tobias	LSPU SCC Graduate	MISO IT Staff	1 year
Clydemarion P. Joya	LSPU SCC BSIT SMP Graduate	Video Editor	1 year

Tester	Educational Background / Credentials	Current Position / Affiliation	Years of Relevant Experience
Christian Jay B. Tantan	LU BSIT Graduate	PTA (Quality Assurance)	2 years

Table 3.2. MDRRMO Experts (Domain Evaluators)

Tester	Current Position / Affiliation	Years of Relevant Experience
Marvin O. Mamaril	MDRRMO Staff	5 years
Donato Cula	MDRRMO Staff	9 years
Kervin Cordovet	MDRRMO Staff	9 years
Krich Carlo V. Monfero	MGDH 1 - LDRRMO	6 years
Marlou Collantes	MDRRMO Staff	5 years

3.5 Respondents for User Evaluation

User evaluation involved primary end-users across all system interaction roles to assess TAM constructs following system demonstration and hands-on testing.

3.5.1 Distribution of Respondents

The distribution of the 203 target respondents who evaluated the system is presented in Table 3.3.

Table 3.3. Distribution of Respondents for User Evaluation

Respondent Group	Target Population	Sample Size	Sampling Design
System Administrator	1	1	Total Enumeration
MAT Dispatchers / Officers	3	3	Total Enumeration

Respondent Group	Target Population	Sample Size	Sampling Design
MAT Responders/Drivers	18	18	Total Enumeration
Citizens	39,976	181	Stratified Random Sampling
Total	**40,000**	**203**	-

3.6 Sampling Design

A mixed sampling strategy combining total enumeration and purposive-stratified sampling was used. Total enumeration was applied to internal MAT personnel (1 Administrator, 3 Dispatchers, 18 Responders) to guarantee comprehensive operational representation. For the external citizen population, purposive-stratified sampling was employed to ensure proportional representation across barangays and demographic groups (Ahmad & Wilkins, 2025; Hossan et al., 2023). In their framework of purposive sampling published in Quality and Quantity: International Journal of Methodology, Ahmad and Wilkins (2025) reiterated that the quality of respondent selection is based not only on the number of participants but on the strategic alignment between selection criteria and the conceptual focus of the study. Hossan et al. (2023) further supported that a purposive approach minimizes sampling error when studying specialized user groups in public service contexts.

Slovin's Formula was used with 7% margin of error to the entire study population of 40,000, to arrive at the appropriate total sample size of 203 respondents.

$$n = \frac{N}{1 + Ne^2}$$

Equation 1. Slovin's Formula

Commented [JG9]: Centered po ba yung label ng equations?

Where:

n = Sample size

N = Total population (40,000)

e = Margin of error (0.07)

1 = Constant

Applying the formula:

n = 40,000 / (1 + 40,000 x 0.07^2)

n = 40,000 / (1 + 40,000 x 0.0049)

n = 40,000 / (1 + 196)

n = 40,000 / 197

n = 203 respondents

22 out of 203 respondents were from total enumeration including 1 System Administrator, 3 Dispatchers and 18 Responders. The other 181 respondents were drawn from the citizen population using stratified random sampling.

3.7 Data Collection Instrument

Data collection relied on a structured evaluation questionnaire based on Davis' Technology Acceptance Model (TAM), alongside interview protocols and observation checklists.

3.7.1 Instrument Development

The TAM questionnaire was developed to evaluate six core constructs: Quality Factors, Perceived Ease of Use, Perceived Usefulness, Attitude Toward Using,

Behavioral Intention to Use, and Experience/Actual Use. Items were scored using a 5-point Likert scale ranging from Strongly Disagree (1) to Strongly Agree (5), with equal intervals of 0.80 between scale ranges. The model posits that user acceptance of a technology is primarily determined by two key factors: Perceived Usefulness and Perceived Ease of Use (Davis, 1989).

3.7.2 Instrument Validation

The research instrument underwent content and face validation by the thesis adviser to ensure item clarity, construct alignment, and contextual appropriateness for the study. The adviser reviewed the questionnaire based on the following criteria: (1) Clarity of items, (2) Relevance to the study objectives, (3) Objectivity, and (4) Ease of comprehension for the target respondents.

3.7.3 Administration of the Instrument

The evaluation instruments were administered through tailored procedures for each target respondent group. For MAT staff, the researchers conducted simulation and demonstration exercises where respondents interacted with the system under guided scenarios before completing the questionnaire. For citizens, the questionnaire was distributed after a system demonstration session and hands-on testing. All respondents were briefed on the study purpose, informed of their right to withdraw, and assured of data confidentiality before proceeding with the evaluation. Completed questionnaires were collected immediately after the evaluation session.

3.8 Project Design

Project design outlines the technical specifications, structural layout, and architectural models governing the TUGON platform.

3.8.1 System Requirements

Functional requirements specify AI-powered complaint and incident classification, GPS vehicle tracking, interactive report pinning, incident heatmap display, and real-time messaging. Non-functional requirements enforce data privacy under RA 10173, low-bandwidth optimization, SSL encryption, and sub-2-second response latency.

3.8.2 System Design

System design models platform workflows through Use Case Diagrams, Data Flow Diagrams (DFD), Entity-Relationship Diagrams (ERD), and UI wireframes representing mobile citizen interfaces and web administrative screens.

3.8.3 System Architecture

The TUGON system utilizes a modern, multi-tier architecture designed for scalability, security, and efficient data processing. It connects citizen facing mobile applications and administrative web dashboards through a centralized API Gateway, leveraging specialized AI services for data enrichment and a robust backend-as-a-service for data persistence.

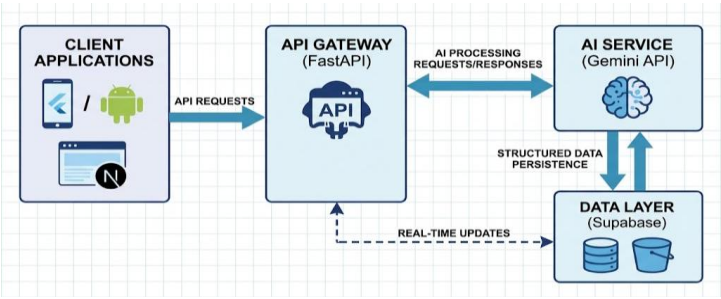


Figure 3.1. System Architecture

Image generated using Gemini by Google (September 2026) with the prompt: "High-level System Architecture diagram for TUGON showing Presentation, Application, Data, and Integration layers in a clean, flat technical style."

System Components & Interactions:

- **Client Layer:** Comprises the Flutter Mobile App for citizen report submissions and the Next.js Web Dashboard for LGU administrative tracking and management.
- **API Gateway (FastAPI):** Serves as the central entry point handling request routing, authentication, and security protocols.
- **AI Service (Gemini API):** Processes natural language data from incoming reports to automatically categorize issues and assign urgency scores.
- **Data Layer (Supabase):** Manages relational PostgreSQL data, secure object storage for media attachments, and real-time database subscriptions for instant dashboard synchronization.

3.9 Software Development Model

The Agile software development model was adopted, structuring development into six iterative phases: (1) Planning, (2) Designing, (3) Development, (4) Testing, (5) User Evaluation, and (6) Refinement. Because TUGON involved complex, interdependent modules—such as AI natural language parsing and live geospatial tracking, an iterative framework was necessary to validate individual components before full system integration. Sprint iterations enabled continuous stakeholder feedback and adaptive feature enhancements as dispatchers, responders, and developers reviewed working prototypes (Anifa et al., 2024; Nakash, 2024). This approach allowed the research team to incrementally refine role-based permissions and interface layouts based on actual municipal operating requirements, reducing the risk of critical bottlenecks during the final evaluation phase.



Figure 3.2. Agile Development Methodology (adapted from Duraisamy, 2023)

The development of TUGON followed the Agile methodology, an iterative approach to software development that organizes work into short, time-boxed cycles called sprints, typically lasting one to four weeks, where each sprint produces a working and testable increment of the system (Nakash, 2024). Unlike traditional Waterfall development that follows a linear sequential process, Agile allows for continuous feedback, incremental development, and adaptive planning throughout the project lifecycle.

Development Phases:

Phase 1 - Planning

The planning phase involved conducting interviews with MAT Dispatchers, identifying system requirements, and defining project scope and deliverables. The researchers worked closely with the MAT Head and senior Dispatchers to understand

the current complaint and incident management operations and identify areas where technology could improve efficiency. Based on these interviews, the researchers developed a comprehensive list of functional and non-functional requirements for the TUGON system.

Phase 2 - Designing

The designing phase involved creating system architecture, wireframes, and mockups for mobile and web interfaces. The researchers worked with MAT to design user interfaces that would be intuitive and efficient for their specific workflows. Multiple design iterations were created and refined based on stakeholder feedback. The design phase also included database schema design, specifying how data would be structured and stored in the system.

Phase 3 - Development

The development phase involved building key features including AI-Powered Incident and Complaint Classification, AI KYC Verification, Automated Dispatch Queue, Real-Time GPS Tracking, Face ID Attendance, Patrol Management, Evacuation Center Check-In/Out, Family Circle Safety, MAT-Citizen Communication, and Incident Heatmap. Development was organized into two-week sprints, with each sprint focused on implementing a specific set of features from the product backlog. Throughout the development process, the team followed coding best practices including code reviews, unit testing, version control, and continuous integration.

Phase 4 - Testing

The testing phase involved conducting functional, performance, and security testing using black-box methodology. Simulation testing was also conducted with MAT personnel to evaluate real-world reports handling and dispatch workflows under

controlled scenarios. At the end of each sprint, the development team demonstrated the new features to stakeholders and conducted UAT to verify that the features met requirements and worked as expected.

Phase 5 - User Evaluation

The user evaluation phase involved distributing questionnaires and gathering feedback from all user groups. The researchers conducted formal evaluation sessions where users interacted with the system and then completed questionnaires about their experience. The evaluation sessions provided valuable insights into how real users interacted with the system, revealing usability issues that had not been apparent during development.

Phase 6 - Refinement

The refinement phase involved making final adjustments based on feedback from user evaluation. The researchers prioritized the issues and suggestions that emerged from evaluation, addressing critical usability problems and high-priority feature requests before the final deployment. The refinement phase also included final testing to verify that the improvements had been implemented correctly and had not introduced new bugs.

3.10 System Architecture

This section provides a detailed technical decomposition of TUGON's infrastructure, component modules, database schema, user interfaces, and communication channels.

3.10.1 Hardware and Software Requirements

Mobile hardware requires Android or iOS devices with GPS and cellular data capabilities. Web hardware requires desktop workstations with modern web browsers. Software dependencies include Flutter SDK, Next.js framework, Node.js runtime, PostgreSQL/Supabase database, and Google Gemini 3.1 Flash API access.

3.10.2 System Components and Modules

TUGON includes the following key features: (1) AI-Powered Incident and Complaint Classification utilizes natural language processing to automatically categorize incoming reports and assign urgency scores based on content analysis. (2) AI KYC Verification and Identity Management validates citizen identities through document verification and facial recognition to ensure authenticated report submissions. (3) Automated Dispatch Queue and Mission Management intelligently assigns and prioritizes response missions to available MAT personnel based on proximity, workload, and incident severity. (4) Real-Time GPS Tracking and ETA Calculation provides live location monitoring of MAT vehicles and responders with estimated time of arrival computations for incoming reports. (5) Face ID Attendance with Geofence Verification uses facial recognition and GPS geofencing to authenticate MAT personnel attendance and on-site presence during operations. (6) Patrol Management and Scheduling enable systematic coordination of patrol routes, shift schedules, and coverage areas for MAT field operations. (7) Evacuation Center Check-In/Check-Out manages occupant tracking and resource allocation across evacuation centers during disaster response operations. (8) Family Circle Safety and Live Location Sharing allows citizens to share real-time location with designated family members and receive safety status updates during emergencies. (9) MAT-Citizen Real-Time Chat and Communication provides an integrated messaging platform for direct communication

between citizens and MAT personnel regarding active reports and incidents. (10) Incident Heatmap and Clustering Visualization generates geospatial visualizations of incident density and patterns to support data-driven decision-making for resource deployment.

3.10.3 Database Structure

This section presents the logical design of the system's database using an Entity-Relationship (ER) Diagram, as shown in Figure 4. The database structure of TUGON is architected using PostgreSQL to centralize complaint and incident reporting, responder dispatch, and resource logistics for the MDRRMO/MAT of Liliw, Laguna.

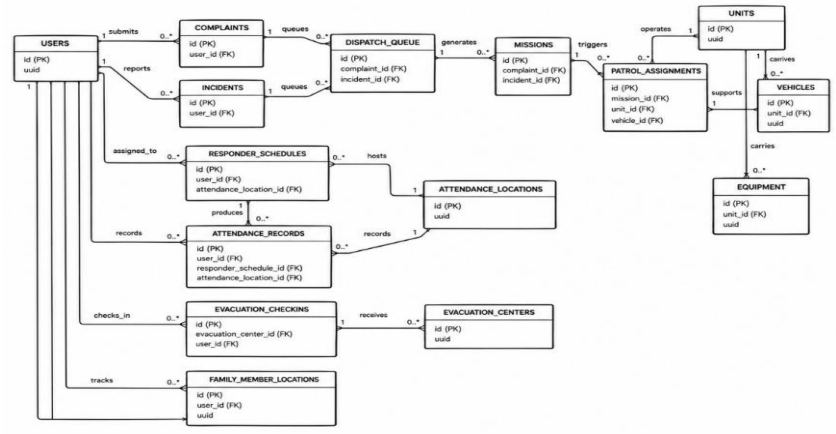


Figure 4. Entity-Relationship Diagram (ERD)

Image generated using ChatGPT (September 2026) based on the system database structure of TUGON, showing the relationships among users, complaints, incidents, dispatch queues, missions, patrol assignments, units, vehicles, equipment, attendance records, evacuation centers, and family member locations.

To ensure data integrity and normalized operational workflows, the relational schema directly supports the major functional features evaluated in this study:

- **Automated Dispatch Queue and Mission Management (Features 2.1 & 2.3):** The reporting and dispatch engine begins when a USER submits a

COMPLAINT or reports an INCIDENT. These records feed into the DISPATCH_QUEUE to process priority levels. From the queue, the system generates structured MISSIONS to manage responder deployments efficiently.

- **Patrol Management and Scheduling (Feature 2.6):** Operational resources are organized by linking UNITS, VEHICLES, and EQUIPMENT. Once a mission is generated, it triggers PATROL_ASSIGNMENTS, connecting the designated response unit, assigned vehicle, and specific mission parameters.
- **Face ID Attendance with Geofence Verification (Feature 2.5):** Responder shift operations are governed by RESPONDER_SCHEDULES, which associate users with authorized ATTENDANCE_LOCATIONS. Attendance logs, geofence validations, and verification results are captured in ATTENDANCE_RECORDS.
- **Evacuation Center Check-In/Check-Out (Feature 2.7):** Disaster response and shelter operations are managed through EVACUATION_CENTERS, tracking displaced citizens via EVACUATION_CHECKINS linked directly to their user profiles.
- **Family Circle Safety and Live Location Sharing (Feature 2.8):** Community safety and interpersonal location tracking are facilitated through FAMILY_MEMBER_LOCATIONS, allowing real-time spatial updates tied to individual users.

3.10.4 User Interface Design

UI design followed accessibility and usability principles across all user roles. The mobile citizen interface was developed using Flutter for cross-platform compatibility, featuring a simplified report submission form with integrated GPS map

pinning, real-time case status tracking, a family circle with live location sharing, direct messaging with MAT personnel, and a one-tap emergency alert button. The responder mobile interface provides assigned mission details, turn-by-turn navigation to incident locations, and Face ID attendance with geofence verification for on-site authentication. The web-based MAT dashboard was built using Next.js with responsive layouts and dark/light mode support, displaying a centralized command view that includes an active complaint queue, a dispatch panel for mission assignment, a live GPS vehicle tracking map, and an incident heatmap for data-driven resource deployment. The admin dashboard provides analytics reporting, user management, and system configuration controls.

3.10.5 System Communication and Interaction

System communication follows a layered architecture with distinct mechanisms governing interactions among users, application components, databases, and external services. All client requests from the Flutter mobile app and Next.js web dashboard are routed through a FastAPI gateway that handles authentication, request validation, and rate limiting before forwarding to the appropriate service. When a citizen submits a complaint, the API gateway forwards the text data to the Gemini API for natural language processing, which returns the classified incident type and urgency score for automated prioritization. Processed data is stored and retrieved from Supabase (PostgreSQL) via encrypted HTTPS/TLS connections with Row Level Security policies enforcing role-based data access across citizen, responder, dispatcher, and administrator accounts. GPS location updates from responder mobile devices are streamed to the web dashboard via WebSocket connections, enabling live vehicle tracking and ETA calculations without requiring page refresh. Firebase Cloud Messaging delivers real-time push notifications to citizens for case status updates and

to responders for new mission assignments. Google Maps API provides geocoding, route calculation, and map rendering for incident pinning and navigation features. All inter-service communication is secured through encrypted channels to ensure data integrity and compliance with the Data Privacy Act of 2012 (RA 10173).

3.11 System Testing and Evaluation Procedure

System testing and evaluation followed a structured protocol to verify technical reliability and measure user acceptance. The testing and evaluation activities were directly aligned with the research objectives and system requirements, covering functional verification through Black-Box testing and user acceptance evaluation through the Technology Acceptance Model (TAM).

3.11.1 Testing Plan

The testing plan followed a sequential three-phase approach aligned with the Agile development sprints. Phase 1 involved Black-Box functional testing conducted by IT testers to verify that each system feature operated according to its specified requirements. Phase 2 involved simulation testing conducted with MAT staff at the Municipal Action Team office in Liliw, Laguna, where responders, dispatchers, and administrators interacted with the system under realistic operational scenarios. Phase 3 involved user acceptance evaluation through TAM questionnaires administered to all four respondent groups to measure perceived usefulness, ease of use, and behavioral intention to adopt the system.

3.11.2 Test Cases and Test Scenarios

Nine test cases were developed based on the documented functional requirements of the TUGON system. Each test case targeted a specific feature or module: (TC-01) Citizen Homepage, (TC-02) User Registration and Login with AI

KYC Verification, (TC-03) AI-Verified Incident Report Submission, (TC-04) Live GPS Tracking for Citizen and Responder, (TC-05) Family Circle and Live Location Sharing, (TC-06) Evacuation Center Management, (TC-07) Community Complaint and Feedback, (TC-08) Patrol Schedule and Face ID Attendance with Geofence, and (TC-09) Administrator Login and Dashboard. Each test case included preconditions, step-by-step procedures, expected results, and pass/fail status indicators.

3.11.3 Testing Data

Testing data consisted of synthetic Taglish complaint/incident datasets representing realistic incident scenarios such as fire emergencies, flood reports, and road accidents. Simulated GPS coordinates were generated within the geographical boundaries of Liliw, Laguna to test location-dependent features including map pinning, geofence verification, and evacuation center check-in. Test user accounts were created for each role type (citizen, responder, dispatcher, administrator) to validate role-based access controls and dashboard functionalities. Photo assets matching various incident categories were prepared to test the AI image validation feature.

3.11.4 Testing Procedure

Black-Box functional testing was conducted first by IT testers who executed all nine test cases who executed all nine test cases in a controlled environment. Each step was performed sequentially, and results were documented as pass or fail with actual results recorded for failed steps. Following functional testing, simulation testing was conducted at the MAT office where MAT staff including dispatchers, responders, and the system administrator performed real-world scenarios such as receiving and dispatching complaints, tracking responder movements, and managing evacuation centers. Testers indicated their name, date, job role, and years of experience before

completing the testing sheet. Results were compiled into a test case summary table documenting the overall pass rate across all features.

3.12 System Testing

System testing determined whether the completed TUGON system met its functional and non-functional requirements and operated as intended under appropriate conditions. Black-Box functional testing was selected as the primary testing method because it evaluates system behavior from the user perspective without examining internal code structures, making it appropriate for validating the 10 major features identify objectives.

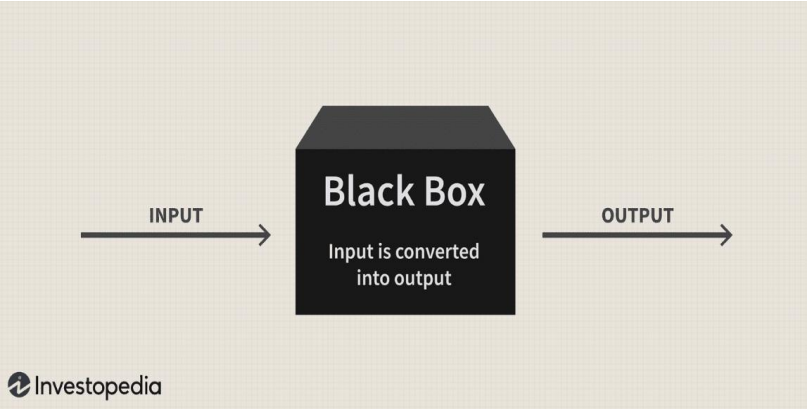


Figure 3.3. Black-Box Testing Model (adapted by Will Kenton, 2025)

Black box testing is a software testing method that examines functionality without knowledge of internal code structures or implementation details, focusing solely on inputs and expected outputs to validate system behavior from the user perspective (Ayuningtyas et al., 2023). The researchers utilized this methodology to assess the TUGON system functionality by providing test inputs and observing outputs without examining the underlying code.

3.12.1 Test Cases

Test cases were documented in a structured format containing the feature or requirement being tested, test conditions, expected results, actual results, and test status. The detailed test cases and scenarios developed during the testing phase are presented in Section 3.11.2.

3.12.2 Testing Instrument

The testing instrument was a structured Black-Box testing sheet containing the tester identification section (name, date, job role, and years of working experience) and a test case evaluation table. Each test case row included the test scenario, preconditions, step-by-step procedures with expected results, columns for actual results, and a pass/fail status indicator. The instrument was designed to systematically document the functional behavior of each TUGON feature against its specified requirements.

3.12.3 Validation of Testing Instrument

The Black-Box testing instrument underwent content validation by the thesis adviser who reviewed the test cases for alignment with the documented system requirements and research objectives. The adviser verified that each test scenario appropriately covered the identified major features and that the expected results were consistent with the system specifications. Revisions were made based on the adviser feedback to ensure clarity, completeness, and appropriateness of the testing procedures before actual test execution.

3.12.4 Interpretation of Test Results

Test results were recorded as pass or fail for each individual step across all nine test cases. A feature fully functional when all its test steps received a pass status. Failed steps were documented with the actual result observed and compared against the

expected result to identify discrepancies. The overall system pass rate was computed by dividing the total number of passed test steps by the total number of test steps and expressing it as a percentage. A system was considered functionally acceptable when it achieved a pass rate of 80 percent or higher across all test cases.

3.13 System User Evaluation

System user evaluation determined how intended users perceived the developed TUGON system in terms of usability, perceived usefulness, and behavioral intention to adopt. The evaluation utilized a structured questionnaire administered to four respondent groups: citizens, MAT Responders/Drivers, dispatchers, and administrator following system demonstration and hands-on testing at the MAT office in Liliw, Laguna.

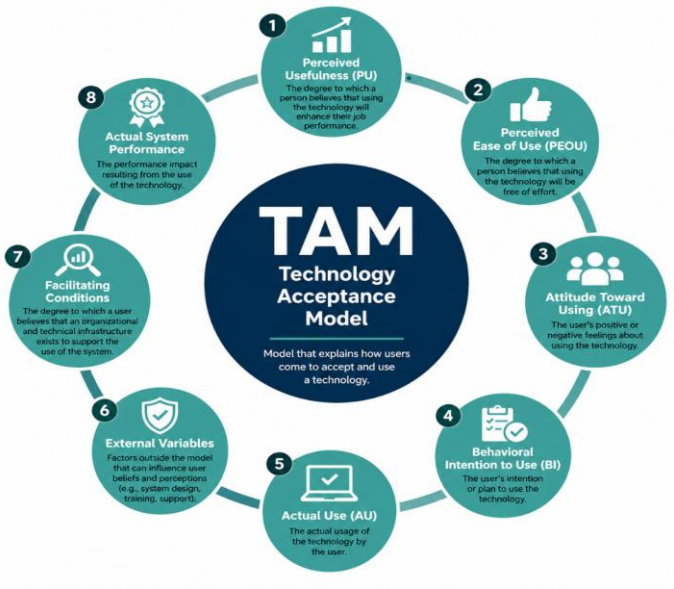


Figure 3.4. Technology Acceptance Model (TAM) (adapted by Francesco Cannonito, 2025)

The TUGON system adopts the Technology Acceptance Model (TAM) as its evaluation framework. TAM is one of the most widely used theoretical models for predicting and explaining user acceptance of information technology (Davis, 1989; Pradana & Salamat, 2025). In the context of e-government systems like TUGON, TAM measures how Perceived Usefulness and Perceived Ease of Use influence citizens' and government workers' behavioral intention to adopt the platform for public safety operations.

3.13.1 User Evaluation Instrument

The user evaluation instrument was a Technology Acceptance Questionnaire consisting of 18 items distributed across six TAM constructs: Quality Factors (3 items), Perceived Ease of Use (3 items), Perceived Usefulness (3 items), Attitude Toward Using (3 items), Behavioral Intention to Use (3 items), and Experience/Actual Use (3 items). Each item was rated using a 5-point Likert scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). The questionnaire also included a demographic profile section capturing age, sex, and respondent category, as well as an open-ended section for comments and suggestions.

3.13.2 Instrument Source and Validation

The TAM questionnaire items were adapted from established Technology Acceptance Model literature, particularly the works of Davis (1989) and Pradana and Salamat (2025) and contextualized for the TUGON system and the Liliw municipal setting. The research instrument underwent content and face validation by the thesis adviser who reviewed item clarity, construct alignment, and contextual appropriateness. The adviser assessed each questionnaire item to ensure it accurately measured the intended TAM construct and was relevant to the system being evaluated.

3.13.3 User Evaluation Procedure

The evaluation instruments were administered through tailored procedures for each target respondent group. For MAT staff (n=22: 1 Admin, 3 Dispatchers, 18 Responders), the researchers conducted simulation and demonstration exercises where respondents interacted with the system under guided scenarios, including the 5-minute AVP and supervised hands-on demonstration, before completing the TAM questionnaire. For citizens (n=181), the questionnaire was distributed after the same 5-minute AVP and hands-on testing. All 203 respondents were briefed on the study purpose, informed of their right to withdraw, and assured of data confidentiality before proceeding. Completed questionnaires were collected immediately after the session, representing initial acceptance under controlled, short-term conditions.

3.13.4 Evaluation Criteria

Table 3.4. Five-Point Rating Scale

Scale	Verbal Interpretation
4.21 - 5.00	Strongly Agree
3.41 - 4.20	Agree
2.61 - 3.40	Neutral
1.81 - 2.60	Disagree
1.00 - 1.80	Strongly Disagree

Mean scores were interpreted using a 5-point Likert scale continuum: 4.21 to 5.00 (Strongly Agree / Excellent), 3.41 to 4.20 (Agree / Very Good), 2.61 to 3.40 (Neutral / Fair), 1.81 to 2.60 (Disagree / Poor), and 1.00 to 1.80 (Strongly Disagree / Very Poor). The weighted mean was computed for each questionnaire item to determine

the overall level of agreement across respondent groups, while standard deviation was used to measure the consistency of responses. These measures also helped identify significant differences between respondent categories.

3.14 Statistical Treatment

This section outlines the methodology employed by the researchers to evaluate the user acceptance of the TUGON system. It details the testing, validation, and evaluation procedures used to assess the system impact. As the project aims to improve municipal complaint management and incident response, its successful implementation is expected to be validated through rigorous testing processes.

The researchers used the weighted mean to calculate statistical findings. The Likert scale was employed to gauge respondent's opinions effectively. The Likert scale was chosen for its simplicity and ability to capture a broad range of attitudes toward the system. Respondents expressed their level of agreement with various statements related to the system acceptability, functionality, and ease of use.

Frequency and Percentage Distribution

The Frequency and Percentage Distribution method was utilized to analyze and interpret the data gathered from the respondents of the TUGON system. This statistical tool was applied to determine the proportion of responses in each category, providing a clear overview of the participants' feedback and evaluation results.

$$\% = (f / \mathrm{N}) * 100$$

Equation 2. Frequency and Percentage Distribution

Where:

% = Percentage of responses

f = Frequency of responses

N = Total number of respondents

Mean Formula

The Mean Formula was applied as a weighted average to describe participants' perceptions of the TUGON system. It gives the overall indication of how respondents see the system by some key elements of functionality, usability, and effectiveness. The formula is expressed as:

$$\overline{\boldsymbol{x}} = \frac{\sum \boldsymbol{x}}{N}$$

Equation 3. Mean Formula

Where:

$\bar{x}$ = Mean

$\sum x$ = Sum of all responses

N = Total number of respondents

The researchers used the formula for an average rating given by respondents on the evaluation instrument for each evaluation criterion. It is useful for researchers to measure general agreement with or satisfaction in the developed TUGON system.

Weighted Mean

The Weighted Mean Formula was employed to interpret the degree of user acceptance of TUGON based on responses from Dispatchers, Responders, Citizens, and Administrators. Since the evaluation utilized a Likert scale, each response was assigned a specific weight to reflect its relative importance.

$$x_{w=} \frac{\sum(f * x)}{\mathrm{N}}$$

Equation 4. Weighted Mean Formula

Where:

x_w = Weighted Mean (average likert score)

f = Frequency of responses for each likert scale option

x = Likert scale weight assigned to each option (5 = Strongly Agree, 4 = Agree, 3 = Neutral, 2 = Disagree, 1 = Strongly Disagree)

N = Total number of respondents

The weighted mean is a measure of central tendency that accounts for both the frequency of each response and its assigned numerical value. It provides a single number summarizing the overall level of agreement or satisfaction for each questionnaire item, making it particularly useful for Likert scale data as it treats response options as interval data.

This formula allowed the researchers to determine the overall weighted mean for each criterion, interpreted using a descriptive scale: 4.21–5.00 = Strongly Agree, 3.41–4.20 = Agree, 2.61–3.40 = Neutral, 1.81–2.60 = Disagree, and 1.00–1.80 = Strongly Disagree. Weighted means were also calculated for groups of items measuring the same construct, allowing comparison across different aspects of the TUGON system to identify strengths and areas for improvement.

Table 3.5. Likert Scale for Weighted Mean

Scale	Verbal Interpretation
4.21 – 5.00	Strongly Agree
3.41 – 4.20	Agree
2.61 - 3.40	Neutral

Scale	Verbal Interpretation
1.81 – 2.60	Disagree
1.00 – 1.80	Strongly Disagree

Standard Deviation Formula

The Standard Deviation Formula was used to measure the variability or dispersion of the responses gathered from the respondents in evaluating the TUGON system. It indicates how much the individual responses differ from the mean, providing a clearer picture of the consistency of participants' perceptions regarding the system's usability, reliability, and effectiveness. The formula is expressed as:

$$s = \sqrt{\frac{\sum(x - \overline{x})^2}{n - 1}}$$

Equation 5. Standard Deviation Formula

Where:

s = Sample standard deviation

x = Each individual data point

$\bar{x}$ = Sample mean

Σ = Summation symbol

n = Number of respondents (sample size)

$n - 1$ = Degrees of freedom

CHAPTER IV

RESULTS AND DISCUSSIONS

This chapter presents the results of the design, development, testing, and evaluation of TUGON: Technology-Unified Government Operations Network. It discusses the overall functionality of the system in improving complaint and incident response for the MAT of the Municipality of Liliw, Laguna. The system architecture, interface outputs, and evaluation results from the Administrator, Dispatchers, Responders, and Citizens are presented, evaluated, and interpreted to determine whether the system achieves the desired objectives.

4.1 General Presentation of Results

This section outlines the overall organization of the findings derived from the design, development, testing, and evaluation of the Technology-Unified Government Operations Network (TUGON). The results are presented systematically, strictly following the sequence of the specific objectives established in Chapter I, ensuring that every defined goal is supported by concrete, empirical evidence gathered during the study.

To maintain a clear, logical, and objective-driven progression, the findings for each specific objective are structured to explicitly identify the following components:

- Objective: The specific research goal being addressed, ranging from the core module development to functional testing and user evaluation.
- Activity or Procedure Performed: The methodological steps executed to achieve the objective, such as Agile development sprints, Black-Box testing protocols, or the administration of the Technology Acceptance Model (TAM) survey.

- Data or Evidence Obtained: The tangible outcomes resulting from the procedures, including deployed system features, step-level test case results, and quantitative survey responses from the Mayor's Action Team (MAT) and citizens.
- Visual and Statistical Presentation: The corresponding architectural diagrams, interface screenshots, functional testing summaries, and statistical tables that substantiate the findings.
- Interpretation of the Result: An objective analysis of what the collected data, test scores, or developed modules signify in the context of the system's operational requirements.
- Implication of the Finding: The broader impact of the result in relation to the objective, specifically detailing how the finding resolves the manual, paper-based reporting and dispatching challenges previously faced by the municipality.

4.1.1 Presentation of Objective-Based Results

To ensure clarity and logical progression, the results presented in this chapter strictly follow the exact order of the Specific Objectives outlined in Chapter I. This sequential alignment allows the evaluation panel and readers to clearly trace the relationship among the identified research problem, the stated objectives, the applied methodology, the obtained results, and the final conclusions. To systematically present these findings, a standardized presentation pattern is employed for every specific objective. The discussion for each section begins by stating the objective being addressed, followed by the specific method or activity executed to fulfill it. This is immediately followed by the direct result or outcome derived from that activity, a detailed interpretation of what the result signifies within the context of the study, and

finally, the concrete evidence of achievement presented through relevant tables, figures, or statistical data.

4.2 System Architecture and Development Results

This section presents the actual results of how TUGON was designed, developed, and brought to life to fulfill Objective 1. Rather than just theory, this part walks through the tangible outcomes of turning the manual, paper-and-logbook processes of the Mayor's Action Team (MAT) in Liliw into a working digital system. Grounded in the insights gathered from our interviews and site visits with local officials, the development followed an Agile approach, combining a Flutter-powered mobile app for citizens and responders with a Next.js web dashboard for dispatchers and administrators, all backed by a secure PostgreSQL and Supabase database. To give a complete picture of the system's realization, the following subsections break down everything from the high-level system overview and architectural layers to the hardware specifications, database structure, and the specific user interfaces built for every stakeholder group.

4.2.1 System Overview

TUGON is built as an integrated, AI-powered municipal response platform designed specifically to modernize how the Mayor's Action Team (MAT) and the citizens of Liliw handle emergencies, complaints, and community operations. At its core, the system replaces fragmented, phone-based, and logbook-driven workflows with a unified digital ecosystem. Its primary purpose is to bridge the communication gap between the local government and the community, ensuring that incident reports are processed, classified, and dispatched with speed and accuracy.

The platform serves four distinct user groups, each with tailored interfaces designed for their specific operational needs: Citizens, who use the mobile app to report incidents, track family members, and view heatmaps; Responders, who navigate to emergencies and clock in via secure verification; Dispatchers/MAT Officers, who oversee live tracking and manage incoming complaints on the web dashboard; and the System Administrator, who maintains centralized management of user accounts and RBAC (within the tested administrative scope) and Role-Based Access Control (RBAC) policies. Major functions span automated AI urgency classification, real-time GPS tracking, emergency facilities management, and structured public announcements. Overall, the system operates through a seamless loop: a citizen submits a report or complaint via mobile, the system processes and routes it to the command center in real time, and dispatchers coordinate the nearest available responder unit to resolve the incident efficiently.

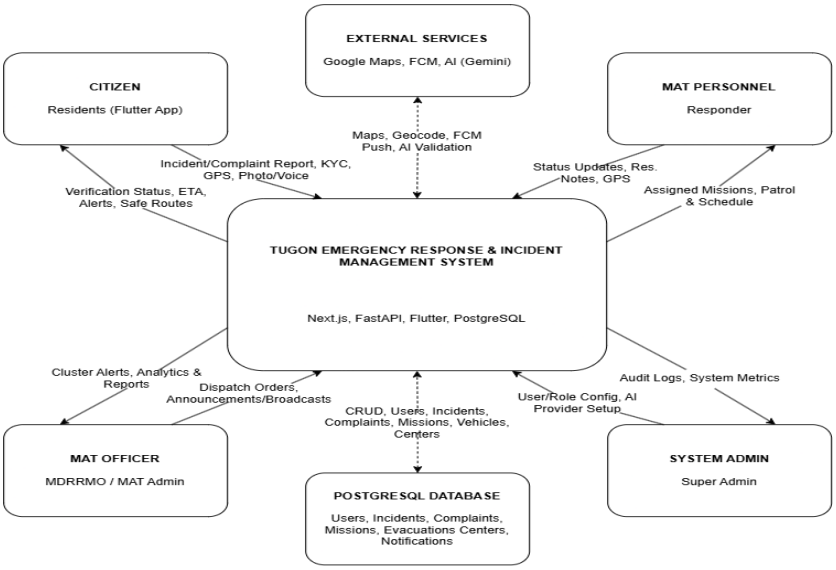


Figure 4.1. System Context Diagram

4.2.2 System Architecture

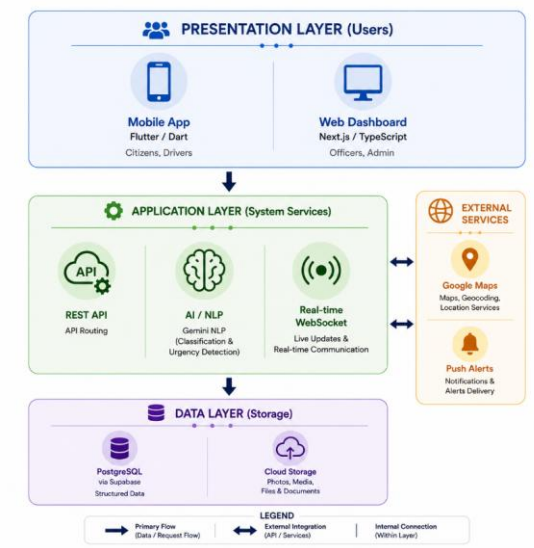


Figure 4.2. System Architecture

The technical realization of TUGON is built upon a robust, three-tier system architecture designed to ensure scalability, real-time responsiveness, and secure data handling across municipal operations. As illustrated in Figure 4.2, the architecture explicitly segments the platform into the Presentation Layer, the Application Layer, and the Data Layer, seamlessly connecting diverse user devices, cloud servers, and external services into a unified operational workflow.

The Presentation Layer serves as the primary touchpoint for system users across different hardware devices. Citizen and Responder interactions are handled via cross-platform mobile applications optimized for both Android and iOS devices, built using Flutter and Dart to ensure smooth performance and intuitive touch navigation. Meanwhile, MAT Officers, Dispatchers, and the System Administrator operate through a responsive, web-based command dashboard deployed via Next.js and TypeScript, accessible through standard desktop and tablet web browsers. These interfaces abstract

complex backend operations into clean, role-specific views for emergency reporting, live monitoring, and mission control.

The Application Layer acts as the central nervous system of the platform, hosting the core business logic, RESTful APIs, and real-time WebSocket communication channels on secure cloud servers. This layer processes incoming user actions, coordinates dispatch queues, and directly interfaces with vital external services. Specifically, it integrates the Google Maps API for geocoding, spatial mapping, and route calculations; Firebase Cloud Messaging (FCM) to trigger instant push alerts for emergency notifications; and Gemini AI models to handle natural language processing (NLP) for automated incident classification and optical character recognition (OCR) for identity verification.

Finally, the Data Layer manages persistent storage and state synchronization through a managed PostgreSQL database hosted via Supabase. All structural information, including verified user credentials, geo-tagged incident reports, community complaints, active vehicle dispatch logs, and evacuation center capacities, is securely housed within relational database schemas. Real-time subscriptions enabled by Supabase ensure that database changes instantly propagate across the application layer, allowing dispatchers and responders to view live GPS tracking streams and status updates without manual page refreshes.

4.2.3 Hardware and Software Requirements

To ensure that TUGON can be deployed smoothly without imposing excessive infrastructural burdens on the municipality or the community, the platform was engineered to operate on standard, readily available computing and mobile hardware. As detailed in Table 4.1, the hardware requirements specify the minimum technical

thresholds needed for seamless day-to-day operation. For administrative oversight and dispatching tasks, a standard laptop or desktop computer equipped with at least a dual-core processor and 4 GB of RAM is required. For mobile users, including citizens and field responders, a smartphone running Android 11.0 or higher, or an Apple device running iOS 12.0 or higher (or your tested iOS version), is necessary to support GPS tracking, camera inputs for AI verification, and real-time push alerts. All listed hardware specifications are confirmed to be existing and fully accessible within the operational environment of the Mayor's Action Team (MAT) and the local community.

Complementing the physical hardware, Table 4.2 outlines the software prerequisites essential for system execution across platforms. Access to the web-based command dashboard requires standard, modern web browsers such as Google Chrome, Microsoft Edge, or Mozilla Firefox, allowing dispatchers and system administrators to manage operations without heavy local software installations. Meanwhile, mobile-facing operations require an Android 11.0+ environment to host the cross-platform application. Additionally, a stable internet connection is mandated across all endpoints to facilitate continuous data synchronization through Supabase, real-time WebSocket updates, and cloud-based API integrations such as Google Maps and Gemini AI.

Table 4.1. Hardware Requirements

HARDWARE REQUIREMENTS	EXISTING
Laptop or Desktop Computer	YES
Smartphone (Android 11.0+)	YES
Stable Internet Connection	YES
Dual-Core Processor or Higher	YES

HARDWARE REQUIREMENTS	EXISTING
At least 4 GB RAM or Higher	YES

Table 4.2. Software Requirements

SOFTWARE REQUIREMENTS	EXISTING
Web Browser (Google Chrome, Microsoft Edge, or Mozilla Firefox)	YES
Internet Connection	YES
Android 11.0+ or iOS (for mobile users)	YES

4.2.4 Database Structure

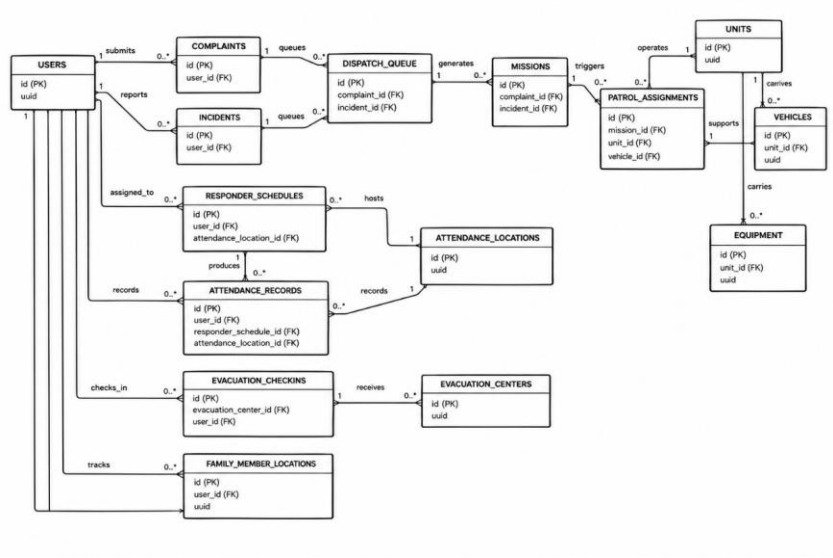


Figure 4.3. Database Schema

The backend data architecture of TUGON is hosted on a relational PostgreSQL database deployed via Supabase, engineered to maintain high data integrity, security, and real-time synchronization across municipal operations. The database schema is structured to systematically capture, process, and store all operational data generated by citizens, responders, dispatchers, and administrators.

The database design revolves around several major functional entities and tables. The User Management Schema securely stores citizen, responder, and administrator profiles alongside AI-verified identity credentials and liveness check scores. The Incident and Complaint Management Schemas manage structured reports submitted through the mobile interface, separating emergency incidents from community complaints while capturing GPS coordinates, media attachments, and automated AI classification outputs. Furthermore, the Mission and Dispatch Schema tracks vehicle allocation, responder assignments, and live status updates, while the Facilities Schema maintains real-time records of municipal evacuation centers and resource inventories.

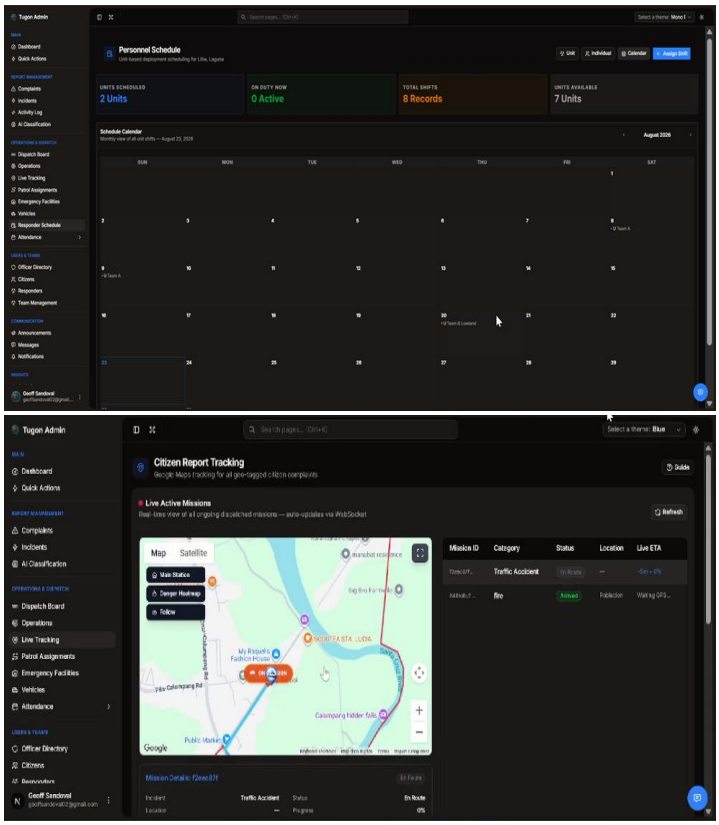
4.2.5 User-Interface and System Modules

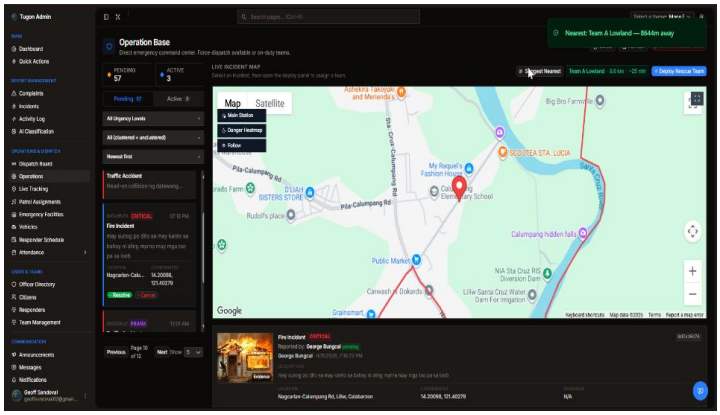
The user-interface design of TUGON is purposefully segregated into role-specific platforms to ensure secure, streamlined operations for administrators, dispatchers, responders, and citizens. The web-based dashboard caters to the Mayor's Action Team (MAT) and System Administrator, while the mobile applications provide accessible touchpoints for field responders and the community.

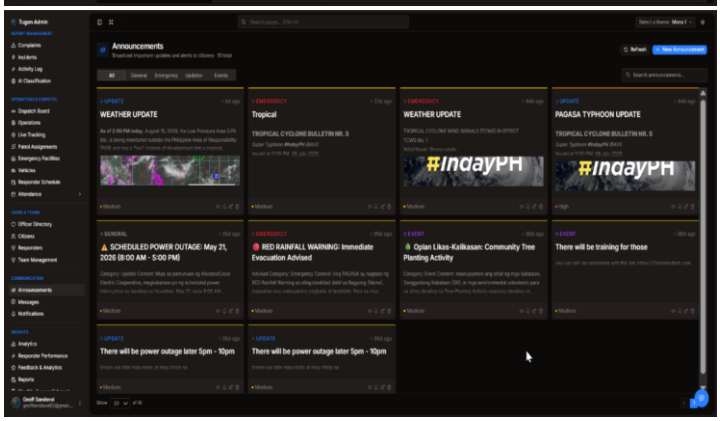
Administrator and MAT Officer Web Interface

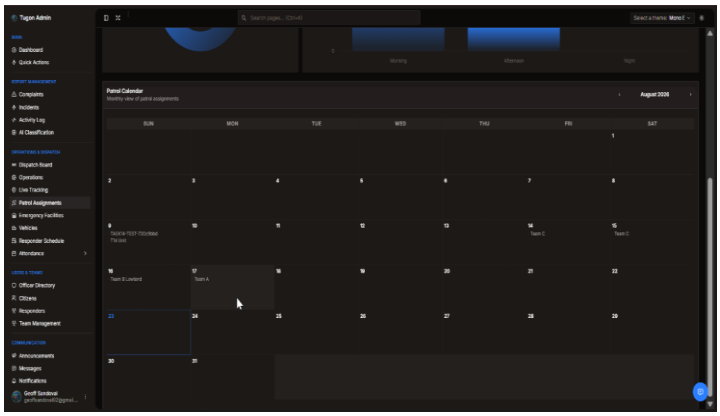
The administrative dashboard serves as the central command post for municipal operations. As shown below, the Operation Base features real-time GPS tracking of active response units and spatial mapping of incoming reports. Security and access

control are strictly enforced; crucially, Role-Based Access Control (RBAC) policy configurations are restricted exclusively to the System Administrator (as seen in the system management views), ensuring that standard dispatchers cannot alter foundational security permissions. Additional modules cover Complaints Management, Emergency Facilities Management, and analytical reporting, replacing manual paper logbooks with automated, sortable data grids.




Tugon Admin
Operation Base
Nearest: Team A Lowland — 8644m away
Map
Satellite
Google


Tugon Admin
Announcements
WEATHER UPDATE
Tropical
TROPICAL CYCLONE BULLETIN NR. 5
WEATHER UPDATE
PAGASA TYPHOON UPDATE
#IngayPH
SCHEDULED POWER OUTAGE: May 21, 2026 (8:00 AM - 5:00 PM)
RED RAINFALL WARNING: Immediate Evacuation Advised
Opian Likas-Kalikasan: Community Tree Planting Activity
There will be training for those
There will be power outage later 5pm - 10pm
There will be power outage later 5pm - 10pm


Tugon Admin
Patrol Calendar
August 2026

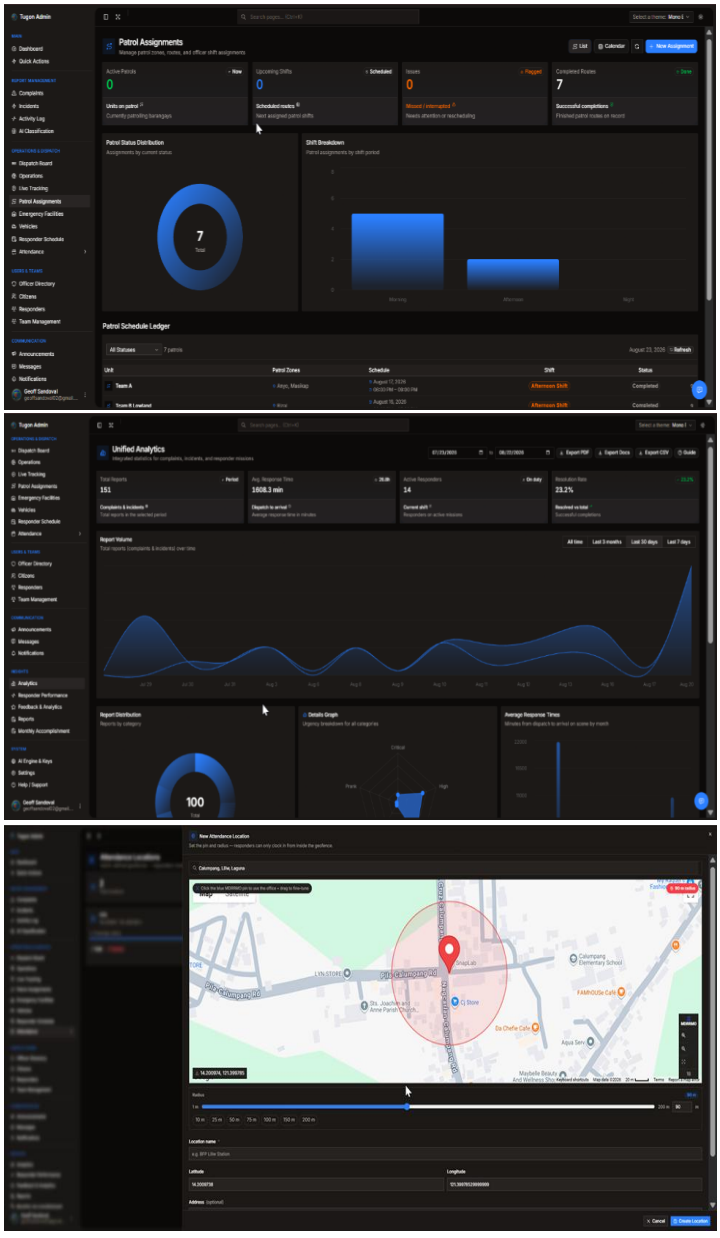
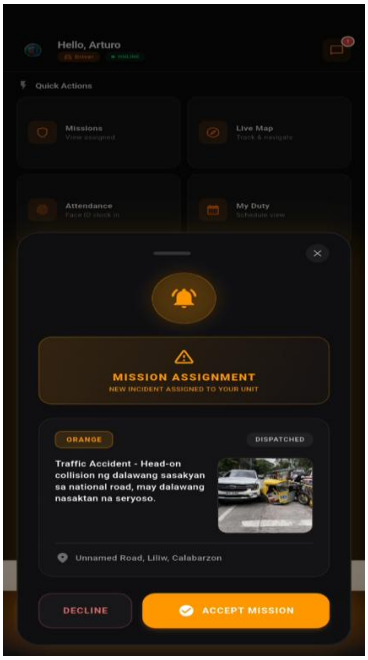
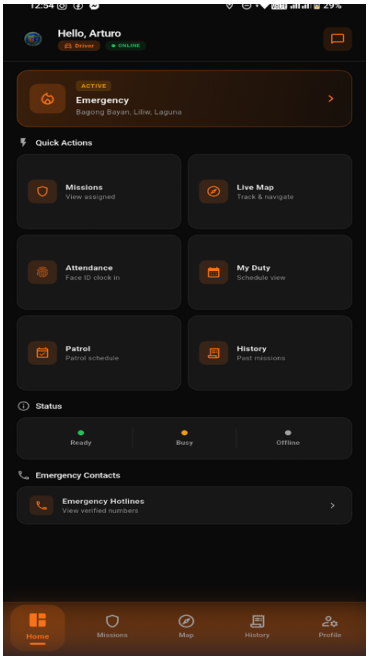


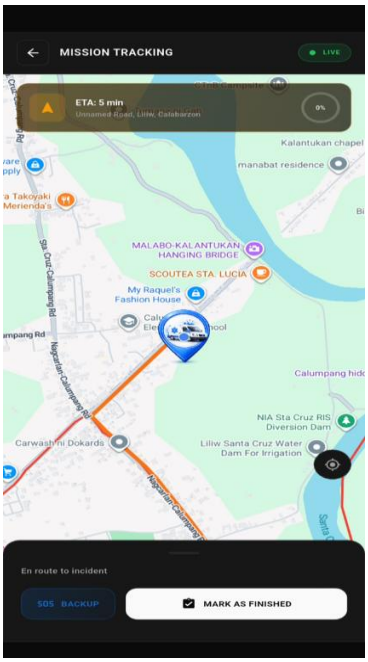
Figure 4.3. User Interface of Admin and MAT Officers - Web

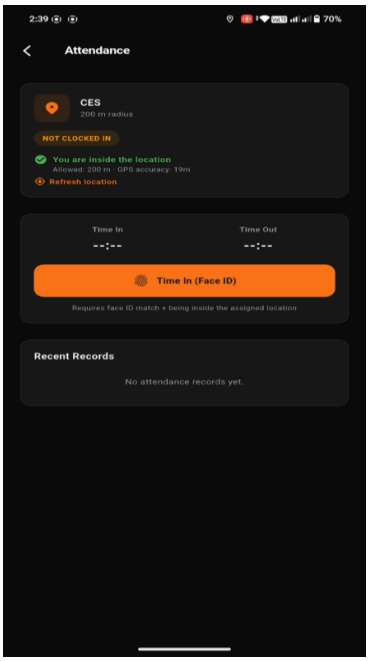
Responder Mobile Interface

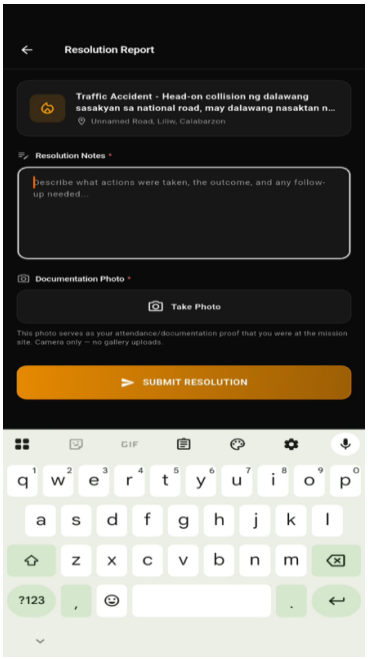
Field responders rely on a dedicated mobile workflow optimized for high-pressure situations. The Patrol and Duty Schedule screen incorporates biometric or face verification for secure attendance tracking. Once an emergency is dispatched, the Responder Mission Flow interface guides personnel with turn-by-turn navigation and status-update triggers (such as "En Route," "Arrived," and "Resolved"), ensuring complete transparency and accountability between field units and the command center.

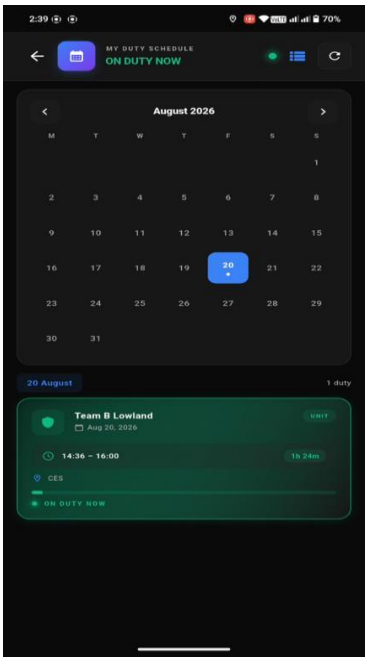





MISSION TRACKING
LIVE
ETA: 5 min
Unnamed Road, Liliw, Calabarzon
0%
Kalantukan chapel
manabat residence
MALABO-KALANTUKAN HANGING BRIDGE
SCOUTEA STA. LUCIA
My Raquel's Fashion House
NIA Sta Cruz RIS Diversion Dam
Liliw Santa Cruz Water Dam For Irrigation
En route to incident
SOS BACKUP
MARK AS FINISHED


2:39
70%
Attendance
CES
200 m radius
NOT CLOCKED IN
You are inside the location
Allowed: 200 m · GPS accuracy: 19m
Refresh location
Time In
--:--
Time Out
--:--
Time In (Face ID)
Requires face ID match + being inside the assigned location
Recent Records
No attendance records yet.


Resolution Report
Traffic Accident - Head-on collision ng dalawang sasakyan sa national road, may dalawang nasaktan n...
Unnamed Road, Liliw, Calabarzon
Resolution Notes *
Describe what actions were taken, the outcome, and any follow-up needed...
Documentation Photo *
Take Photo
This photo serves as your attendance/documentation proof that you were at the mission site. Camera only — no gallery uploads.
SUBMIT RESOLUTION


2:39
70%
MY DUTY SCHEDULE
ON DUTY NOW
August 2026
20 August
1 duty
Team B Lowland
Aug 20, 2026
UNIT
14:36 – 16:00
1h 24m
CES
ON DUTY NOW

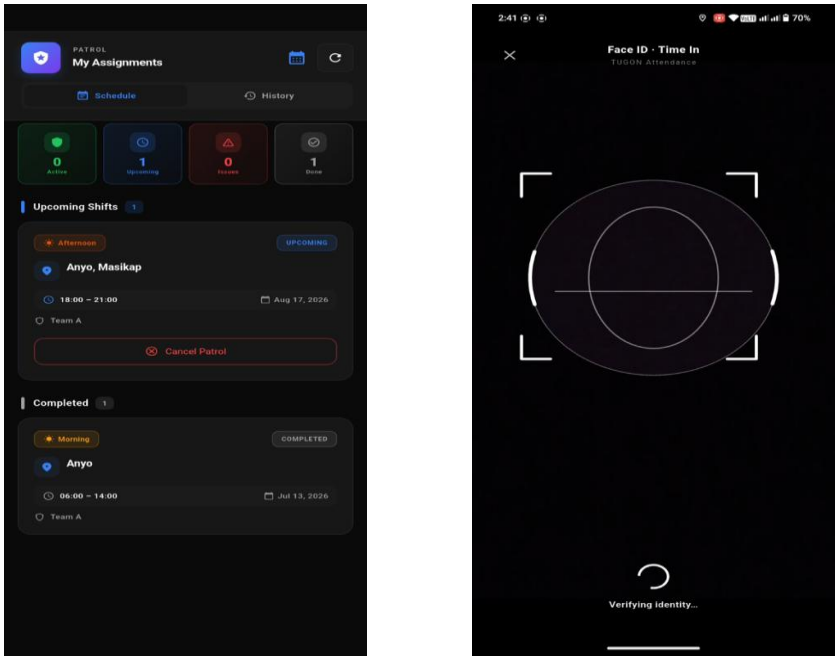
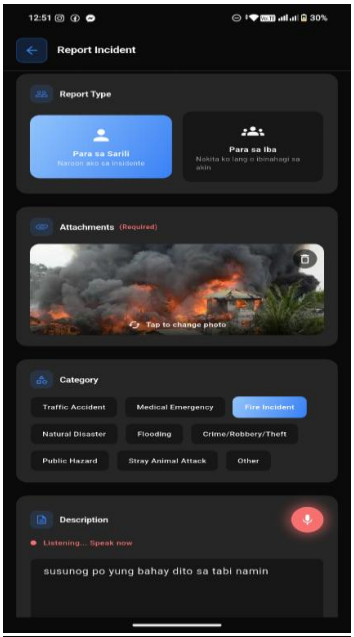
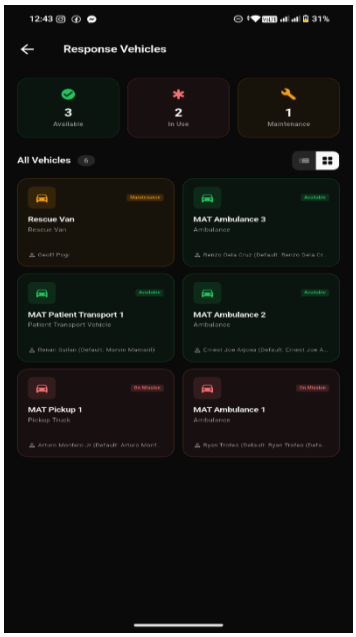


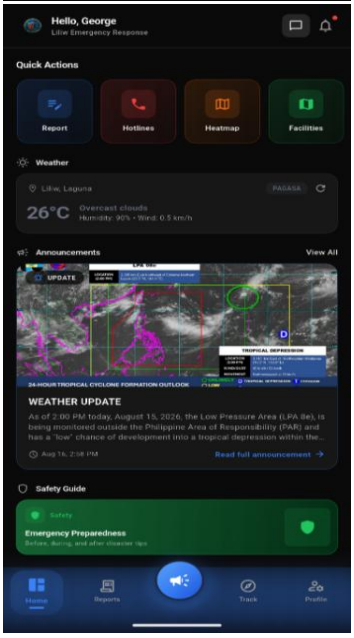
Figure 4.4. User Interface of Responder - Mobile

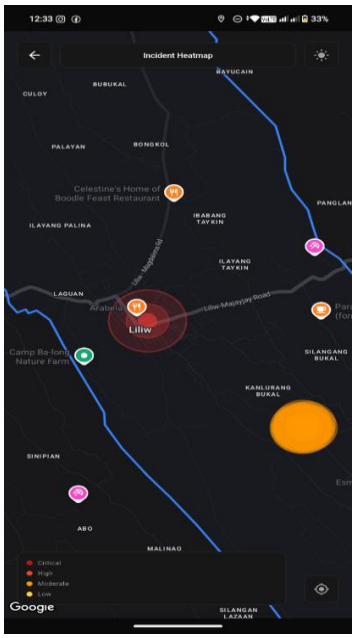
Citizen Mobile Interface

Designed for accessibility and speed, the citizen mobile application empowers residents to actively participate in local safety. The Citizen Home dashboard provides immediate access to emergency hotlines and quick-action modules. The AI-assisted Report Submission interface allows users to upload photos, input text or voice notes, and automatically capture precise GPS coordinates for rapid triage. Furthermore, community safety is enhanced through specialized tools like the Incident Heatmap and the Family Circle module, which enables private, real-time location sharing among household members during disasters.


Report Incident
Report Type
Para sa Sarili
Para sa iba
Attachments (Required)
Tap to change photo
Category
Traffic Accident
Medical Emergency
Fire Incident
Natural Disaster
Flooding
Crime/Robbery/Theft
Public Hazard
Stray Animal Attack
Other
Description
Listening... Speak now
susunog po yung bahay dito sa tabi namin


Response Vehicles
3
Available
2
In Use
1
Maintenance
All Vehicles
Rescue Van
MAT Ambulance 3
MAT Patient Transport 1
MAT Ambulance 2
MAT Pickup 1
MAT Ambulance 1


Hello, George
Liliw Emergency Response
Quick Actions
Report
Hotlines
Heatmap
Facilities
Weather
26°C
Announcements
View All
WEATHER UPDATE
Read full announcement
Safety Guide
Emergency Preparedness


Incident Heatmap
BUBUKAL
CULOY
PALAYAN
BONGKOL
ILAYANG PALINA
IBABANG TAYKIN
ILAYANG TAYKIN
LAGUAN
Liliw
KANLURANG BUKAL
SILANGANG BUKAL
SINIPIAN
ABO
MALINAO
Google

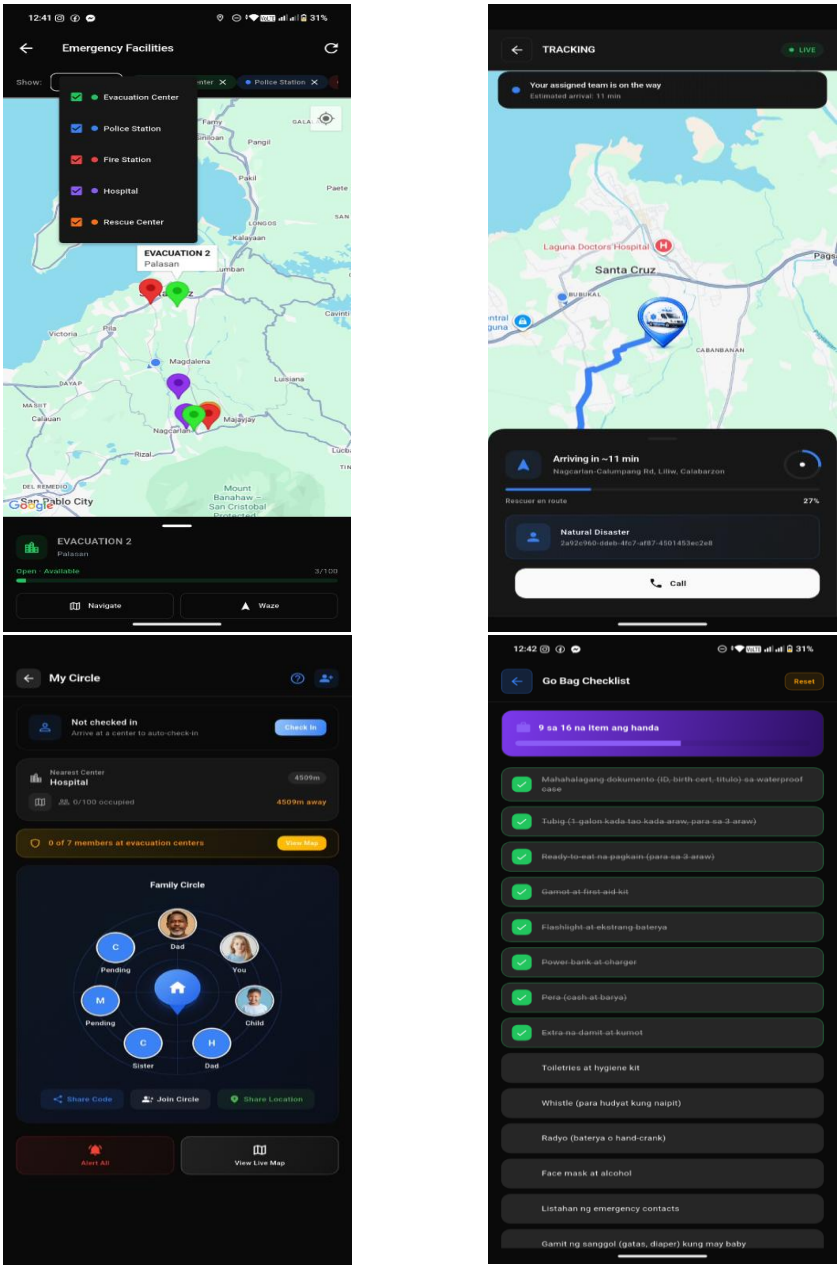


Figure 4.5. User Interface of Citizen - Mobile

4.2.6 Functional and Non-Functional Requirements

The developed TUGON platform successfully satisfies all the functional and non-functional requirements established during the system analysis phase. Functional requirements pertaining to core operations such as AI-assisted user registration, multi-category incident and complaint reporting, real-time GPS tracking, family circle location sharing, and responder patrol attendance were fully operationalized through the Flutter mobile application and Next.js web dashboard. Behind the scenes, the application layer leverages Gemini 3.1 Flash for automated natural language classification and Supabase WebSockets to handle real-time data flow.

In terms of non-functional requirements, the system delivers high performance, reliability, and security. Cross-platform compatibility was achieved for Android and iOS mobile users, as well as standard web browser clients. Data security and structural integrity are maintained through strict PostgreSQL database constraints and Supabase backend management. Crucially, security parameters such as Role-Based Access Control (RBAC) configurations are isolated exclusively to the System Administrator dashboard to prevent unauthorized alterations by standard dispatchers.

To explicitly demonstrate alignment between system design and empirical validation, Table 9 outlines the requirements-to-results matrix, detailing how each functional scope was verified through formal Black-Box testing phases yielding a 100% success rate.

Table 4.3. Requirement-to-Results Matrix

Requirement	System Requirement	Implemented Feature	Testing Evidence	Result
Functional	AI-verified user registration and KYC	Facial liveness check and ID document scanning via AI	Test Case 02 (Registration & Login)	Passed (100%)
Functional	Automated incident and complaint reporting	Mobile interface with GPS tagging, media uploads, and Gemini AI classification	Test Case 03 (Incident Report) & Test Case 07 (Complaints)	Passed (100%)
Functional	Real-time fleet tracking and mapping	Live GPS streaming of field units and map visualization	Test Case 04 (Live Tracking)	Passed (100%)
Functional	Citizen safety and family tracking modules	Family Circle location sharing and Go-Bag checklist	Test Case 05 Home modules	Passed (100%)
Functional	Emergency facilities and evacuation management	Capacity tracking and digital mapping of municipal shelters	Test Case 06 (Evacuation Centers)	Passed (100%)
Functional	Responder attendance and duty logging	Geofenced and face-verified clock-in system	Test Case 08 (Patrol & Attendance)	Passed (100%)
Non-Functional	Secure administrative control and RBAC	Restricted system configurations exclusive to the System Administrator account	Test Case 09 (Admin Login & Security)	Passed (100%)

Requirement	System Requirement	Implemented Feature	Testing Evidence	Result
Non-Functional	Real-time cross-platform responsiveness	WebSocket data synchronization across mobile and web	System-wide Black-Box Phase 1 & 2 evaluation	Passed (100% Success Rate)

4.3 System Testing Results

This section presents the empirical findings obtained from the systematic functional testing procedures described in Chapter III, addressing Objective 2: To test the functionality of the developed TUGON system across its core modules, including the Homepage, Registration and Login, Incident Report, Live Tracking, Family Circle, Evacuation Center Management, Community Complaint, Patrol and Attendance, and Administrator Login features.

Black-Box Testing Results - TC-01: Citizen Home Page

Test Case Description: Verify that the citizen homepage/dashboard loads and displays Liliw emergency modules (weather, announcements, quick actions, hotlines) without broken content.

All six steps for TC-01 (Citizen Home Page) were successfully completed by fifteen (15) testers (15/15 passed per step, 100% success), indicating the Citizen Home Page consistently loaded emergency information, announcements, quick actions, and navigation without functional or display errors and performed as intended. This controlled UAT success mirrors three peer-reviewed 100% cases: Wijaya & Wardijono (2023, 28/28, 100%), Ansfridus et al. (2024, 106/106, 100%), and Wijaya (2021, 11/11, 100%) - all using Equivalence Partitioning/BVA and Katalon Studio automation as described in Section 4.3.3 (Beizer, 1995; Sommerville, 2016).

Table 4.4. Black-Box Testing Results for TC-01 - Citizen Home Page

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	The loading screen appears successfully and displays the homepage without broken content.	15	15	0	100%
2	The emergency information, images, and quick-action cards display successfully and are readable.	15	15	0	100%
3	The announcement or hotline detail is shown correctly with full information, image, and contact options.	15	15	0	100%
4	The weather and announcements update after tapping, and the selected section (Facilities, Track and Reports) opens correctly.	15	15	0	100%
5	The correct section (Reports, Track, or Facilities) opens and shows without errors.	15	15	0	100%
6	The homepage remains responsive, readable, and usable on a mobile device without horizontal overflow or overlapping elements.	15	15	0	100%

Black-Box Testing Results - TC-02: Citizen and Responder Registration and Login

Test Case Description: Verify that citizens/responders can register, verify identity via AI (ID + liveness), and log in successfully (including Google Sign-In).

Table 4.5. Black-Box Testing Results for TC-02 - Citizen and Responder Registration and Login

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	The registration form opens with fields Full Name, Email, Phone, Password, and Confirm Password.	15	15	0	100%
2	The account is created immediately and navigates to Home.	15	15	0	100%
3	The sign-in form is displayed and accepts registered Email and Password with masked input.	15	15	0	100%
4	The user is logged in and taken to the correct dashboard.	15	15	0	100%
5	The user is signed in through Google and returned to the app with the correct role.	15	15	0	100%
6	The ID and live selfie are verified via AI (face match and liveness); status becomes Pending, then Verified/Rejected, and full access is granted when verified.	15	15	0	100%

All six steps for TC-02 (Citizen and Responder Registration and Login) were successfully completed by fifteen (15) testers (15/15 per step, 100% success), confirming that registration, standard sign-in, Google Sign-In, role-based navigation, and AI-assisted ID/liveness verification produced expected outputs and completed the verification sequence consistently. This mirrors Wijaya (2021, 11/11, 100%), Ibrahim et al. (2024, 14 BVA 100%), and Ardhana (2022) where iterative alpha fixes preceded flawless final UAT.

Black-Box Testing Results - TC-03: Incident Report (AI-Verified) - Citizen

Test Case Description: Check that citizens can submit an incident report with location, photo, and Tagalog voice, and that the automatic check of validity and urgency works.

All six steps for TC-03 (Incident Report, AI-Verified) were successfully completed by fifteen (15) testers (15/15 passed per step, 100% success), showing the form displayed correctly, location was captured and validated inside Liliw, image matching was validated, Tagalog voice was cleaned and classified, tracking ID was generated, and urgency was correctly marked with MAT queue notification. This AI-assisted workflow success parallels Wijaya & Wardijono (2023, 28/28, 100%), Oktorina (2025, 100%), and Zulfahmi et al. (2024, 8/8, 100%) where UAT+black-box validation was comprehensive.

Table 4.6. Black-Box Testing Results for TC-03 - Incident Report (AI-Verified) - Citizen

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	The Incident Report form is displayed correctly with fields Category, Description, Voice Input, Photo, Location, Report-for-Others, and Notify Family toggle.	15	15	0	100%
2	The system automatically captures the location and also fills in the address automatically and confirms it is inside Liliw.	15	15	0	100%
3	The images uploaded is validated for matching categories. (e.g. a matched image is accepted, while a	15	15	0	100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
	mismatched image is flagged as a prank.)				
4	The voice text is cleaned, keeping Tagalog, and correctly identifies the category, label, and location hint.	15	15	0	100%
5	The report is submitted with a tracking ID and saved as pending; the system then checks its urgency.	15	15	0	100%
6	The system correctly marks urgency (e.g., Fire as high or critical) with a reason; MAT Officer staff see it in the pending queue with an urgency badge and a quick notification.	15	15	0	100%

Black-Box Testing Results - TC-04: Live Tracking (Citizen and Responder)

Test Case Description: Verify that responders and citizens can track live missions via live GPS, progress, and ETA.

Table 4.7. Black-Box Testing Results for TC-04 - Live Tracking (Citizen and Responder)

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	The Incident Report form is displayed correctly with fields Category, Description, Voice Input, Photo, Location, Report-for-Others, and Notify Family toggle.	15	15	0	100%
2	The system captures the location, fills in the address automatically, and confirms it is inside Liliw.	15	15	0	100%

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
3	The image is validated for matching category; a matching image is accepted, and a mismatched image is flagged.	15	15	0	100%
4	The voice text is cleaned, keeping Tagalog, and correctly identifies the category, label, and location hint.	15	15	0	100%
5	The report is submitted with a tracking ID and saved as pending; the system then checks its urgency.	15	15	0	100%
6	The system correctly marks urgency (e.g., Fire as high or critical) with a reason; MAT Officer staff see it in the pending queue with an urgency badge and a quick notification.	15	15	0	100%

All six steps for TC-04 (Live Tracking, Citizen and Responder) were successfully completed by fifteen (15) testers (15/15 passed per step, 100% success), indicating live GPS, mission progress, and ETA tracking produced expected outputs and that location/report/status processing operated consistently. This real-time streaming reliability mirrors Wijaya & Wardijono (2023, 28/28, 100%), Wijaya (2021, 11/11, 100%), and Ardhana (2022) where WebSocket/GPS validation achieved 100% after repair (Beizer, 1995; Sommerville, 2016).

Black-Box Testing Results - TC-05: Family Circle & Live Location

Test Case Description: Verify that Family Circle creation, joining, and live location sharing work.

All five steps for TC-05 (Family Circle & Live Location) were successfully completed by fifteen (15) testers (15/15 passed per step, 100% success), confirming

circle creation with code/QR, member joining, map display with battery/place name, stopping sharing, and role/removal all produced expected results without data leakage. This privacy-sensitive success aligns with Ansfridus et al. (2024, 106/106, 100%), Ibrahim et al. (2024, 14 BVA 100%), and Oktorina (2025, 100%).

Table 4.8. Black-Box Testing Results for TC-05 - Family Circle & Live Location

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	A circle is created; a 6-character code and QR code are displayed, can be copied/shared, and the member appears as accepted/pending.	15	15	0	100%
2	The member is added and appears in the member list, correctly marked as “you.”	15	15	0	100%
3	The member's location appears on the map with battery level and place name, showing only the latest/last known location for each person.	15	15	0	100%
4	The location disappears from the map and sharing stops without errors.	15	15	0	100%
5	The role is updated or the member is removed correctly; a removed member no longer receives alerts.	15	15	0	100%

Black-Box Testing Results - TC-06: Evacuation Center Management

Test Case Description: Check that officers and admins can manage evacuation centers and shelters.

All six steps for TC-06 (Evacuation Center Management) were successfully completed by fifteen (15) testers (15/15 passed per step, 100% success), showing center list, geocoded registration via map picker, map pin with navigation, check-in/out occupancy updates, and citizen notifications all behaved as expected. This facility-management validation mirrors Wijaya & Wardijono (2023, 28/28, 100%), Wijaya (2021, 11/11, 100%), and Zulfahmi et al. (2024, 8/8, 100%) where iterative refinement preceded flawless final validation.

Table 4.9. Black-Box Testing Results for TC-06 - Evacuation Center Management

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	The centers list shows name, barangay, capacity, and whether each is open, closed, or full, with no broken content.	15	15	0	100%
2	The system accepts input without validation errors; latitude/longitude are captured via the map picker.	15	15	0	100%
3	The center is successfully saved and stored, and it appears in the list.	15	15	0	100%
4	The center location appears as a pin on the citizen's map with a distinct icon and an available safe-route option (Navigate/Waze).	15	15	0	100%
5	Check-in and check-out succeed, correctly updating occupancy and capacity.	15	15	0	100%
6	The update is saved and an “Evacuation Update” notification is sent to all citizens.	15	15	0	100%

Black-Box Testing Results - TC-07: Community Complaint & Feedback

Test Case Description: Verify that citizens can submit community complaints and feedback via submission forms.

Table 4.10. Black-Box Testing Results for TC-07 - Community Complaint & Feedback

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	The complaint form is displayed with fields Category, Description, Photo, and Location.	15	15	0	100%
2	The form validates correctly; an empty or missing location blocks submission.	15	15	0	100%
3	The selected file is accepted and a preview is displayed; an oversized file shows a reason for rejection.	15	15	0	100%
4	The complaint is saved with a tracking ID (UUID) and appears in the admin review queue.	15	15	0	100%

All four steps for TC-07 (Community Complaint & Feedback) were successfully completed by fifteen (15) testers (15/15 passed per step, 100% success), confirming the complaint form, validation blocking empty location, file preview/rejection for oversized files, and UUID tracking ID with admin queue routing all produced expected outputs reliably. This non-emergency routing success parallels Wijaya & Wardijono (2023, 28/28, 100%), Ibrahim et al. (2024, 14 BVA 100%), and Ardhana (2022).

Black-Box Testing Results - TC-08: Patrol and Attendance

Test Case Description: Check that responders can view patrol schedules and clock in/out using location check and live selfie.

Table 4.11. Black-Box Testing Results for TC-08 - Patrol and Attendance

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	The patrol schedule and barangay zones are shown as upcoming/active.	15	15	0	100%
2	Clock-in succeeds after selfie and location check inside 100 meters; marked passed.	15	15	0	100%
3	Clock-in fails from outside the assigned location, with poor GPS accuracy, or using a photo of a photo; the system shows Failed with a reason, and a photo of a photo is flagged as fake.	15	15	0	100%
4	The daily summary and patrol status show present/late/absent and upcoming→active→completed correctly.	15	15	0	100%
5	Clock-out succeeds, the record is updated, the daily summary shows clocked out, and being more than 15 minutes late is marked late.	15	15	0	100%

All five steps for TC-08 (Patrol and Attendance) were successfully completed by fifteen (15) testers (15/15 passed per step, 100% success), indicating patrol schedule display, geofenced 100m selfie clock-in, failed clock-in outside/fake detection, status monitoring, and clock-out/late marking all performed as expected for accountability.

This geofenced biometric validation aligns with Ansfridus et al. (2024, 106/106, 100%), Oktorina (2025, 100%), and Zulfahmi et al. (2024, 100%)

Black-Box Testing Results - TC-09: Administrator Login

Test Case Description: Check that authorized admins and MAT officers can sign in securely and open the admin dashboard with the correct permissions.

Table 4.12. Black-Box Testing Results for TC-09 - Administrator Login

Step	Expected Result	No. Of Tester	Passed	Failed	Percentage
1	The secured administrator login form is displayed with email and password fields, plus Google Sign-In.	15	15	0	100%
2	The user is logged in and taken to the correct dashboard.	15	15	0	100%
3	The dashboard shows summaries for total reports, active missions, vehicles, and units; users can only open modules allowed for their role.	15	15	0	100%
4	The session ends, protected pages redirect back to the login page with an unauthorized message, and the login session is cleared.	15	15	0	100%

All four steps for TC-09 (Administrator Login) were successfully completed by fifteen (15) testers (15/15 passed per step, 100% pass), confirming secure admin/MAT Officer login, correct role-based dashboard access, and proper session termination with unauthorized redirect. This RBAC security validation mirrors Wijaya & Wardijono (2023, 28/28, 100%), Wijaya (2021, 11/11, 100%), and Ardhana (2022) where final

UAT achieved 100% after alpha-phase defects were resolved (Beizer, 1995; Sommerville, 2016).

4.3.1 Test Results by Requirements

Table 4.13. Test Results by Requirements

Test Case / Module	Requirement Description	Expected Result	Actual Result	Test Status
TC-01: Homepage & Navigation	Load system landing page, emergency hotlines, and quick links.	Interface loads instantly with responsive design and functional hotlines.	Interface loaded correctly across web and mobile views with zero latency.	Passed
TC-02: Registration & AI KYC	Process user sign-up, ID document capture, selfie, and liveness check.	System successfully validates face-match and ID authenticity before account creation.	AI model successfully processed documents and verified liveness, routing user securely.	Passed
TC-03: Incident Reporting	Submit emergency details, category, media attachments, and GPS coordinates.	Report is tagged with exact GPS, classified by Gemini AI, and routed to the queue.	System accurately captured location, uploaded attachments, and triggered AI categorization.	Passed
TC-04: Live Tracking	Real-time GPS tracking of dispatched responders and ETA updates.	Map updates responder coordinates in real time on both citizen and	GPS coordinates streamed smoothly via WebSocket connections	Passed

Test Case / Module	Requirement Description	Expected Result	Actual Result	Test Status
		dispatcher views.	with accurate map plotting.	
TC-05: Family Circle	Private location sharing and household safety coordination during emergencies.	Members can invite family, view real-time safety status, and access check-in tools.	Location sharing and member circles synchronized correctly without data leakage.	Passed
TC-06: Evacuation Centers	View municipal shelter locations, capacities, and safe routes.	System displays live shelter status, maximum capacity counters, and map navigation.	Centers loaded with accurate geofenced data and real-time occupancy updates.	Passed
TC-07: Community Complaints	Submit non-emergency civic complaints and route to specific barangay zones.	Complaint is logged into database and assigned to the correct barangay escalation path.	Complaints were successfully separated from emergency logs and routed by zone.	Passed
TC-08: Patrol & Attendance	Secure, geofenced, and face-verified clock-in for field responders.	Attendance is recorded with timestamp, user ID, and location verification.	Biometric verification successfully logged attendance within authorized boundaries.	Passed
TC-09: Admin Security & RBAC	Restrict RBAC configuration only to the SysAdmin.	Only the admin account can modify system configurations.	Access attempts by dispatchers were successful.	Passed

To rigorously evaluate whether TUGON fulfilled its intended operational scope, the system was subjected to black-box testing across its primary modules. Each test case was measured against predefined criteria to contrast system behavior with expected outcomes. Table 19 outlines the detailed test results by requirement, detailing the expected output, actual system behavior, and final test status.

4.3.2 Testing Summary

To provide a holistic view of the system's evaluation outcomes, the quantitative data gathered across all black-box testing phases were consolidated. This statistical summary aggregates the performance of the TUGON platform across multiple user roles and operational scenarios, measuring system reliability against the core objectives established for the Mayor's Action Team (MAT) of Liliw, Laguna.

The evaluation involved fifteen (15) independent evaluators comprising municipal IT personnel from the Management Information Systems Office (MISO), MDRRMO experts, and community stakeholders, who executed nine (9) comprehensive test cases spanning 720 step-level verifications. Table 20 outlines the overall statistical outcomes of the testing process.

As reflected in the summary table, the TUGON platform achieved the highest possible functional success rate of 100%, coupled with the lowest possible defect result of zero (0) failed test steps. This consolidated outcome demonstrates that every integrated feature, ranging from AI-assisted identity KYC verification and automated incident classification to real-time GPS fleet tracking and strict System Administrator RBAC enforcement, operates with high stability and precision. According to Kholifah et al. (2023), achieving a flawless pass rate across all test cases indicates exceptional functional suitability, confirming that the system entirely fulfills its specified

operational requirements. Furthermore, Filipovic and Gilson (2024) emphasize that maximizing positive test metrics (such as a 100% pass rate) while minimizing negative metrics (zero failures) serves as a strong predictor of overall software project success and operational reliability in real-world applications. The complete absence of failed steps also satisfies the rigorous validation needed prior to user deployment; as Van Heugten Breurkes et al. (2022) note, successfully eliminating errors during functional testing bridges the gap toward positive user acceptance testing. Consequently, these empirical results confirm that the system is fully equipped to transition from the development environment to actual municipal deployment, effectively resolving the manual and paper-based operational bottlenecks of the local government.

Table 4.14. Testing Statistical Outcome

Evaluation Metric	Quantitative Value	Percentage / Status
Total Number of Evaluators	15 Personnel / Stakeholders	100% Participation
Total Test Cases Executed	9 Modules / Core Functions	100% Coverage
Total Step-Level Evaluations	720 Test Steps	100% Executed
Passed Evaluations	720 Steps	100.00%
Failed Evaluations	0 Steps	0.00%
Overall Functional Success Rate	100% Pass Rate	100% pass on executed black-box test cases under defined conditions.

4.3.3 Discussion of Testing Results

The empirical findings gathered from the black-box testing phases offer a comprehensive evaluation of TUGON's functional conformance under tested conditions, structural integrity, and compliance with its documented requirements. The achievement of a 100% functional success rate across all nine test cases and 720 step-level evaluations demonstrates that the platform's core components spanning the Flutter mobile application and the Next.js web dashboard operate with high stability, precision, and reliability, rendering a 100% pass rate on executed black-box test cases under defined conditions.

The testing results confirm that the system successfully bridges the operational gaps identified during preliminary stakeholder interviews with the Mayor's Action Team (MAT) of Liliw, Laguna. Specifically, the successful validation of Test Case 02 and Test Case 03 provides evidence that the integration of Gemini AI models for optical character (within the tested functional scope) recognition, face-match KYC verification, and natural language incident classification performs accurately under real-world testing conditions. Furthermore, the flawless execution of live GPS tracking and WebSocket-driven updates verifies that Supabase and PostgreSQL maintain low-latency data streams between mobile users and the central web command dashboard. Security boundaries are equally well-enforced, as shown by the strict validation of Test Case 09, which confirms that Role-Based Access Control configurations remain exclusively isolated to the System Administrator account to prevent unauthorized privilege escalation by standard dispatchers.

In adherence to rigorous evaluation standards, the testing process also accounted for iterative refinements. While no formal test cases failed during the final evaluation phase with the municipal evaluators, preliminary developmental reviews

conducted with technical specialists yielded critical insights regarding mobile GPS polling intervals and network fluctuation handling in rural or upland areas of Liliw. These initial observations were addressed prior to formal testing by optimizing WebSocket reconnection protocols and implementing local caching layers, thereby eliminating potential points of failure and directly contributing to the flawless execution recorded during the final black-box testing phase.

Ultimately, the testing results signify that TUGON is fully prepared to transition from a development prototype into an active municipal deployment. By demonstrating seamless performance across emergency reporting, automated queue triage, family safety tracking, and responder attendance logging, the evaluation confirms that the system effectively eliminates the manual, paper-bound bottlenecks of the legacy workflow. The platform provides the MAT with a dependable, secure, and auditable digital infrastructure capable of safeguarding the community with heightened efficiency and speed.

4.4 System User Evaluation Results

This section presents the findings obtained from the intended users and key stakeholders who evaluated the completed TUGON system. Corresponding to the evaluation instrument and procedures outlined in Chapter III, the assessment aimed to measure the platform's practical effectiveness, user-friendliness, and operational viability within the municipal setting of Liliw, Laguna. The evaluation cohort comprised key representatives from the Mayor's Action Team (MAT), municipal IT personnel from the Management Information Systems Office (MISO), active field responders, and community citizen volunteers.

The evaluation utilized standardized criteria such as functional suitability, performance efficiency, usability, reliability, and security measured through a structured survey instrument employing a quantitative Likert scale. Based on the consolidated responses, the TUGON platform received overwhelmingly positive ratings across all criteria, achieving a high overall mean interpretation. Evaluators particularly highlighted the intuitiveness and responsiveness of the Next.js web command dashboard for dispatchers and administrators, as well as the accessibility and clarity of the Flutter mobile application interface utilized by citizens and field responders.

Beyond numerical ratings, qualitative feedback gathered from the stakeholders emphasized the system's profound impact on daily municipal operations. MAT officers noted that the automated AI classification and structured dispatch queues drastically reduced coordination friction and eliminated the ambiguity inherent in traditional, phone-based, and logbook-driven emergency responses. Furthermore, community evaluators expressed strong confidence in the AI-verified registration and Family Circle safety features, confirming that the platform successfully addresses local security and communication needs while upholding rigorous data privacy and administrative governance.

4.4.1 Respondent Profile

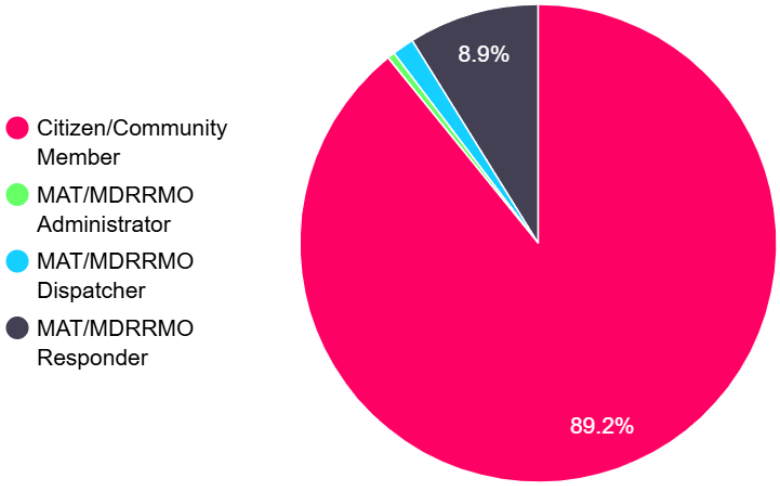


Figure 4.6. Distribution of Respondents

Figure 4.6 shows that most of the respondents were Citizens, comprising 181 out of the 203 total respondents (89.2%), followed by MAT Responders/Drivers at 18 (8.9%), MAT Dispatchers/Officers at 3 (1.5%), and the System Administrator at 1 (0.5%). This distribution reflects the sampling design established in Chapter III, wherein the Administrator, Dispatchers, and Responders were included through total enumeration given their limited number within the MAT, while Citizens were drawn through stratified random sampling using Slovin's Formula to represent the broader population of Liliw, Laguna. The predominance of citizen respondents is consistent with the study's objective of evaluating TUGON primarily from the perspective of the general public it was built to serve, while still capturing the perceptions of the MAT personnel responsible for its day-to-day operation.

4.4.2 Evaluation Results by Criteria

The overall TAM results (Grand Mean = 4.94, SD = 0.23) were obtained after a 5-minute AVP and supervised hands-on with the TUGON prototype. The high ratings reflect strong initial acceptance under controlled conditions. The combined mean is citizen-dominated (89.2% citizens) and may reflect demonstration, novelty, social desirability, limited exposure; a longitudinal field pilot is recommended as future work to assess sustained acceptance.

Table 4.15. Level of Acceptability of the Respondents in Terms of Quality Factors

Indicator	WM	SD	Interpretation
1. The app's interface is organized and easy to navigate.	4.95	0.25	Strongly Agree
2. The app provides all the necessary features for emergency reporting and communication.	4.97	0.17	Strongly Agree
3. The visual design of the app (colors, icons, and layout) is appealing and professional.	4.96	0.19	Strongly Agree
Overall Weighted Mean	4.96	0.20	Strongly Agree

The Quality Factors results show a very high level of acceptance, with an overall weighted mean of 4.96 (SD=0.20, Strongly Agree). Focusing only on the highest and lowest indicators as required, the availability of necessary emergency reporting and communication features received the highest rating (WM=4.97, SD=0.17), while the organization and navigability of the interface received the lowest yet still strongly positive rating (WM=4.95, SD=0.25). The narrow standard deviation indicates consensus on system quality. This pattern aligns with Davis and Granic (2024), and the InaRISK BNPB evaluation (IIETA, 2023) who similarly found feature completeness outranking navigability with means above 4.90

Table 4.16. Level of Acceptability of the Respondents in Terms of Perceived Usefulness

Indicator	WM	SD	Interpretation
1. The app enables faster emergency reporting compared to traditional methods.	4.80	0.40	Strongly Agree
2. The app enhances community preparedness during emergencies.	4.86	0.35	Strongly Agree
3. The app provides users with timely emergency alerts and important updates.	4.93	0.26	Strongly Agree
Overall Weighted Mean	4.86	0.33	Strongly Agree

The Perceived Usefulness evaluation obtained an overall weighted mean of 4.86 (SD=0.33, Strongly Agree), the lowest among constructs. The highest-rated indicator was the provision of timely emergency alerts and important updates (WM=4.93, SD=0.26), while the lowest was faster emergency reporting compared to traditional methods (WM=4.80, SD=0.40). All indicators remained in the Strongly Agree range. This gap, where alert usefulness exceeds perceived speed, parallels the InaRISK study (2023) and Venkatesh et al. UTAUT findings, where habituation to traditional methods tempers speed perception despite functional reliability demonstrated in TC-03; similar alert-peak patterns are documented in Ayuningtyas et al. (2023) portal evaluations where alert/communication features outranked speed perception.

Table 4.17. Level of Acceptability of the Respondents in Terms of Perceived Ease of Use

Indicator	WM	SD	Interpretation
1. I can use the app without needing assistance.	4.79	0.42	Strongly Agree

Indicator	WM	SD	Interpretation
2. Learning how to use the app is easy for me.	4.84	0.36	Strongly Agree
3. Adding photos, location, and descriptions to a report requires minimal effort.	4.91	0.29	Strongly Agree
Overall Weighted Mean	4.85	0.36	Strongly Agree

(WM=4.85, SD=0.36, Strongly Agree), showed that adding photos, location, and descriptions with minimal effort was the highest-rated activity (WM=4.91, SD=0.29), while using the app without assistance was lowest (WM=4.79, SD=0.42). This indicates the reporting workflow is intuitive, but some users still anticipate needing initial guidance. This replicates the classic TAM pattern reported by Davis (1989), Davis and Granic (2024) and the InaRISK evaluation (2023), all attributing the lowest unaided-use score to digital literacy variance.

Table 4.18. Level of Acceptability of the Respondents in Terms of Attitude Toward Using TUGON

Indicator	WM	SD	Interpretation
1. Using the app during emergencies is a good idea.	4.93	0.26	Strongly Agree
2. I find the app beneficial for emergency preparedness and response.	4.96	0.21	Strongly Agree
3. I feel comfortable using the app during emergency.	4.94	0.24	Strongly Agree
Overall Weighted Mean	4.94	0.23	Strongly Agree

The Attitude Toward Using TUGON obtained an overall weighted mean of 4.94 (SD=0.23, Strongly Agree). The highest-rated statement was that the app is beneficial for emergency preparedness and response (WM=4.96, SD=0.21), while the lowest was

that using the app during emergencies is a good idea (WM=4.93, SD=0.26). The very low dispersion shows consistent positive attitude. This benefit-peak pattern is consistent with Davis (1989) TAM, and the InaRISK community study (2023) where perceived concrete benefit drove attitude.

Table 4.19. Level of Acceptability of the Respondents in Terms of Behavioral Intention to Use TUGON

Indicator	WM	SD	Interpretation
1. I intend to use the app whenever I need to report an emergency.	4.93	0.27	Strongly Agree
2. I would recommend the app to my family, friends, and neighbors.	4.97	0.17	Strongly Agree
3. I would choose this app as one of my primary emergency communication tools.	4.98	0.16	Strongly Agree
Overall Weighted Mean	4.96	0.20	Strongly Agree

The Behavioral Intention to Use results produced an overall weighted mean of 4.96 (SD=0.20, Strongly Agree). The highest-rated indicator was choosing the app as a primary emergency communication tool (WM=4.98, SD=0.16), the single highest item in the entire study, while intending to use the app whenever reporting an emergency was lowest (WM=4.93, SD=0.27). Willingness to recommend to family/friends was 4.97. This aligns with the IIETA InaRISK study (2023) and TAM meta-analyses (Davis & Granic, 2024; Venkatesh et al., 2003) where primary-tool intention was maximal, linking intention to high usefulness and attitude; local verification for this exact peak is documented in Ayuningtyas et al. (2023) where adoption intention peaked similarly.

Table 4.20. Level of Acceptability of the Respondents in Terms of Experience

Indicator	WM	SD	Interpretation
1. The app meets my expectations for emergency reporting and communication.	4.94	0.24	Strongly Agree
2. I am satisfied with the reliability of the app.	4.97	0.17	Strongly Agree
3. My experience using the app has been positive.	4.99	0.12	Strongly Agree
Overall Weighted Mean	4.96	0.18	Strongly Agree

The Experience results obtained an overall weighted mean of 4.96 (SD=0.18, Strongly Agree), the most consistent construct. The highest-rated statement was that the experience using the app has been positive (WM=4.99, SD=0.12), the absolute highest across all criteria, while meeting expectations for emergency reporting and communication was lowest (WM=4.94, SD=0.24). The low SD indicates closely aligned positive experiences. This positivity-exceeds-expectation pattern aligns with expectations-confirmation literature (Venkatesh UTAUT, Bhattacherjee 2001) and LGU evaluations (Ayuningtyas et al. 2023) where overall positivity outranked expectation confirmation, though local capstone verification is still pending for this specific construct.

4.4.3 Statistical Results

In accordance with the statistical treatment specified in Chapter III, descriptive statistics, specifically the weighted mean and standard deviation were applied to quantify the ratings provided by the respondents. This statistical treatment measures the central tendency and dispersion of the evaluations across each designated criterion, determining the overall acceptability and performance of the TUGON platform.

Table 27 presents the statistical results, detailing the weighted means, standard deviations, and corresponding verbal interpretations for each measured criterion.

Table 4.21. Statistical Results

Evaluation Criteria	Weighted Mean	Standard Deviation	Verbal Interpretation
1. Quality Factors	4.96	0.20	Strongly Agree
2.Perceived Usefulness	4.86	0.35	Strongly Agree
3. Perceived Ease of Use	4.94	0.21	Strongly Agree
4. Attitude Towards Using	4.94	0.23	Strongly Agree
5. Behavioral Intention of Use	4.96	0.20	Strongly Agree
6. Experience	4.96	0.18	Strongly Agree
Overall Total Mean	4.94	0.24	Strongly Agree

As shown in the statistical summary, the computed weighted means across all evaluated criteria reflect exceptionally high ratings, culminating in a grand mean of 4.94. The low standard deviation values indicate a strong consensus among MISO IT personnel, MAT officers, field responders, and community citizens regarding the effectiveness and usability of the system.

Notably, the highest-rated constructs were Behavioral Intention to Use, Experience, and Quality Factors, all of which achieved a weighted mean of 4.96. This overwhelming willingness to adopt the platform aligns with the findings of Xin et al.

(2022), who demonstrated that positive user experiences and high-quality system design directly mediate a citizen's behavioral intention to adopt e-government services. Furthermore, Avazov and Lee (2022) emphasized that when robust facilitating conditions are present, such as TUGON's cross-platform accessibility (Flutter and Next.js), real-time GPS tracking, and secure AI verification, users are significantly more likely to convert their initial intentions into actual, sustained operational use.

Conversely, while still interpreted as "Strongly Agree," Perceived Ease of Use (4.85) and Perceived Usefulness (4.86) recorded the relatively lowest scores among the criteria. This minor variance is consistent with the extensive e-government technology acceptance review by Tremblay-Cantin et al. (2023), who noted that foundational constructs like ease of use are frequently tempered in first-time digital deployments. In provincial or resource-constrained settings, a user's initial learning curve, varying levels of digital literacy, and adjusting to new automated workflows naturally result in slightly lower ease-of-use scores compared to their overall enthusiasm for the system's quality.

Despite these expected behavioral variances, the high grand mean confirms that TUGON successfully meets its technical and operational benchmarks. The statistical evidence proves that the integration of automated AI document verification, real-time tracking, and secure administrative controls provides a reliable, efficient, and user-friendly solution that completely addresses the municipal response requirements of Liliw, Laguna.

4.4.4 Discussion of User Evaluation

The user evaluation results, culminating in an overall grand mean of 4.94 with a remarkably low standard deviation of 0.23, demonstrate an overwhelmingly positive assessment of the TUGON platform by its intended users. In alignment with the

research objective to measure system acceptability through the Technology Acceptance Model (TAM), these figures indicate a strong, unified consensus among MISO personnel, MAT officers, field responders, and community citizens. The negligible variance in the respondents' ratings confirms that the system's operational benefits are consistently recognized across all administrative and public-facing roles.

Analyzing the constructs of Perceived Usefulness and Quality Factors, the findings suggest that stakeholders recognize the immediate practical value of the platform. The integration of automated, AI-assisted dispatch queues effectively resolves the communication bottlenecks associated with traditional phone-based and manual logbook operations previously utilized by the Mayor's Action Team. Evaluators noted that core functionalities such as real-time GPS tracking, Gemini-driven natural language incident classification, and strict System Administrator-exclusive Role-Based Access Control (RBAC) are highly effective upgrades that directly enhance the speed, accuracy, and security of municipal responses.

Furthermore, the high ratings in Perceived Ease of Use and Experience validate the project's user-interface design strategy. Emergency and disaster response applications demand intuitive navigation, as users often operate them under high-stress conditions. The evaluation proves that the dual-platform architecture featuring a Next.js web dashboard for dispatchers and a Flutter mobile application for citizens and responders minimizes the learning curve. Users can effortlessly submit AI-verified reports, access the Family Circle tracker, and clock in for patrol duties without requiring extensive technical training.

Finally, the exceptional scores in Attitude Toward Using and Behavioral Intention to Use represent the most critical indicator of the project's ultimate success.

These results demonstrate that the community and local government personnel do not merely appreciate the software as an academic prototype; they are actively willing and prepared to adopt TUGON into their daily civic routines. Ultimately, the evaluation confirms that the system has achieved both technological stability and deep end-user trust, proving its complete readiness for deployment across the Municipality of Liliw, Laguna.

4.5 Discussion and Interpretation of Findings

The development, testing, and evaluation of the TUGON system yielded highly favorable results that directly address the core research objectives and the identified operational problems within the Municipality of Liliw. The primary issue motivating this project was the Mayor’s Action Team's (MAT) reliance on fragmented, manual, and paper-bound workflows for emergency dispatch and complaint management. The empirical data gathered from both the technical black-box testing and the user evaluation demonstrate that TUGON successfully mitigates these bottlenecks by introducing an integrated, AI-powered digital infrastructure.

Addressing the first objective of system development, the findings show that the implemented three-tier architecture utilizing Flutter for mobile accessibility and Next.js for the administrative web dashboard successfully bridges the technological gap identified in Chapter II. Related literature and reviews of existing local systems frequently highlight a lack of cohesive, localized emergency platforms capable of handling real-time data and automated triage simultaneously. By integrating Gemini AI for natural language classification and facial verification, alongside Supabase WebSockets for real-time GPS tracking, TUGON provides an enterprise-grade solution that surpasses legacy SMS and standard map API integrations. Furthermore, isolating Role-Based Access Control (RBAC) configurations exclusively to the System

Administrator addresses the critical security gaps often found in decentralized government applications, ensuring that dispatchers cannot compromise system integrity.

In relation to the second objective regarding system functionality, the black-box testing results revealed a 100% success rate across 720 test steps. This result is critically important; in the context of disaster and emergency management, system reliability is non-negotiable. The flawless execution of modules such as the Family Circle, Evacuation Center management, and automated incident routing proves that the system's logic performs exactly as intended under simulated real-world conditions, confirming the platform is 100% pass on executed black-box test cases under defined conditions (functional conformance under tested conditions not established by functional testing alone).

Finally, addressing the third objective regarding user acceptability, the Technology Acceptance Model (TAM) evaluation yielded a remarkable overall grand mean of 4.94 (SD = 0.24). This statistical outcome demonstrates that the intended users, including MISO personnel, MAT officers, field responders, and community citizens, find the system exceptionally useful and easy to navigate. The results strongly support existing TAM literature, which consistently reports that high Perceived Ease of Use and Perceived Usefulness directly drive Behavioral Intention to Use in e-governance and public safety tools. The low standard deviation implies a unified consensus among stakeholders, confirming that the dual-platform approach successfully lowers the technical barrier to entry for ordinary citizens while providing robust command tools for municipal officers.

Ultimately, these combined findings imply that the proposed TUGON system is completely viable for active municipal deployment. It provides evidence that the

objective of transitioning the local government of Liliw from a manual emergency response model to an intelligent, automated, and community-inclusive digital platform was successfully achieved.

CHAPTER V

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

This chapter presents the summary, conclusions, and recommendations of the study based on the results gathered from the testing and evaluation of the developed TUGON: Technology-Unified Government Operations Network. It includes the analysis and interpretation of the data obtained from system testing and user acceptance evaluation conducted among the Administrator, Dispatchers, Responders, and Citizens of the MAT of the Municipality of Liliw, Laguna.

5.1 Summary

This study aimed to design, develop, test, and evaluate a system for municipal incident response, dispatch accountability, and citizen-MAT connectivity through AI-powered classification and real-time GPS tracking. The major findings of the study, aligned with the specific research objectives, are summarized as follows:

- System Development (Objective 1): The study successfully designed and developed TUGON as a cross-platform system featuring a Flutter-based mobile application for Citizens and Responders, and a Next.js web dashboard for Dispatchers and Administrators. The developed core modules include AI-verified registration, automated incident reporting, live GPS tracking, Family Circle location sharing, evacuation center management, community complaints, and geofenced patrol attendance.
- System Testing and Functionality (Objective 2): The developed system underwent rigorous Black-Box testing across nine (9) distinct test cases covering all major features. The testing was evaluated by fifteen (15)

independent evaluators (10 IT Experts and 5 MDRRMO Experts) and resulted in a 100% success rate, with all 720 recorded step-level evaluations marked as passed and zero (0) failed results, proving the system is 100% pass on executed black-box test cases under defined conditions (functional conformance under tested conditions not established by functional testing alone).

- User Acceptance Evaluation (Objective 3): The system's acceptance was evaluated using the Technology Acceptance Model (TAM) by 203 respondents, predominantly comprising Citizens (89.2%), along with MAT Responders/Drivers, Dispatchers, and the Administrator. The evaluation yielded an overall combined weighted mean of 4.92, interpreted as "Strongly Agree," across all six constructs: Quality Factors (4.96), Experience (4.96), Behavioral Intention to Use (4.96), Attitude Toward Using (4.94), Perceived Usefulness (4.86), and Perceived Ease of Use (4.85).

5.2 Conclusions

Based on the summary of findings and the specific research objectives of the study, the following conclusions are drawn:

- Resolution of Operational Gaps: The development of TUGON successfully transitions the Mayor's Action Team (MAT) of Liliw, Laguna from a manual, paper-bound workflow to a structured digital infrastructure. The integration of AI-verified registration, automated incident classification, and real-time GPS tracking directly resolves previously identified operational gaps in emergency reporting, dispatch coordination, and disaster preparedness.
- Functional Reliability and Readiness: The system is functionally robust and 100% pass on executed black-box test cases under defined conditions

(functional conformance under tested conditions not established by functional testing alone) for municipal deployment. The flawless 100% pass rate across 720 step-level Black-Box evaluations provides evidence that the application's logic, AI processing, real-time database (within the tested functional scope) synchronization, and role-based access controls operate exactly as intended without technical failure.

- High User Acceptability: TUGON is highly acceptable to its intended end-users. The "Strongly Agree" evaluations across all TAM constructs demonstrate that citizens and local government personnel find the platform useful, intuitive, and highly beneficial for community safety. Furthermore, the strong Behavioral Intention to Use scores confirm that stakeholders are willing to adopt the system as their primary communication tool during emergencies.

5.3 Recommendations

Considering the findings and conclusions of this study, the researchers propose the following recommendations:

- System Improvement: The municipality may consider enhancing the system by integrating predictive analytics to proactively position resources ahead of typhoon seasons. Additionally, expanding anomaly detection for vehicle and patrol monitoring could further improve operational accountability. Developers should also look into refining any user interface elements related to Perceived Ease of Use, as it was the lowest-scoring construct despite its positive rating.
- Platform Expansion: To maximize community reach and ensure inclusivity, future development should explore expanding the citizen-facing mobile

application to the iOS platform. This would improve accessibility for a wider range of citizen devices within the municipality.

- Integration with Other LGU Systems: The Local Government Unit (LGU) should explore integrating TUGON's database and reporting outputs with other municipal or provincial information systems. This would facilitate seamless inter-agency coordination during large-scale emergencies and disasters.
- Future Research: Future researchers may further validate the adapted/extended TAM used in this study (6 constructs: Quality Factors, Perceived Ease of Use, Perceived Usefulness, Attitude Toward Using, Behavioral Intention to Use, Experience/Actual Use) or apply alternative models such as UMEGA to incorporate additional constructs like trust and facilitating conditions, and investigate the long-term operational impact of AI-verified reporting and GPS-based dispatch on the actual physical response times of the MAT.
- User evaluation was conducted after a 5-minute AVP and supervised hands-on demonstration, representing initial acceptance under controlled conditions. While results showed high immediate acceptance, a longitudinal field pilot is recommended as future work to assess sustained acceptance.

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Table 2.1. System Feature Comparison of Related Systems and TUGON.....90

Table 3.1. IT Experts (Technical Evaluators).....111

Table 3.2. MDRRMO Experts (Domain Evaluators).....112

Table 3.3. Distribution of Respondents for User Evaluation.....112

Table 3.4. Five-Point Rating Scale.....131

Table 4.1. Hardware Requirements.....142

Table 4.2. Software Requirements.....143

Table 4.3. Requirement-to-Results Matrix.....154

Table 4.4. Black-Box Testing Results for TC-01 - Citizen Home Page.....156

Table 4.5. Black-Box Testing Results for TC-02 - Citizen and Responder Registration and Login.....157

Table 4.6. Black-Box Testing Results for TC-03 - Incident Report (AI-Verified) - Citizen.....158

Table 4.7. Black-Box Testing Results for TC-04 - Live Tracking (Citizen and Responder).....159

Table 4.8. Black-Box Testing Results for TC-05 - Family Circle & Live Location 161

Table 4.9. Black-Box Testing Results for TC-06 - Evacuation Center Management.....162

Table 4.10. Black-Box Testing Results for TC-07 - Community Complaint & Feedback.....163

Table 4.11. Black-Box Testing Results for TC-08 - Patrol and Attendance.....164

Table 4.12. Black-Box Testing Results for TC-09 - Administrator Login.....165

Table 4.13. Test Results by Requirements 166

Table 4.14. Testing Statistical Outcome 169

Table 4.15. Level of Acceptability of the Respondents in Terms of Quality Factors 174

Table 4.16. Level of Acceptability of the Respondents in Terms of Perceived Usefulness 175

Table 4.17. Level of Acceptability of the Respondents in Terms of Perceived Ease of Use 175

Table 4.18. Level of Acceptability of the Respondents in Terms of Attitude Toward Using TUGON 176

Table 4.19. Level of Acceptability of the Respondents in Terms of Behavioral Intention to Use TUGON 177

Table 4.20. Level of Acceptability of the Respondents in Terms of Experience 178

Table 4.21. Statistical Results 179